The Flying Screen: Coupled Sizing, Flight Dynamics and Control of a Human-Adjacent Hovering Display

Derivation, computational treatment and design study

Abstract

A display that hovers beside a standing person is not an aircraft-sizing problem with a payload attached. It is a coupled problem: the mass of the machine determines the power it needs, the power determines the mass of the battery, the battery is part of the mass, and the trajectory the controller flies determines the power in the first place. This paper derives the whole system and solves it. The derivation combines analytical models (momentum and blade-element theory, thin-wall beam theory, rigid-body dynamics), empirical fits where theory runs out (motor mass, the acoustic anchor), numerical optimisation and simulation, and says which is which.

The paper has three parts. Part I develops the algebraic layer. From the actuator-disk control volume we obtain the induced velocity and ideal hover power, add a blade-element profile term to produce rather than assume a figure of merit, and close the mass budget. The closure reduces to $\alpha m - \beta m^q = M_0$ with $q = (3-p)/2$, where p is the exponent in a disk-area scaling law $A \sim m^p$. Theorem 6.2 shows that for $q > 1$ this equation admits exactly two, one, or zero positive solutions according to the sign of $M_0^{\max} - M_0$, that the transition is a saddle-node bifurcation, and that the lighter root is precisely the attracting fixed point of the naive sizing iteration. Theorem 7.3 gives the logarithmic sensitivities of $M_0^{\max}$ in closed form; endurance enters with exponent -2 and specific energy, figure of merit and electrical efficiency each with $+2$.

Part II derives the flight dynamics on SE(3): Newton-Euler translation and rotation with the full inertia tensor of one shared geometry, rotor thrust and reaction torque through an allocation map, first-order rotor-speed dynamics, a battery energy state, an orientation-dependent flat-plate model for the screen that couples attitude to translation and back, and a gimbal. It then builds the cascaded controller: a geometric attitude law with angular-rate feedforward from differential flatness, a gain heuristic sized from the machine's own control authority, which is computed by linear programming, and a reference governor whose keep-out constraint moves with the user.

Part III states the co-design problem as a differential-algebraic system with a fixed-point constraint, gives conditions for the inner map to be well posed and for the simulation-in-the-loop battery iteration to contract, describes the engine and its numerical methods, reports verification against the closed forms and calibration against measured hardware, and presents the resulting design.

Four findings invert the naive expectation. First, for an indoor human-adjacent machine the binding constraint is acoustic, not energetic. Second, a declared sound-power level is not a pressure at one metre, and the eight-decibel confusion between them, compounded by an uncalibrated anchor, was worth fourteen decibels of error. Third, the standoff a user specifies is measured to the display, while the rotors reach most of a metre back toward them; a safety criterion evaluated at the vehicle centre is wrong by the whole reach of the machine. Fourth, the standoff that makes the blades safe makes ordinary text too small: at 1.2 m a 1920-pixel laptop panel needs about 278 % interface scaling to meet a $16'$ character height, which leaves a desktop of about 692×389 logical pixels. The resolution is there; the usable workspace is that of a tablet.

Contents

1 Introduction

1.1 The question

Consider a display that hovers beside a person while they work standing, and follows them when they move. The engineering question is not whether a multirotor can carry a screen; multirotors carry cameras of comparable mass routinely. The question is whether the coupled system closes, and if it does, whether the machine it produces is one a person would tolerate at arm's length for half an hour.

The coupling is the whole difficulty. Write it as a cycle:

$$m \longrightarrow T \longrightarrow P \longrightarrow E \longrightarrow m_{\text{bat}} \longrightarrow m, \tag{1}$$

gross mass sets the thrust required to hover, thrust sets the power through momentum theory, power integrated over the mission sets the energy, energy divided by specific energy sets the battery mass, and the battery is part of the gross mass. A second cycle runs through the dynamics,

$$m, J \longrightarrow \text{plant} \longrightarrow u(t) \longrightarrow P(t) \longrightarrow E \longrightarrow m_{\text{bat}} \longrightarrow m, J, \tag{2}$$

because the power actually consumed depends on what the controller does, which depends on the inertia, which depends on the mass distribution, which depends on the battery. Figure 1 draws both cycles. They share the arc from energy to battery to gross mass, which is why neither can be closed without the other.

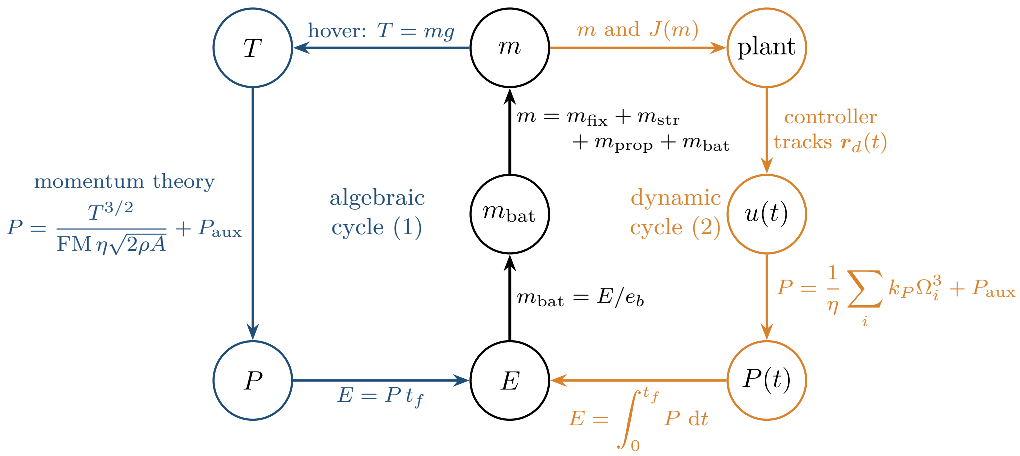


Figure 1: The two coupling cycles of the problem. The algebraic cycle (1) (blue) runs through hover thrust and momentum theory; the dynamic cycle (2) (orange) runs through the plant, the controller and the instantaneous rotor power. Both close through the same spine (black): energy sets the battery mass, and the battery is part of the gross mass, which in turn sets the thrust and the inertia. Every edge carries the relation that implements it.

Neither cycle is a mere iteration to be carried out until convergence. The first has, under a fixed rotor geometry, a genuine bifurcation: past a critical payload there is no fixed point at all. That is the mathematical content of section 6, and it is the reason a paper is warranted rather than a spreadsheet.

1.2 What this paper does

We construct the problem in three layers and then solve them together. Figure 2 shows how the sections depend on one another.

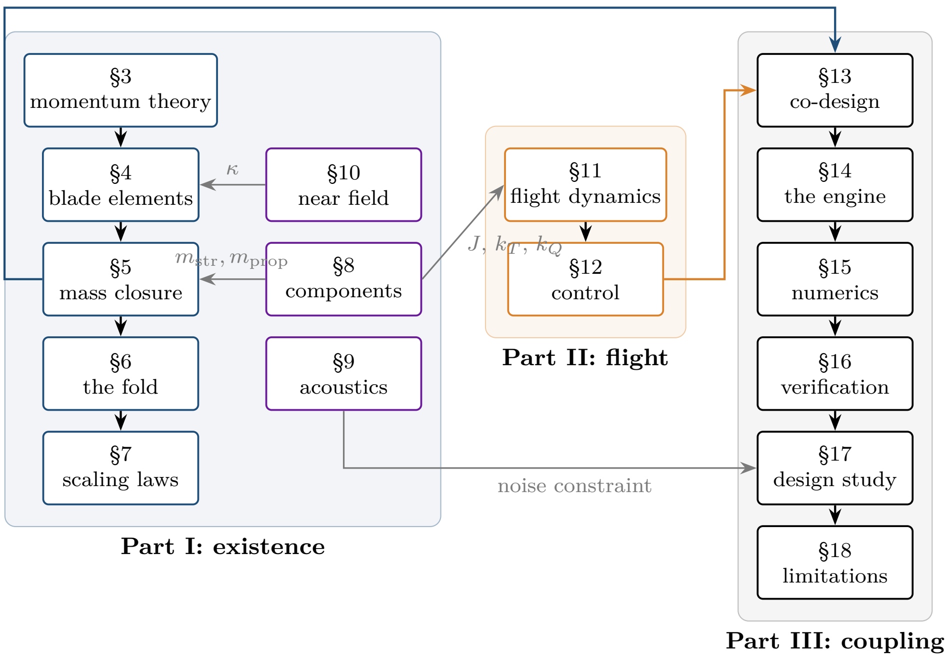


Figure 2: Dependencies between the sections. Part I establishes when a design exists; the component models (purple) replace its lumped constants and supply the corrections of section 10. Part II describes how the machine flies. The two parts meet in the co-design problem of section 13, which takes the gross mass from the closure (blue arrow) and the flight power from the controller (orange arrow); the rest of Part III solves, verifies and applies it.

Layer one: algebra. Momentum theory over an actuator disk (section 3) gives hover power as a function of thrust and disk area. A blade-element profile term (section 4) turns the figure of merit from an assumption into a consequence of blade geometry. Substituting into a mass budget (section 5) produces a single scalar equation whose solvability is the existence of the design. Section 6 analyses that equation completely: a trichotomy theorem, the closed form of the critical payload, the identification of the transition as a saddle-node bifurcation, and a proof that the naive sizing iteration converges precisely to the physically meaningful root and diverges from the other.

Layer two: differential equations. Once the machine exists its mass is constant and the flight dynamics begin (section 11). We derive the Newton-Euler equations on SE(3), the rotor wrench and its allocation map, first-order rotor dynamics, the battery energy state, and an orientation-dependent aerodynamic model for the screen. The screen is what makes this vehicle different from a camera drone: it is a large flat plate rigidly coupled to the airframe through a gimbal, so attitude modulates translational drag and drag applied off the centre of mass feeds back as a pitching moment.

Layer three: control. Section 12 constructs the cascade from position to velocity to attitude to body rate to rotor speed, states the exponential stability result of Lee, Leok and McClamroch for the geometric attitude law on SO(3) together with the ways the implemented law departs from it, and treats control allocation under saturation. Two items there are specific to this vehicle. The control authority of each axis is computed as a linear program over the rotor speed limits, and the attitude gains are bounded by it, which matters because a rotor optimised for quietness is a rotor with very little reaction torque and therefore very little yaw authority. Proposition 12.11 states what the reference governor with its moving keep-out constraint guarantees and what it does not; rate limiting alone, introduced to stop the display being whipped around the user's head, will on a curved path steer the corner-cutting reference through the user.

Coupling and computation. Section 13 states the full problem as a differential-algebraic system with a fixed-point constraint and gives conditions for the inner mass map to be well posed and for the battery iteration to contract. Sections 14 and 15 describe the engine that solves it and the numerical methods: a bounded root search for the closure that reports its own failure, secant-accelerated fixed-point iteration for the battery loop with the simulator inside it and a verification flight at the end, fixed-step Runge-Kutta for the trajectory, and differential evolution over the design vector.

1.3 Contributions

The mathematical contributions are theorems 6.2, 6.6, 6.8, 6.11 and 7.3 and propositions 7.8 and 13.3: a complete description of the sizing equation including its bifurcation structure and the convergence behaviour of the natural iteration; exact logarithmic sensitivities of the critical payload; the condition on the disk-area exponent under which the fold exists at all, and the separate statement that an endurance ceiling survives the fold's disappearance at the constant-disk-loading boundary; and the slope condition under which the component mass closure is uniquely solvable. The control results (proposition 12.11, the authority computation and the gain heuristic) are stated with their conditions and are verified by simulation, not proved.

The modelling contributions are the calibration of the acoustic and motor submodels against measured hardware (section 16), and the near-field corrections of section 10, which close a stated gap and, more usefully, bound its size.

The engineering contribution is the design study of section 17, whose main conclusion is that for this class of machine the binding constraint is acoustic rather than energetic, and that stating a noise figure without stating whether it is a sound power or a pressure, and whether it is free-field or in a room, is worth more decibels than any design decision available.

1.4 What this paper does not do

No hardware was built. Every number reported is the output of a model whose assumptions are stated where they are introduced, and section 18 collects the places where those assumptions are known to be inadequate. In particular the rotor aerodynamics are momentum theory with a single profile term and three near-field corrections; there is no blade-element momentum theory, no wake model, and no treatment of the recirculating flow that a machine of this disk loading establishes in a closed room within a minute of taking off. The acoustic model is a broadband scaling law fitted to four aircraft, coupled to a single-coefficient Sabine room; tonal content is computed but not summed into the level.

2 Notation, frames and standing assumptions

2.1 Frames

Let $\mathcal{I}$ be an inertial frame with orthonormal basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, $\boldsymbol{e}_3$ pointing vertically upward, so that gravity acts along $-\boldsymbol{e}_3$. Let $\mathcal{B}$ be a frame fixed in the airframe with origin at its centre of mass, $\boldsymbol{b}_3$ along the nominal thrust axis and $\boldsymbol{b}_1$ pointing toward the user.

The attitude is the rotation $R \in \mathrm{SO}(3)$ mapping body coordinates to inertial coordinates: a vector with body coordinates $\boldsymbol{u}$ has inertial coordinates $R\boldsymbol{u}$. Position and velocity of the

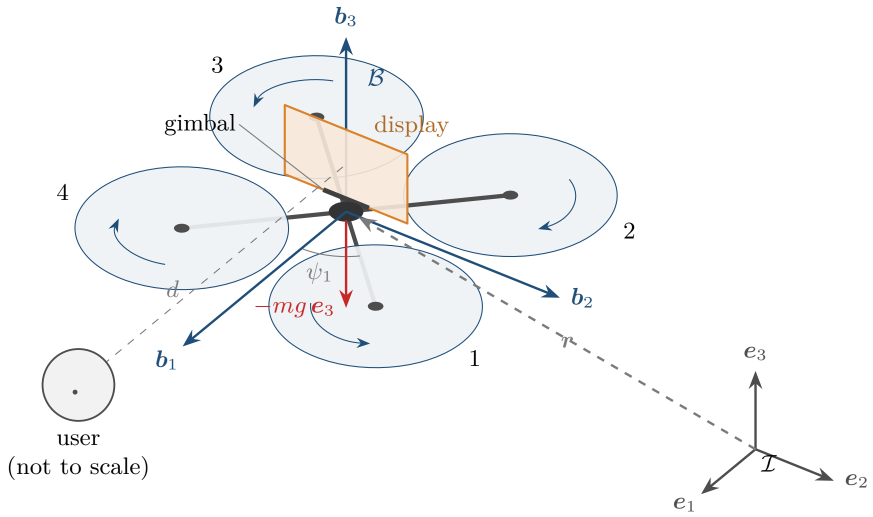


Figure 3: Frames and layout, to scale for the design of section 17 except for the user. $\mathcal{I}$ is inertial with $\boldsymbol{e}_3$ up; $\mathcal{B}$ is attached at the centre of mass with $\boldsymbol{b}_1$ along the screen normal toward the user, $\boldsymbol{b}_3$ along the nominal thrust and $\boldsymbol{b}_2 = \boldsymbol{b}_3 \times \boldsymbol{b}_1$. The four rotors sit on arms at $\psi_i = \pi/4 + (i-1)\pi/2$ from $\boldsymbol{b}_1$ (the $\times$ layout) with alternating spin senses, marked by the arrows: rotors 1 and 3 turn anticlockwise seen from above, 2 and 4 clockwise. The display stands above the rotor plane, normal to $\boldsymbol{b}_1$, on a two-axis gimbal hinged at its lower edge. $\boldsymbol{r}$ locates the centre of mass in $\mathcal{I}$, and d is the standoff to the user's eye.

centre of mass in $\mathcal{I}$ are $\boldsymbol{r}, \boldsymbol{v} \in \mathbb{R}^3$, and the angular velocity of $\mathcal{B}$ relative to $\mathcal{I}$, expressed in $\mathcal{B}$, is $\boldsymbol{\omega} \in \mathbb{R}^3$. Figure 3 fixes the geometry that the rest of the paper uses.

2.2 Algebraic preliminaries

The hat map $\widehat{\cdot} : \mathbb{R}^3 \to \mathfrak{so}(3)$ is the isomorphism

$$\widehat{\boldsymbol{u}} = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}, \qquad \widehat{\boldsymbol{u}}\boldsymbol{w} = \boldsymbol{u} \times \boldsymbol{w} \quad \forall\, \boldsymbol{u}, \boldsymbol{w} \in \mathbb{R}^3, \tag{3}$$

with inverse the vee map $(\cdot)^\vee : \mathfrak{so}(3) \to \mathbb{R}^3$. We use repeatedly the identities

$$\widehat{(R\boldsymbol{u})} = R\,\widehat{\boldsymbol{u}}\,R^\top, \qquad \operatorname{tr}(\widehat{\boldsymbol{u}}^\top \widehat{\boldsymbol{w}}) = 2\,\boldsymbol{u}^\top \boldsymbol{w}, \qquad R \in \mathrm{SO}(3). \tag{4}$$

A unit quaternion $q = (q_0, \boldsymbol{q}_v) \in \mathbb{S}^3$ represents the same rotation through

$$R(q) = (q_0^2 - \boldsymbol{q}_v^\top \boldsymbol{q}_v)\, I + 2\,\boldsymbol{q}_v \boldsymbol{q}_v^\top + 2q_0\, \widehat{\boldsymbol{q}_v}, \tag{5}$$

a two-to-one covering, $R(q) = R(-q)$. We integrate the quaternion rather than R because eq. (5) carries no coordinate singularity, unlike any three-parameter attitude representation.

2.3 Symbols

Table 13 in appendix A lists every symbol. The ones used constantly are: m gross mass, g gravitational acceleration, ρ air density, N rotor count, D rotor diameter, $A = N\pi D^2/4$ total actuator disk area, T total thrust, v_i induced velocity, FM figure of merit, η electrical efficiency, e_b usable battery specific energy, t_f mission duration, λ thrust margin, J inertia tensor.

2.4 Standing assumptions

The following hold throughout unless explicitly suspended. They are stated here so that later results can be read as conditional on them rather than as claims about reality.

Assumption 2.1 (Incompressible, steady, inviscid outer flow)**.** The air is incompressible with constant density ρ, the flow through the rotor disk is steady in the mean, and outside thin boundary layers on the blades it is inviscid. Tip Mach numbers stay below 0.3, which the designs of section 17 satisfy by a wide margin.

Assumption 2.2 (Rigid airframe)**.** The airframe is a rigid body. Section 8 sizes the arms so that the first structural bending mode lies at least an order of magnitude above the attitude-loop bandwidth, which is the condition under which this assumption is self-consistent, and the engine reports the ratio so that it can be checked rather than assumed.

Assumption 2.3 (Constant mass)**.** $\dot{m} = 0$ in flight. The battery loses energy, not mass. The relativistic mass defect $\dot{m} = -P/c^2 \approx -1 \times 10^{-15}\,\mathrm{kg/s}$ at $P = 100\,\mathrm{W}$ is ignored.

Assumption 2.4 (Uniform disk loading)**.** Within the momentum-theory layer each rotor is modelled as an actuator disk carrying uniform loading. Non-uniformity and tip loss are absorbed into an empirical induced-power factor $\kappa > 1$ (section 4).

Assumption 2.5 (Rotors share thrust equally in trim)**.** In hover each of the N rotors carries mg/N. Differential thrust for attitude control is a perturbation about this trim.

Assumption 2.6 (Quasi-steady rotor aerodynamics)**.** Rotor thrust and torque respond instantaneously to rotor speed; all actuator lag is placed in the rotor-speed dynamics of section 11. Inflow dynamics, whose time constant is of order $R/v_i \approx 0.1\,\mathrm{s}$, are not modelled.

Assumption 2.7 (Perfect state estimation)**.** The controller has exact knowledge of $\boldsymbol{r}, \boldsymbol{v}, R, \boldsymbol{\omega}$ and of the human's position. This is the assumption most obviously false in practice and is revisited in section 18.

Remark 2.8. Assumptions 2.1, 2.4 and 2.6 belong to the sizing layer; assumptions 2.2, 2.3, 2.5 and 2.7 to the dynamic layer. The separation matters: the algebraic results of sections 6 and 7 do not depend on the dynamic assumptions at all, and remain valid statements about the closure equation whatever one believes about the controller.

Part I
The Algebraic Layer: Existence of a Design

3 Momentum theory over an actuator disk

Everything downstream rests on one result: the power required to hold a mass in the air, as a function of how much air the machine can reach. We derive it from a control volume rather than quote it, because the exponents that appear are the entire argument of section 6 and it matters that they are earned.

3.1 The model

Model 3.1 (Rankine-Froude actuator disk)**.** The rotor is replaced by a permeable disk of area A across which the static pressure jumps by Δp while the velocity is continuous. The flow is axisymmetric, the fluid is at rest far upstream, and the slipstream is a well-defined streamtube. Rotation imparted to the wake, viscous losses and the finite number of blades are all excluded; they return in section 4.

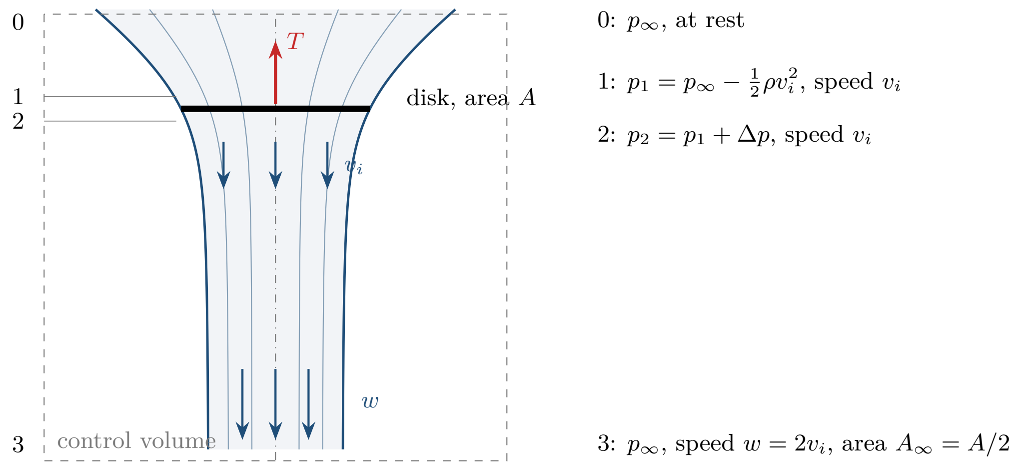


Figure 4: The actuator disk in hover. Air at rest far upstream (station 0) is drawn into a contracting streamtube, crosses the disk (stations 1 and 2) at the induced velocity v_i with a jump Δp in static pressure, and leaves in a wake of speed w and area A_∞ at ambient pressure (station 3). The momentum theorem is applied to the dashed control volume. The tube is drawn from continuity, $r^2 u = R^2 v_i$, with the axial velocity u of the vortex-cylinder solution on the axis; it contracts to $A_\infty = A/2$, as theorem 3.2 requires.

This is the classical model of Rankine and W. Froude, given in the form used here by Glauert [7] and, in the rotorcraft setting, by Leishman [14, Ch. 2] and Johnson [10, Ch. 2].

Let v_i denote the axial velocity induced *at the disk* and w the axial velocity in the far wake, both measured relative to the undisturbed fluid. Let A_∞ be the cross-sectional area of the far wake. Figure 4 shows the control volume and the four stations used below.

3.2 The three conservation laws

Mass. The streamtube carries a constant mass flow. Evaluating at the disk,

$$\dot{m}_{\text{air}} = \rho A v_i = \rho A_\infty w. \tag{6}$$

Momentum. Applying the momentum theorem to a control volume enclosing the streamtube between a station far upstream, where the fluid is at rest and the pressure is p_∞, and a station far downstream, where the pressure has returned to p_∞ and the velocity is w, the only external axial force is that exerted by the disk on the fluid, equal and opposite to the thrust:

$$T = \dot{m}_{\text{air}}\,(w - 0) = \rho A v_i w. \tag{7}$$

Energy. Bernoulli's equation holds along streamlines on each side of the disk separately, but not across it, because the disk does work on the fluid. Upstream, from the far field to the disk face,

$$p_\infty = p_1 + \tfrac{1}{2}\rho v_i^2, \tag{8}$$

and downstream, from the disk face to the far wake,

$$p_2 + \tfrac{1}{2}\rho v_i^2 = p_\infty + \tfrac{1}{2}\rho w^2. \tag{9}$$

Subtracting eq. (8) from eq. (9) eliminates p_∞ and v_i:

$$\Delta p \coloneqq p_2 - p_1 = \tfrac{1}{2}\rho w^2, \qquad \text{whence} \qquad T = A\,\Delta p = \tfrac{1}{2}\rho A w^2. \tag{10}$$

3.3 The central result

Theorem 3.2 (Froude's theorem and the hover power law)**.** *Under model 3.1 the far-wake velocity is exactly twice the velocity induced at the disk,*

$$w = 2v_i, \tag{11}$$

the thrust is

$$T = 2\rho A v_i^2, \qquad \textit{equivalently} \qquad v_i = \sqrt{\frac{T}{2\rho A}}, \tag{12}$$

the far wake contracts to half the disk area, $A_\infty = A/2$, and the power delivered to the fluid is

$$P_{\text{ideal}} = T v_i = \frac{T^{3/2}}{\sqrt{2\rho A}}. \tag{13}$$

Proof. Equate the two expressions for thrust, eqs. (7) and (10):

$$\rho A v_i w = \tfrac{1}{2}\rho A w^2.$$

Since $T > 0$ forces $w > 0$ and $A > 0$, division by $\rho A w$ is legitimate and gives $v_i = w/2$, which is eq. (11). Substituting $w = 2v_i$ into eq. (7) yields $T = 2\rho A v_i^2$, and solving for v_i gives eq. (12); the positive root is selected because the wake is directed downward for positive thrust. Wake contraction follows from eq. (6): $A_\infty = A v_i / w = A/2$.

For the power, the rate of work done by the disk on the fluid equals the flux of kinetic energy out of the control volume, the fluid entering at rest:

$$P = \tfrac{1}{2}\dot{m}_{\text{air}} w^2 = \tfrac{1}{2}(\rho A v_i)(2v_i)^2 = 2\rho A v_i^3 = (2\rho A v_i^2)\, v_i = T v_i,$$

so $P = T v_i$, which is also what one obtains directly as force times the velocity of the disk relative to the fluid it acts on. Substituting eq. (12) gives $P_{\text{ideal}} = T\sqrt{T/(2\rho A)} = T^{3/2}(2\rho A)^{-1/2}$. □

Remark 3.3. The factor of two in eq. (11) is the reason the disk sees only half the dynamic pressure of the wake, and is the origin of the 3/2 exponent that drives the entire feasibility analysis. Half the velocity increment happens upstream of the disk and half downstream; fig. 5 shows both halves.

Corollary 3.4 (Hover)**.** *In hover, $T = mg$, so*

$$v_i = \sqrt{\frac{mg}{2\rho A}}, \qquad P_{\text{ideal}} = \frac{(mg)^{3/2}}{\sqrt{2\rho A}}. \tag{14}$$

At fixed disk area, $P_{\text{ideal}} \propto m^{3/2}$.

3.4 Disk loading

Definition 3.5 (Disk loading)**.** $\mathrm{DL} \coloneqq T/A$, with units of pressure.

Rewriting eqs. (12) and (13) in terms of disk loading isolates the physically meaningful group:

$$v_i = \sqrt{\frac{\mathrm{DL}}{2\rho}}, \qquad \frac{P_{\text{ideal}}}{T} = \sqrt{\frac{\mathrm{DL}}{2\rho}}, \qquad \frac{P_{\text{ideal}}}{m} = g\sqrt{\frac{\mathrm{DL}}{2\rho}}. \tag{15}$$

Observation 3.6. Power per unit mass in hover depends on the disk loading alone, not on mass and area separately. A hovering machine is therefore not primarily asking for powerful motors; it is asking for low disk loading. The quantity $T/P_{\text{ideal}} = \sqrt{2\rho/\mathrm{DL}}$, the power loading, is the correct figure of merit for a hovering vehicle and improves as disk loading falls.

Figure 6 places the design of section 17 on this law.

This single observation determines the shape of every machine in section 17: large, slowly turning rotors, because eq. (15) says that is the only lever with any authority, and section 9 will show that the same choice is what makes the machine quiet.

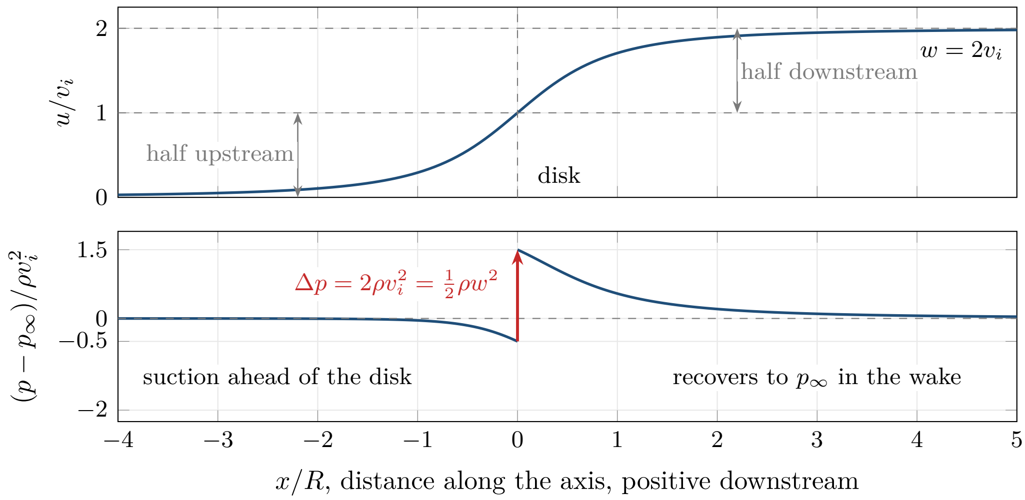


Figure 5: Axial velocity and static pressure along the rotor axis, from Bernoulli's equation applied separately on each side of the disk with the vortex-cylinder velocity $u/v_i = 1 + \xi/\sqrt{1+\xi^2}$, $\xi = x/R$. The velocity is continuous through the disk and rises from 0 to v_i upstream and from v_i to $w = 2v_i$ downstream, the two halves of eq. (11). The pressure falls ahead of the disk, jumps by $\Delta p = \frac{1}{2}\rho w^2$ across it, eq. (10), and recovers to p_∞ in the wake.

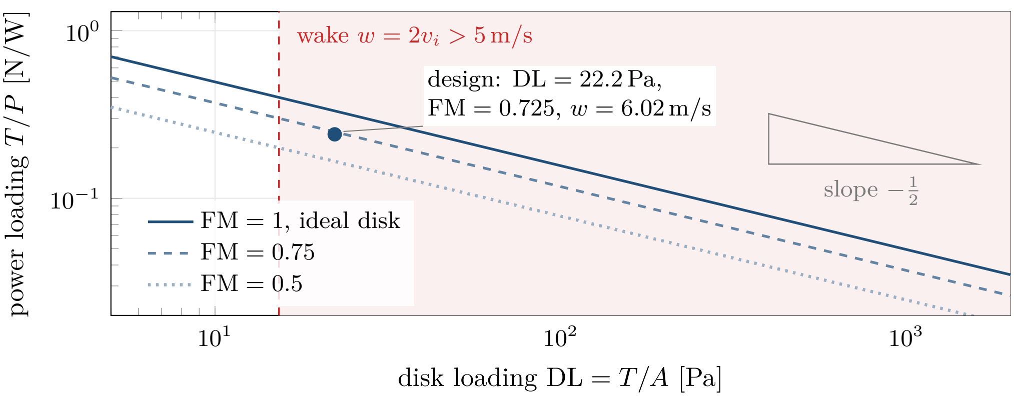


Figure 6: Power loading against disk loading, $T/P = \mathrm{FM}\sqrt{2\rho/\mathrm{DL}}$, from eq. (15) and the definition of FM in section 4. On logarithmic axes every figure of merit gives a line of slope $-\frac{1}{2}$: halving the disk loading buys a factor $\sqrt{2}$ in thrust per watt, whatever the rotor. The shaded region is where the wake exceeds $5\,\mathrm{m/s}$, the speed at which loose paper begins to move (requirement (R3) of section 17); the design sits just inside it, at DL $= 22.2\,\mathrm{Pa}$ and $w = 6.0\,\mathrm{m/s}$.

3.5 Axial flight and the validity of theorem 3.2

For axial motion at climb velocity V_c relative to the air, the same control volume argument with a non-zero inflow gives

$$T = 2\rho A(V_c + v_i)v_i \quad \Longrightarrow \quad v_i = -\frac{V_c}{2} + \sqrt{\left(\frac{V_c}{2}\right)^2 + \frac{T}{2\rho A}}, \tag{16}$$

which reduces to eq. (12) at $V_c = 0$ and is implemented in that form. The power is then $P = T(V_c + v_i)$.

Equation (16) is valid in climb and in the windmill-brake state, but fails in descent for $-2 \lesssim V_c/v_{i,h} \lesssim 0$, where $v_{i,h}$ is the hover induced velocity. This is the vortex ring state, where the streamtube assumption breaks down and the flow is unsteady and recirculating [14, §2.13]. The machines considered here descend slowly and the engine does not enter that regime, but it is a real limitation of the model rather than a technicality: a fast descent to land is precisely the manoeuvre it cannot predict.

Two further caveats are worth recording. Momentum theory gives a lower bound on the power: it is the power required to produce the thrust by the most efficient possible redistribution of momentum, and any real rotor does worse. Section 4 quantifies by how much. Second, the theory says nothing about how the loading is produced, so it cannot predict stall, compressibility or Reynolds effects; those enter through the blade element layer.

4 Blade element corrections and the figure of merit

Momentum theory bounds the power from below. A real rotor pays two further tolls: the induced flow is not uniform and the blades are finite in number, and the blade sections have profile drag. This section derives the second from blade element theory and absorbs the first into a coefficient, with the consequence that the figure of merit becomes an output of blade geometry rather than an input.

4.1 Induced power factor

Model 4.1 (Non-ideal induced power)**.** The induced power is written

$$P_{\text{ind}} = \kappa\, T v_i, \qquad \kappa > 1, \tag{17}$$

where κ collects non-uniform inflow, tip loss and swirl. Typical values for small rotors are $\kappa \in [1.10, 1.25]$ [14, §2.4]; we take $\kappa = 1.15$.

The alternative is Prandtl's tip-loss factor B, with the induced velocity computed over an effective disk of radius BR. The two are related by $\kappa \approx B^{-2}$ for small losses; eq. (17) is used because it keeps the algebra of section 5 in closed form.

4.2 Profile power

Definition 4.2 (Solidity)**.** For a rotor with N_b blades of constant chord c and radius R, the solidity is the ratio of blade planform area to disk area,

$$\sigma := \frac{N_b c R}{\pi R^2} = \frac{N_b c}{\pi R} = \frac{N_b}{\pi\,\mathrm{AR}}, \qquad \mathrm{AR} := \frac{R}{c}. \tag{18}$$

Figure 7 fixes the notation of blade element theory.

Proposition 4.3 (Profile power of a rotor in hover)**.** *Let each blade section have drag coefficient C_{d_0}, taken constant along the span, and let the rotor turn at angular rate Ω with tip speed $V_{\text{tip}} = \Omega R$. Neglecting the inflow angle in the drag direction, the profile power is*

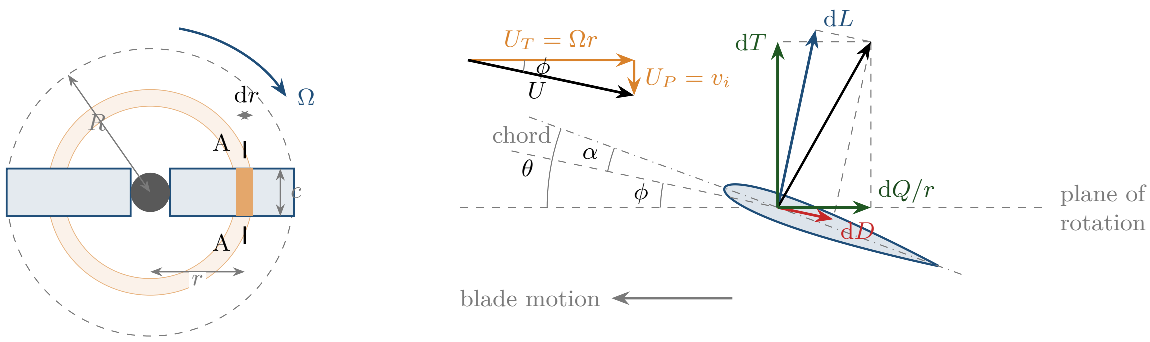
$$P_{\text{prof}} = \frac{1}{8}\,\rho\, A\, V_{\text{tip}}^3\, \sigma\, C_{d_0}. \tag{19}$$

Proof. Consider a blade element at radius r of span $\mathrm{d}r$. Its sectional velocity is Ωr to leading order, so the drag on the element is

$$\mathrm{d}D = \tfrac{1}{2}\rho(\Omega r)^2 c\, C_{d_0}\, \mathrm{d}r,$$

and the power absorbed is $\mathrm{d}P = \Omega r\, \mathrm{d}D$. Integrating over the blade and multiplying by N_b blades,

$$P_{\text{prof}} = N_b \int_0^R \tfrac{1}{2}\rho \Omega^3 r^3 c\, C_{d_0}\, \mathrm{d}r = \tfrac{1}{8}\rho\, \Omega^3 c\, N_b\, C_{d_0} R^4.$$



(a) rotor seen from above

(b) section A-A, seen from the tip

Figure 7: Blade element notation. (a) A two-bladed rotor of radius R and chord c, so $\sigma = N_b c/(\pi R)$; the annulus at radius r of width $\mathrm{d}r$ cuts an element from each blade. (b) The element in section. It meets the air at $U_T = \Omega r$ in the plane of rotation and $U_P = v_i$ through it, so the resultant U is inclined at the inflow angle $\phi = \arctan(U_P/U_T)$ and the angle of attack is $\alpha = \theta - \phi$ for a pitch θ. Lift $\mathrm{d}L$ is normal to U and drag $\mathrm{d}D$ along it. Resolved on the rotor axes they give the thrust $\mathrm{d}T = \mathrm{d}L\cos\phi - \mathrm{d}D\sin\phi$ and the in-plane force $\mathrm{d}Q/r = \mathrm{d}L\sin\phi + \mathrm{d}D\cos\phi$. Proposition 4.3 keeps the drag part of the second in the limit $\phi \to 0$, where $\mathrm{d}P = \Omega r\,\mathrm{d}D$.

Substituting $c = \sigma\pi R/N_b$ from eq. (18),

$$P_{\text{prof}} = \tfrac{1}{8}\rho\,\Omega^3 C_{d_0}\,\sigma\pi R^5 = \tfrac{1}{8}\rho\,C_{d_0}\,\sigma\,(\pi R^2)(\Omega R)^3 = \tfrac{1}{8}\rho\,A\,V_{\text{tip}}^3\,\sigma\,C_{d_0}. \qquad \square$$

4.3 Figure of merit

Definition 4.4 (Figure of merit)**.**

$$\mathrm{FM} := \frac{P_{\text{ideal}}}{P_{\text{sh}}} = \frac{T v_i}{\kappa T v_i + P_{\text{prof}}} \in (0,1). \tag{20}$$

Proposition 4.5 (Figure of merit at fixed blade loading)**.** *Define the blade loading coefficient*

$$\frac{C_T}{\sigma} := \frac{T}{\rho A V_{\text{tip}}^2 \sigma}. \tag{21}$$

Then

$$\mathrm{FM} = \left[\kappa + \frac{C_{d_0}}{8}\,\frac{1}{(C_T/\sigma)}\,\frac{V_{\text{tip}}}{v_i}\right]^{-1}, \qquad \frac{V_{\text{tip}}}{v_i} = \sqrt{\frac{2}{\sigma\,(C_T/\sigma)}}, \tag{22}$$

so that at fixed blade loading the figure of merit depends on the blade geometry only through the group $\sigma\,(C_T/\sigma) = C_T$, and is independent of rotor size and of tip speed separately.

Proof. From eqs. (19) and (21), $\rho A\sigma = T/(V_{\text{tip}}^2(C_T/\sigma))$ and hence

$$P_{\text{prof}} = \tfrac{1}{8}V_{\text{tip}}^3 C_{d_0}\cdot\frac{T}{V_{\text{tip}}^2(C_T/\sigma)} = \frac{C_{d_0}}{8}\,\frac{T\,V_{\text{tip}}}{(C_T/\sigma)}.$$

Dividing eq. (20) through by Tv_i gives the first identity. For the second, eq. (12) gives $v_i^2 = T/(2\rho A)$ and eq. (21) gives $V_{\text{tip}}^2 = T/(\rho A\sigma(C_T/\sigma))$, so $V_{\text{tip}}^2/v_i^2 = 2/(\sigma(C_T/\sigma))$. $\square$

Corollary 4.6 (Design rule)**.** *Holding blade loading at its design value and increasing solidity lowers the tip speed required to carry a given thrust, $V_{\text{tip}} \propto \sigma^{-1/2}$, and lowers the profile power with it, $P_{\text{prof}} \propto V_{\text{tip}} \propto \sigma^{-1/2}$.*

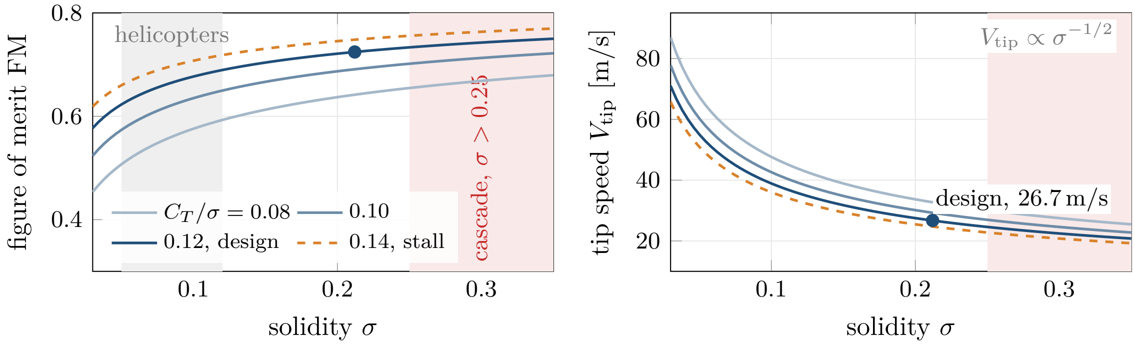


Figure 8: The design rule of corollary 4.6 at the thrust per rotor of the design, $T = 4.23\,\text{N}$, with its induced factor $\kappa = 1.168$ and $C_{d_0} = 0.023$ held fixed. Left: at fixed blade loading the figure of merit eq. (22) rises with solidity, because the profile term falls as $\sigma^{-1/2}$. Right: the tip speed that carries the thrust falls as $\sigma^{-1/2}$, eq. (21), which is what makes the rotor quiet (section 9). Both gains stop at the cascade limit $\sigma = 0.25$ of assumption 4.7 and at the stall limit $C_T/\sigma = 0.14$ of assumption 4.8. The design (•) sits at $\sigma = 0.212$ and $C_T/\sigma = 0.120$; helicopter rotors occupy the grey band.

Proof. Immediate from eq. (21) with T, ρA and C_T/σ held fixed, together with

$$P_{\text{prof}} = \frac{C_{d_0}}{8} \frac{T\,V_{\text{tip}}}{C_T/\sigma}. \qquad \square$$

Corollary 4.6 is not obvious and is used heavily later: a fatter, slower blade is better on *both* counts, quieter by section 9 and cheaper in profile power, until the assumptions break. They break in two places, and both are enforced as constraints. Figure 8 shows both effects at the design's thrust and the two limits that stop them.

4.4 Two limits on solidity and blade loading

Assumption 4.7 (Blade-element independence)**.** Blade element theory treats each section as an isolated aerofoil in a uniform inflow. This fails once the blades operate as a cascade. Helicopter rotors run $\sigma \in [0.05, 0.12]$; high-solidity propellers reach about 0.25. We impose $\sigma \le \sigma_{\max} = 0.25$ and report violations, because beyond it the model flatters the rotor and one no longer has a propeller but a ducted fan without a duct.

Assumption 4.8 (Stall margin)**.** Sectional lift coefficient is bounded by stall, which through $C_T/\sigma \approx C_L/6$ for linearly twisted blades bounds the blade loading. We impose $C_T/\sigma \le 0.14$ and design at 0.12.

Remark 4.9 (Fixed-pitch rotors run at constant C_T). For a fixed-pitch rotor at fixed collective, $T \propto \Omega^2$ so $C_T = T/(\rho A V_{\text{tip}}^2)$ is invariant under changes of rotor speed. The blade-loading constraint therefore binds identically in hover and at maximum thrust, which is why the engine reports the same C_T/σ at both conditions. Stall margin cannot be recovered by spinning faster.

4.5 Reynolds number

Both propositions 4.3 and 4.5 take C_{d_0} constant. A design driven toward low tip speed by corollary 4.6 operates at chord Reynolds numbers

$$\text{Re} = \frac{0.7\,V_{\text{tip}}\,c}{\nu}, \tag{23}$$

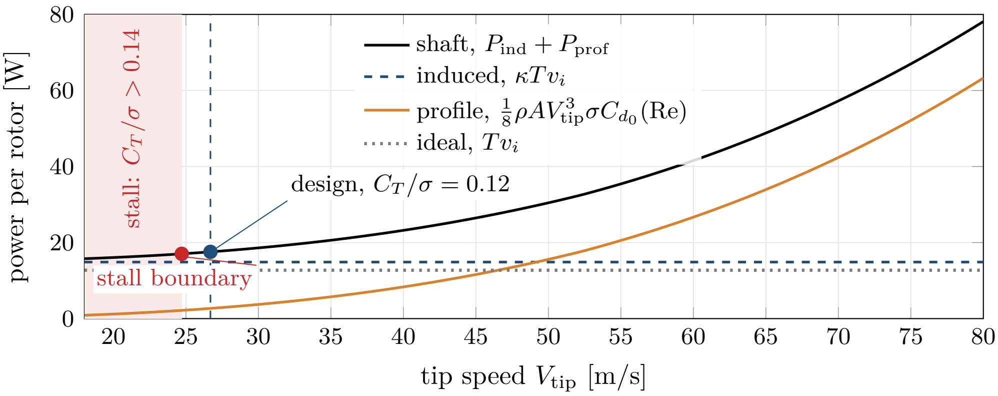


Figure 9: One rotor of the design at its hover thrust, with the geometry fixed and the tip speed varied (engine data). The induced power $\kappa T v_i$ does not depend on tip speed; the profile power grows as $V_{\text{tip}}^3 C_{d_0}(\text{Re})$, slightly slower than V_{tip}^3 at low speed because C_{d_0} falls as the Reynolds number rises, eq. (24). The shaft power is therefore least at the lowest tip speed the blade-loading limit admits, on the stall boundary $C_T/\sigma = 0.14$ at 24.71 m/s. The design runs at $C_T/\sigma = 0.12$ and 26.69 m/s, and pays for its stall margin with the difference between 17.56 W and 17.04 W of shaft power per rotor.

evaluated at the $0.7R$ station, of order 10^5 or below, where profile drag rises sharply. Treating C_{d_0} as constant would hand the optimiser a free lunch: it would drive the tip speed arbitrarily low. We therefore use

$$C_{d_0}(\text{Re}) = C_{d_0}^{\text{ref}} \min\left\{ \left(\frac{\text{Re}_{\text{ref}}}{\text{Re}} \right)^{0.2}, 3 \right\}, \qquad \text{Re}_{\text{ref}} = 2 \times 10^5, \tag{24}$$

an empirical power law of the standard form for attached turbulent flow, capped to keep it from diverging. This is the crudest submodel in the paper and is flagged as such; its only job is to prevent an unphysical optimum. Figure 9 shows the resulting trade for one rotor of the design.

4.6 Electrical power

Motor and electronic speed controller losses are lumped into a single efficiency η, so the electrical power drawn by the propulsion system is

$$P_{\text{el}} = \frac{P_{\text{sh}}}{\eta} = \frac{\kappa T v_i + P_{\text{prof}}}{\eta} = \frac{P_{\text{ideal}}}{\text{FM}\,\eta} = \frac{T^{3/2}}{\text{FM}\,\eta\,\sqrt{2\rho A}}. \tag{25}$$

The final form is the one carried into section 5; the intermediate forms are what the engine actually evaluates, so that FM is computed from eq. (20) rather than assumed.

Adding the non-propulsive load P_{aux} (display, computer, sensors), the total electrical power in hover is

$$P_{\text{tot}}(m) = \frac{(mg)^{3/2}}{\text{FM}\,\eta\,\sqrt{2\rho A}} + P_{\text{aux}}. \tag{26}$$

Remark 4.10. Unlike a camera drone, this vehicle carries a payload that consumes power continuously. P_{aux} is not negligible: at the designs of section 17 it is 14 to 16 W against roughly 80 W of propulsion, and because it is constant it converts directly into battery mass, appearing in section 5 as an addition to the fixed load rather than as a term scaling with m.

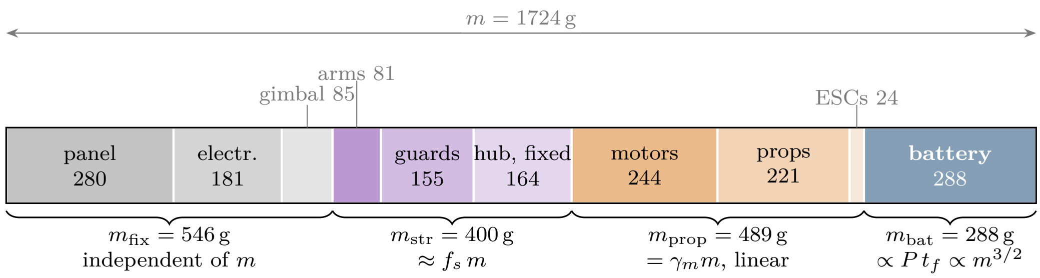


Figure 10: The mass budget eq. (27) of the design of section 17, to scale, from the component models of section 8; masses in grams. The four groups enter the closure differently. The fixed load does not depend on m; structure and propulsion grow in proportion to it (models 5.3 and 5.4); the battery grows with the hover power, as $m^{3/2}$ at fixed rotor area (model 5.5 and corollary 3.4). Only the last is nonlinear, and it is the one that makes eq. (36) fold.

5 The mass closure

5.1 Decomposition

Definition 5.1 (Mass budget)**.** The gross mass is partitioned as

$$m = m_{\mathrm{fix}} + m_{\mathrm{str}} + m_{\mathrm{prop}} + m_{\mathrm{bat}}, \tag{27}$$

where m_{fix} is the payload and avionics that do not scale with aircraft size (display, driver electronics, computer, camera, gimbal), m_{str} is the structure, m_{prop} is motors, speed controllers and propellers, and m_{bat} is the battery pack.

Equation (27) is an identity. It becomes an equation, and a nontrivial one, once the three terms on the right that depend on m are made explicit. The essential point is that they depend on m in different ways, and lumping them together, as one is tempted to do by writing a single "structural fraction", destroys the structure of the problem. Figure 10 shows the budget of the design of section 17 with the dependence of each group on m that the rest of this section derives.

5.2 Propulsion mass: sized by peak thrust

Definition 5.2 (Thrust margin)**.** The propulsion system is sized so that the maximum total thrust is

$$T_{\max} = \lambda\, mg, \qquad \lambda > 1. \tag{28}$$

λ is the ratio of maximum thrust to weight; $\lambda = 2$ means the vehicle hovers at half throttle.

Model 5.3 (Specific thrust)**.** Let $S_m \coloneqq T_{\max}/m_{\mathrm{prop}}$ be the specific thrust of the complete propulsion system, in N/kg. Then

$$m_{\mathrm{prop}} = \frac{T_{\max}}{S_m} = \frac{\lambda g}{S_m}\, m \eqqcolon \gamma_m\, m. \tag{29}$$

Propulsion mass is therefore *linear* in gross mass. This is worth emphasising because it is the coupling most people worry about first, and it is benign: it consumes a fixed fraction of the mass budget and produces a runaway only in the degenerate case $\gamma_m \geq 1$.

Model 5.3 is a lumped model. Section 8 replaces it with motor mass derived from the torque the rotor demands, at which point S_m becomes an output; the engine reports both so the assumption can be audited.

5.3 Structural mass

Model 5.4 (Structural fraction)**.**

$$m_{\mathrm{str}} = f_s\, m, \qquad f_s \in (0,1). \tag{30}$$

This is the weakest model in the algebraic layer. It is not a law; it is an observation that across a class of similar vehicles the structure tends to a roughly constant fraction of gross mass, typically $f_s \in [0.15, 0.25]$. It is retained because it keeps the closure in closed form, which is what makes section 6 possible. Section 8 derives m_{str} from beam theory instead and the two are compared in section 17.

5.4 Battery mass: sized by mission energy

Model 5.5 (Battery)**.** Let e_b be the specific energy of the pack in J/kg that the mission may use: the cell value reduced by the depth of discharge and by the landing reserve ς, so $e_b = e_{\mathrm{cell}}\,\mathrm{DoD}\,(1-\varsigma)$. Where the reserve must be visible, as in section 13, it is written out and e_b denotes the value before the reserve is removed. For a mission of duration t_f flown at total electrical power P_{tot},

$$m_{\mathrm{bat}} = \frac{E_{\mathrm{mission}}}{e_b} = \frac{1}{e_b}\int_0^{t_f} P\big(x(t), u(t)\big)\,\mathrm{d}t \xrightarrow[\text{hover}]{} \frac{P_{\mathrm{tot}}\, t_f}{e_b}. \tag{31}$$

The integral form is the honest one and is what section 13 uses; the right-hand form is its evaluation under the assumption that the whole mission is spent hovering, and is what makes the algebra tractable. The ratio between them is measured in section 17 and is between 1.00 and 1.10 depending on the mission, which retrospectively justifies the substitution for sizing purposes but not for endurance claims.

5.5 The master equation

Substituting eqs. (26) and (29) to (31) into eq. (27):

$$m = m_{\mathrm{fix}} + f_s m + \gamma_m m + \frac{t_f}{e_b}\left[\frac{(mg)^{3/2}}{\mathrm{FM}\,\eta\,\sqrt{2\rho A}} + P_{\mathrm{aux}}\right]. \tag{32}$$

Collecting terms gives the central equation of the algebraic layer.

Definition 5.6 (Closure coefficients)**.**

$$\alpha \coloneqq 1 - f_s - \frac{\lambda g}{S_m} = 1 - f_s - \gamma_m \qquad \text{(thrust coupling, dimensionless)}, \tag{33}$$

$$\beta \coloneqq \frac{t_f\, g^{3/2}}{e_b\,\mathrm{FM}\,\eta\,\sqrt{2\rho A}} \qquad \text{(energy coupling, kg}^{-1/2}\text{)}, \tag{34}$$

$$M_0 \coloneqq m_{\mathrm{fix}} + \frac{P_{\mathrm{aux}}\, t_f}{e_b} \qquad \text{(effective fixed load, kg)}. \tag{35}$$

Theorem 5.7 (The closure equation)**.** *Under models 3.1, 4.1 and 5.3 to 5.5 with fixed disk area, the gross mass of a feasible design satisfies*

$$\boxed{\alpha\, m - \beta\, m^{3/2} = M_0} \tag{36}$$

and conversely any positive root of eq. (36) *determines a mass budget consistent with eq.* (27).

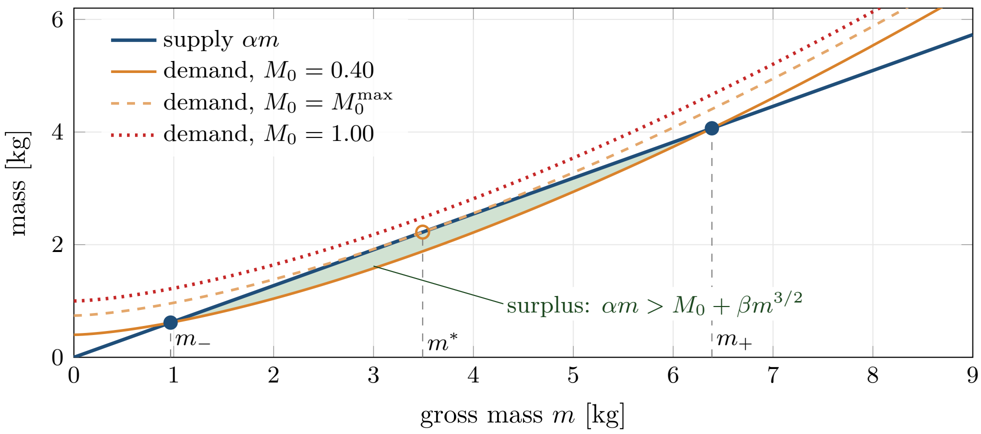


Figure 11: Equation (36) as a balance between the mass left after structure and propulsion, αm, and the mass the fixed load and the battery demand, $M_0 + \beta m^{3/2}$, with the coefficients of the worked example. A design exists where the two meet. For $M_0 = 0.40$ they meet twice, at a light root m_- and a heavy root m_+, with a surplus between them. At $M_0 = M_0^{\max} = 0.741$ the demand curve is tangent to the line at $m^* = 3.49$. Above it, as for $M_0 = 1.00$ and for the worked example's $M_0 = 0.762$, they never meet and no mass closes the budget. Section 6 proves this trichotomy.

Proof. Rearranging eq. (32),

$$m\,(1 - f_s - \gamma_m) - \frac{t_f g^{3/2}}{e_b \mathrm{FM}\eta\sqrt{2\rho A}}\, m^{3/2} = m_{\text{fix}} + \frac{P_{\text{aux}} t_f}{e_b},$$

where the $m^{3/2}$ term arises because $(mg)^{3/2} = g^{3/2} m^{3/2}$. Identifying the three groups with eqs. (33) to (35) gives eq. (36). The converse holds because every step is an equivalence for $m > 0$. □

Remark 5.8 (Reading the coefficients). α is the fraction of gross mass left over after structure and propulsion have taken their cut; it is a pure number, must be positive for any design to exist, and is easy to keep positive. β carries the mission: it is proportional to endurance and inversely proportional to specific energy, figure of merit, electrical efficiency and the square root of disk area. It multiplies the nonlinear term, and it is where the design dies. M_0 is what must be carried: note that the auxiliary power appears here, not in β, because it is independent of aircraft mass, so a display that consumes power converts directly into fixed load through the battery needed to run it.

Example 5.9. For a 13.3-inch panel with $m_{\text{fix}} = 0.762\,\text{kg}$, $f_s = 0.20$, $\lambda = 2$, $S_m = 120\,\text{N/kg}$, $\mathrm{FM} = 0.65$, $\eta = 0.75$, $e_b = 900\,\text{kJ/kg}$, $t_f = 3600\,\text{s}$ and four $0.40\,\text{m}$ rotors ($A = 0.503\,\text{m}^2$), definition 5.6 gives $\alpha = 0.6366$, $\beta = 0.2271\,\text{kg}^{-1/2}$, $M_0 = 0.762\,\text{kg}$. Section 6 shows that this configuration has no solution.

Figure 11 reads eq. (36) as supply against demand.

5.6 Generalised disk-area scaling

Corollary 3.4 assumed the disk area fixed while mass varies. That is an assumption about the design family, not a law. Suppose instead the rotors are allowed to grow with the aircraft,

$$A(m) = A_0 \left(\frac{m}{m_0}\right)^p, \tag{37}$$

anchored so that $A(m_0) = A_0$ reproduces a stated rotor at a stated mass.

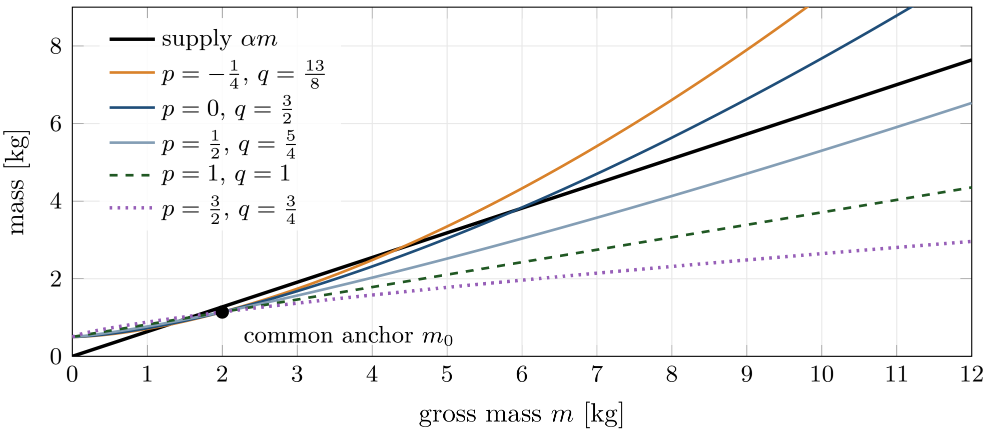


Figure 12: The generalised closure eq. (39) for five disk-area exponents, all anchored to the same rotor at $m_0 = 2\,\text{kg}$ so that the demand curves $M_0 + \beta_q m^q$ pass through one point, with the worked-example α, β and $M_0 = 0.5\,\text{kg}$. For $q > 1$ the demand is convex and eventually overtakes the supply line, so a second root and a fold exist; the smaller $q - 1$, the farther out they lie, and for $q = \frac{5}{4}$ they are beyond the plotted range. At $q = 1$ (constant disk loading, corollary 5.11) the demand is a line, and a root exists exactly when its slope is below α. For $q < 1$ the demand is concave and is overtaken by the supply for good.

Proposition 5.10 (Generalised power law)**.** *Under eq.* (37) *the hover propulsive power obeys*

$$P_{\text{el}} \propto m^q, \qquad q = \frac{3-p}{2}, \tag{38}$$

and the closure becomes

$$\alpha m - \beta_q m^q = M_0, \qquad \beta_q = \frac{t_f\, g^{3/2}\, m_0^{p/2}}{e_b\, \text{FM}\, \eta\, \sqrt{2\rho A_0}}. \tag{39}$$

Proof. Substituting eq. (37) into eq. (25) with $T = mg$,

$$P_{\text{el}} = \frac{(mg)^{3/2}}{\text{FM}\eta\sqrt{2\rho A_0}} \left(\frac{m}{m_0}\right)^{-p/2} = \frac{g^{3/2} m_0^{p/2}}{\text{FM}\eta\sqrt{2\rho A_0}}\, m^{(3-p)/2}.$$

Multiplying by t_f/e_b and inserting into eq. (27) as before gives eq. (39). $\square$

Three cases matter and are treated in sections 6 and 7:

- $p = 0$: rotors fixed, $q = 3/2$, the case of theorem 5.7;
- $p = 1$: constant disk loading, $q = 1$, the closure becomes linear;
- $p > 1$: rotors grow faster than mass, $q < 1$, the nonlinearity is sublinear and inverts its role entirely.

Figure 12 shows the family.

Corollary 5.11 (Constant disk loading)**.** *If the rotors are sized to hold* $\text{DL} = mg/A$ *constant, then by eq.* (15) *the propulsive power is exactly linear in mass,* $P_{\text{el}} = c_P m$ *with*

$$c_P = \frac{g}{\text{FM}\,\eta}\sqrt{\frac{\text{DL}}{2\rho}}, \tag{40}$$

the battery mass fraction is $\gamma_b = c_P t_f/e_b$*, and the closure collapses to*

$$m = \frac{M_0}{1 - f_s - \gamma_m - \gamma_b}, \tag{41}$$

which has a positive solution if and only if

$$f_s + \gamma_m + \gamma_b < 1. \tag{42}$$

Proof. Immediate from eq. (15) with $T = mg$ and eq. (39) at $q = 1$, $\beta_1 = \gamma_b$. □

The contrast between eq. (36) and eq. (41) is the central structural fact of the sizing problem, and it is the subject of the next section. In the first, feasibility fails discontinuously at a fold. In the second, mass diverges continuously as the fractions approach unity. They are different kinds of impossibility.

6 The fold: existence, multiplicity and the sizing iteration

This section analyses eq. (39) completely. The results are elementary but they are the substance of the problem, and they are what separates "the design is heavy" from "the design does not exist".

Throughout, fix $\alpha > 0$, $\beta > 0$, $M_0 > 0$ and $q > 0$, and define

$$F : (0, \infty) \to \mathbb{R}, \qquad F(m) := \alpha m - \beta m^q. \tag{43}$$

$F(m)$ is the fixed load that a machine of gross mass m can support: gross mass less what structure, propulsion and battery consume. Solving the closure means solving $F(m) = M_0$.

6.1 The trichotomy

Lemma 6.1 (Strict concavity)**.** *If $q > 1$ then F is strictly concave on $(0, \infty)$, attains a unique maximum at*

$$m^* = \left(\frac{\alpha}{q\beta}\right)^{\frac{1}{q-1}}, \tag{44}$$

and

$$M_0^{\max} := F(m^*) = \alpha\, m^* \left(1 - \frac{1}{q}\right). \tag{45}$$

Moreover $F(0^+) = 0$, F is strictly increasing on $(0, m^*)$, strictly decreasing on (m^*, ∞), and $F(m) \to -\infty$ as $m \to \infty$.

Proof. F is smooth on $(0, \infty)$ with

$$F'(m) = \alpha - q\beta m^{q-1}, \qquad F''(m) = -q(q-1)\beta m^{q-2}. \tag{46}$$

For $q > 1$ and $m > 0$ we have $q(q-1)\beta m^{q-2} > 0$, hence $F'' < 0$ everywhere on $(0, \infty)$: F is strictly concave. Setting $F'(m) = 0$ gives $m^{q-1} = \alpha/(q\beta)$, and since $q - 1 > 0$ the map $m \mapsto m^{q-1}$ is a strictly increasing bijection of $(0, \infty)$ onto itself, so the stationary point is unique and given by eq. (44). Strict concavity makes it the global maximum. Evaluating,

$$F(m^*) = \alpha m^* - \beta(m^*)^q = \alpha m^* - \beta m^*(m^*)^{q-1} = \alpha m^* - \beta m^* \frac{\alpha}{q\beta} = \alpha m^* \left(1 - \frac{1}{q}\right),$$

which is eq. (45). The sign of F' follows from monotonicity of m^{q-1}, and $F(m) = m(\alpha - \beta m^{q-1}) \to -\infty$ because the bracket tends to $-\infty$. □

Theorem 6.2 (Trichotomy for the closure)**.** *Let $q > 1$, $\alpha, \beta, M_0 > 0$, and let $M_0^{\max}$ be as in eq. (45). Then the equation $F(m) = M_0$ has, on $(0, \infty)$,*

(i) exactly two solutions $0 < m_- < m^* < m_+$ if $M_0 < M_0^{\max}$;

(ii) exactly one solution $m = m^*$, a double root, if $M_0 = M_0^{\max}$;

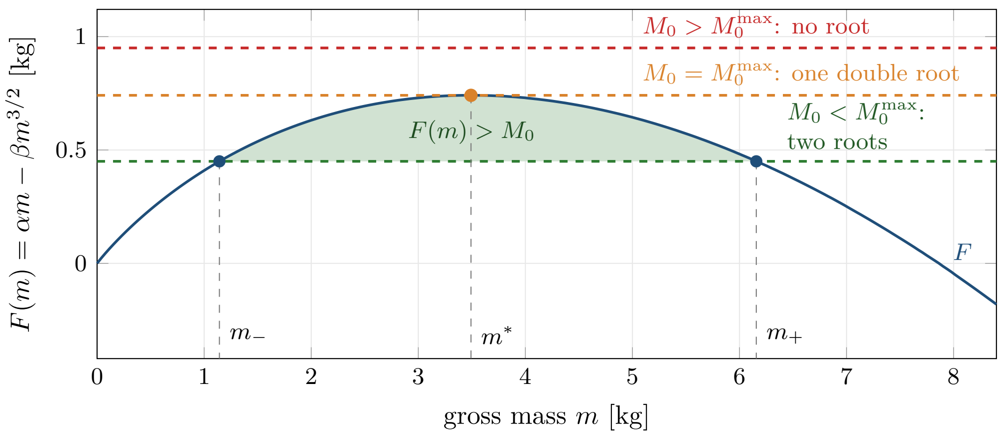


Figure 13: The closure function $F(m) = \alpha m - \beta m^{3/2}$ of eq. (43) for the worked example of section 5, $\alpha = 0.6366$ and $\beta = 0.2271\,\mathrm{kg}^{-1/2}$. F is strictly concave with its maximum $M_0^{\max} = 0.741\,\mathrm{kg}$ at $m^* = 3.49\,\mathrm{kg}$ (lemma 6.1). A horizontal line at height M_0 meets it twice, once or never, which is theorem 6.2; at $M_0 = 0.45\,\mathrm{kg}$ the roots are $m_- = 1.14\,\mathrm{kg}$ and $m_+ = 6.16\,\mathrm{kg}$. On the shaded interval $F(m) > M_0$: a machine of that mass could carry more than it is asked to, and by theorem 6.8 the sizing iteration moves down there.

(iii) no solution if $M_0 > M_0^{\max}$.

Proof. By lemma 6.1, F increases strictly from $F(0^+) = 0$ to $F(m^*) = M_0^{\max}$ on $(0, m^*]$ and decreases strictly from $M_0^{\max}$ to $-\infty$ on $[m^*, \infty)$. Both branches are continuous and strictly monotone.

(iii) If $M_0 > M_0^{\max}$ then $F(m) \le M_0^{\max} < M_0$ for all $m > 0$, so no solution exists.

(ii) If $M_0 = M_0^{\max}$, then $F(m) = M_0$ forces $m = m^*$ by uniqueness of the maximiser. It is a double root because $F'(m^*) = 0$, so $F(m) - M_0$ has a zero of order at least two; the order is exactly two since $F''(m^*) < 0$.

(i) If $0 < M_0 < M_0^{\max}$ then on $(0, m^*]$ the function F is a strictly increasing continuous bijection onto $(0, M_0^{\max}]$, so there is exactly one $m_- \in (0, m^*)$ with $F(m_-) = M_0$. On $[m^*, \infty)$, F is a strictly decreasing continuous bijection onto $(-\infty, M_0^{\max}]$, so there is exactly one $m_+ \in (m^*, \infty)$ with $F(m_+) = M_0$. Strictness gives $m_- < m^* < m_+$ and excludes further roots. □

Figure 13 draws the three cases for the worked example of section 5.

Corollary 6.3 (Closed forms at $q = 3/2$)**.** *For the fixed-rotor case $q = 3/2$,*

$$m^* = \frac{4\alpha^2}{9\beta^2}, \qquad M_0^{\max} = \frac{4\alpha^3}{27\beta^2}. \tag{47}$$

Proof. Put $q = 3/2$ in eq. (44): $1/(q-1) = 2$ and $\alpha/(q\beta) = 2\alpha/(3\beta)$, so $m^* = 4\alpha^2/(9\beta^2)$. Then eq. (45) gives $M_0^{\max} = \alpha m^*(1 - 2/3) = \alpha m^*/3 = 4\alpha^3/(27\beta^2)$. □

Remark 6.4 (Which root is the design). The physically meaningful root is m_-. Both roots satisfy the mass budget, but m_+ is a machine that is heavier than necessary, carries a battery sized for its own excess weight, and is, by theorem 6.8 below, unstable under the natural sizing iteration. It is a mathematical artefact of the quadratic-like structure, not a design.

Remark 6.5 (Cubic reduction). Equation (36) at $q = 3/2$ becomes a cubic under the substitution $u = \sqrt{m}$: $\beta u^3 - \alpha u^2 + M_0 = 0$. The product of its three roots is $-M_0/\beta < 0$, so exactly one root is negative and the other two are the pair $\sqrt{m_\pm}$ of theorem 6.2, real precisely when the discriminant is non-negative. The engine solves the equation by bracketing rather than by radicals, for reasons given in section 15.

6.2 The transition is a saddle-node bifurcation

Regard M_0 as a parameter and consider the one-parameter family of equations $G(m; M_0) := F(m) - M_0 = 0$.

Theorem 6.6 (Saddle-node)**.** *At* $(m, M_0) = (m^*, M_0^{\max})$ *the family* G *undergoes a saddle-node (fold) bifurcation: the defining conditions*

$$G = 0, \qquad \frac{\partial G}{\partial m} = 0 \tag{48}$$

hold, together with the nondegeneracy and transversality conditions

$$\frac{\partial^2 G}{\partial m^2}(m^*, M_0^{\max}) = -q(q-1)\beta(m^*)^{q-2} \neq 0, \qquad \frac{\partial G}{\partial M_0}(m^*, M_0^{\max}) = -1 \neq 0. \tag{49}$$

Consequently, near the bifurcation the two roots behave as

$$m_\pm = m^* \pm \sqrt{\frac{2(M_0^{\max} - M_0)}{q(q-1)\beta(m^*)^{q-2}}} + O(M_0^{\max} - M_0), \tag{50}$$

so they approach one another with a square-root law and annihilate.

Proof. Equation (48) is lemma 6.1 together with the definition of $M_0^{\max}$. Equation (49) follows from eq. (46) and $\partial G/\partial M_0 = -1$. These are exactly the hypotheses of the saddle-node normal form theorem (see e.g. Guckenheimer and Holmes [8, §3.4] or Kuznetsov [12, §3.2]), which asserts the existence of a smooth change of coordinates taking G to $\dot{y} = \mu - y^2$ near the point. For eq. (50), Taylor-expand F about m^*:

$$M_0 = F(m) = M_0^{\max} + \tfrac{1}{2}F''(m^*)(m - m^*)^2 + O((m - m^*)^3),$$

using $F'(m^*) = 0$. Solving for $m - m^*$ and inserting $F''(m^*) = -q(q-1)\beta(m^*)^{q-2}$ gives the stated expansion. $\square$

The bifurcation diagram is fig. 14.

Remark 6.7 (Why this matters practically). Equation (50) says that near the boundary of feasibility the design mass is an infinitely sensitive function of the requirement: $\mathrm{d}m_-/\,\mathrm{d}M_0 \to \infty$ as $M_0 \uparrow M_0^{\max}$. A design placed close to the fold is not merely marginal; its mass is unbounded in its derivative with respect to every assumption. This is a stronger statement than "leave margin" and is the reason the engine reports the utilisation $M_0/M_0^{\max}$ as a first-class quantity.

6.3 The naive sizing iteration

The obvious way to size such a vehicle is to guess a mass, compute what it implies, and repeat:

$$m_{k+1} = \Phi(m_k), \qquad \Phi(m) := M_0 + f_s m + \gamma_m m + \beta m^q. \tag{51}$$

Φ is exactly eq. (27) read as an assignment. Its fixed points are the roots of eq. (39), since $\Phi(m) = m \iff F(m) = M_0$.

Theorem 6.8 (Convergence of the sizing iteration)**.** *Let* $q > 1$ *and* $M_0 < M_0^{\max}$*, with roots* $m_- < m^* < m_+$ *as in theorem 6.2. Then*

(i) $\Phi'(m) = 1 - F'(m)$*, so* $\Phi'(m_-) < 1$ *and* $\Phi'(m_+) > 1$*;*

(ii) m_- *is an attracting fixed point of eq. (51) with asymptotic linear rate* $\Phi'(m_-) = 1 - F'(m_-) \in (0, 1)$*, and* m_+ *is repelling;*

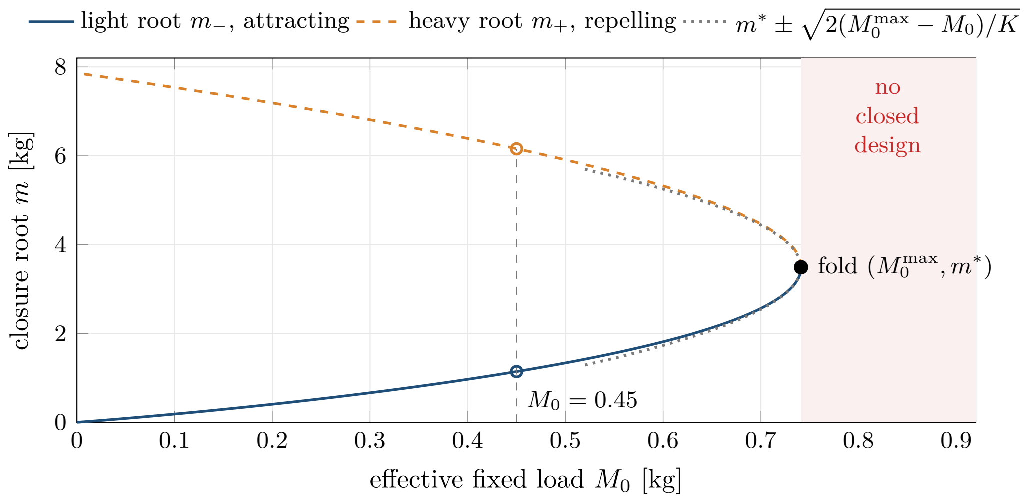


Figure 14: The roots of $F(m) = M_0$ against the load for the worked example, drawn as the parametric curve $(F(m), m)$. The light branch m_- (solid) and the heavy branch m_+ (dashed) meet at the fold $(M_0^{\max}, m^*)$ and annihilate; beyond it no mass closes. Near the fold both branches follow the square-root law eq. (50) (dotted), with $K = q(q-1)\beta(m^*)^{q-2}$, so $\mathrm{d}m_-/\,\mathrm{d}M_0 \to \infty$ as $M_0 \uparrow M_0^{\max}$: the design mass becomes infinitely sensitive to the load just before it ceases to exist.

(iii) the iteration started from any $m_0 \in [0, m_+)$ converges monotonically upward to m_- if $m_0 < m_-$, and monotonically downward to m_- if $m_0 \in (m_-, m_+)$;

(iv) for $m_0 > m_+$ the iterates increase without bound;

(v) if $M_0 > M_0^{\max}$ the iterates increase without bound from every starting point.

Proof. (i) Differentiating eq. (51), $\Phi'(m) = f_s + \gamma_m + q\beta m^{q-1}$. Using $f_s + \gamma_m = 1 - \alpha$ from eq. (33) and $F'(m) = \alpha - q\beta m^{q-1}$ from eq. (46),

$$\Phi'(m) = 1 - \alpha + q\beta m^{q-1} = 1 - F'(m).$$

By lemma 6.1, $F'(m_-) > 0$ and $F'(m_+) < 0$, giving $\Phi'(m_-) < 1$ and $\Phi'(m_+) > 1$. Also $\Phi' > 0$ everywhere since $f_s, \gamma_m, \beta \geq 0$ and not all zero, so Φ is strictly increasing; and $\Phi'(m_-) = 1 - F'(m_-) > 0$ because $F'(m_-) \leq \alpha < 1$.

(ii) Standard: a fixed point of a C^1 map is attracting when $|\Phi'| < 1$ and repelling when $|\Phi'| > 1$, with linear rate $|\Phi'|$.

(iii) Φ is increasing and $\Phi(m) - m = M_0 - F(m)$, which is negative exactly on (m_-, m_+) and positive on $[0, m_-)$. Hence for $m_0 < m_-$ the sequence is increasing and bounded above by m_- (because Φ increasing and $\Phi(m_-) = m_-$ imply $m_k < m_- \Rightarrow m_{k+1} < m_-$), so it converges, necessarily to a fixed point in $[m_0, m_-]$, which must be m_-. The argument on (m_-, m_+) is symmetric with the sequence decreasing.

(iv) For $m_0 > m_+$, $\Phi(m) - m = M_0 - F(m) > 0$ since $F(m) < M_0$ there, so the sequence increases; it cannot converge because there is no fixed point above m_+; hence it diverges.

(v) If $M_0 > M_0^{\max}$ then $M_0 - F(m) > 0$ for every m, so $m_{k+1} > m_k$ always and there is no fixed point to converge to. □

Figure 15 shows the iteration as a cobweb diagram.

Corollary 6.9 (Local stability criterion)**.** *A design is a stable fixed point of the sizing iteration if and only if*

$$\frac{\mathrm{d}}{\mathrm{d}m}\Big[m_{\mathrm{str}}(m) + m_{\mathrm{prop}}(m) + m_{\mathrm{bat}}(m)\Big] < 1, \tag{52}$$

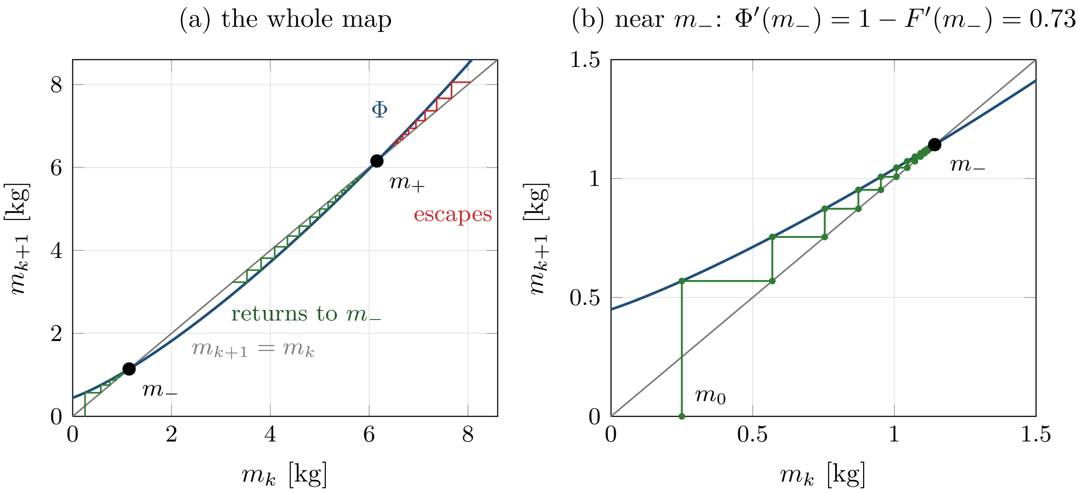


Figure 15: Cobweb diagram of the sizing iteration eq. (51) for the worked example at $M_0 = 0.45\,\text{kg}$. Each vertical step evaluates Φ and each horizontal step returns to the diagonal. (a) Started below m_- or between the roots, the iterates go monotonically to m_-; started above m_+ they escape, as theorem 6.8(iii) and (iv) state. The fixed points are where Φ crosses the diagonal, Φ being below it exactly on (m_-, m_+). (b) Near m_- the slope of Φ is $1 - F'(m_-) = 0.73 < 1$, so the error shrinks by that factor at each step.

which for the model of section 5 reads $f_s + \gamma_m + q\beta m^{q-1} < 1$, equivalently $F'(m) > 0$, equivalently $m < m^*$.

Corollary 6.9 is the precise form of the intuition that "each extra kilogram must require less than a kilogram of extra machine to carry it". The engine evaluates eq. (52) numerically at the solved design point and reports it.

Remark 6.10 (Critical slowing down). By theorem 6.8(ii) the convergence rate is $1 - F'(m_-)$, and $F'(m_-) \to 0$ as $M_0 \uparrow M_0^{\max}$. The iteration therefore becomes arbitrarily slow near the fold: near-infeasible designs are not distinguished from infeasible ones by a fixed iteration budget. This is a real trap. A practical consequence is that a solver which reports "did not converge in K iterations" cannot be used to decide feasibility; the analytic criterion $M_0 \lessgtr M_0^{\max}$ must be used instead. The engine does exactly this and classifies slow convergence separately from divergence. Figure 16 quantifies the effect.

6.4 Behaviour when $q \le 1$

Theorem 6.11 (No fold for $q \le 1$)**.** *Let $\alpha, \beta, M_0 > 0$.*

- *(i) If $q = 1$ and $\alpha > \beta$, then $F(m) = M_0$ has the unique solution $m = M_0/(\alpha - \beta)$; if $\alpha \le \beta$ there is no solution and $F \le 0$.*
- *(ii) If $0 < q < 1$, then $F(m) = M_0$ has exactly one solution for every $M_0 > 0$. In particular no critical payload exists.*

Consequently, under the disk-area scaling eq. (37) a fold exists if and only if $p < 1$.

Proof. (i) $F(m) = (\alpha - \beta)m$ is linear; the claim is immediate.

(ii) For $0 < q < 1$, $F'(m) = \alpha - q\beta m^{q-1}$ with $q - 1 < 0$, so $m^{q-1} \to \infty$ as $m \to 0^+$ and $m^{q-1} \to 0$ as $m \to \infty$. Hence $F'(m) < 0$ for small m and $F'(m) > 0$ for large m, with a single sign change at $m_{\min} = (q\beta/\alpha)^{1/(1-q)}$. Thus F decreases from $F(0^+) = 0$ to a negative minimum and then increases strictly to $+\infty$ (since $F(m) = m(\alpha - \beta m^{q-1})$ and the bracket tends to $\alpha > 0$). On $[m_{\min}, \infty)$, F is a strictly increasing continuous bijection onto $[F(m_{\min}), \infty) \supset (0, \infty)$, so there is exactly one root with $F = M_0 > 0$, and none on $(0, m_{\min})$ where $F \le 0 < M_0$.

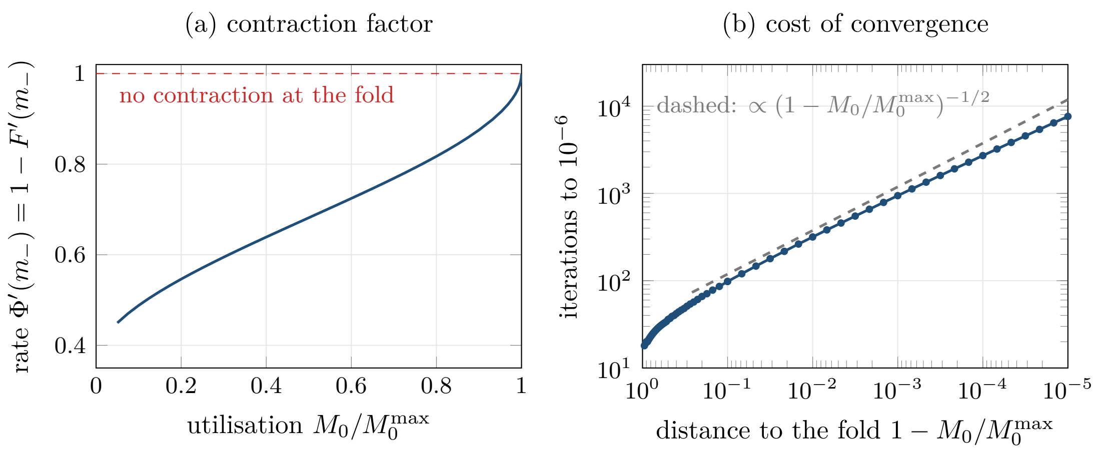


Figure 16: Critical slowing down of the sizing iteration, for the worked example. (a) The contraction factor $\Phi'(m_-) = 1 - F'(m_-)$ of theorem 6.8(ii) tends to one as the utilisation $M_0/M_0^{\max}$ tends to one. (b) The number of plain iterations from $m_0 = 0$ to a relative error of 10^{-6} grows like $(1 - M_0/M_0^{\max})^{-1/2}$, because by eq. (50) $F'(m_-) = K(m^* - m_-) \approx \sqrt{2KM_0^{\max}(1 - M_0/M_0^{\max})}$. A fixed iteration budget therefore cannot separate a design close to the fold from one beyond it.

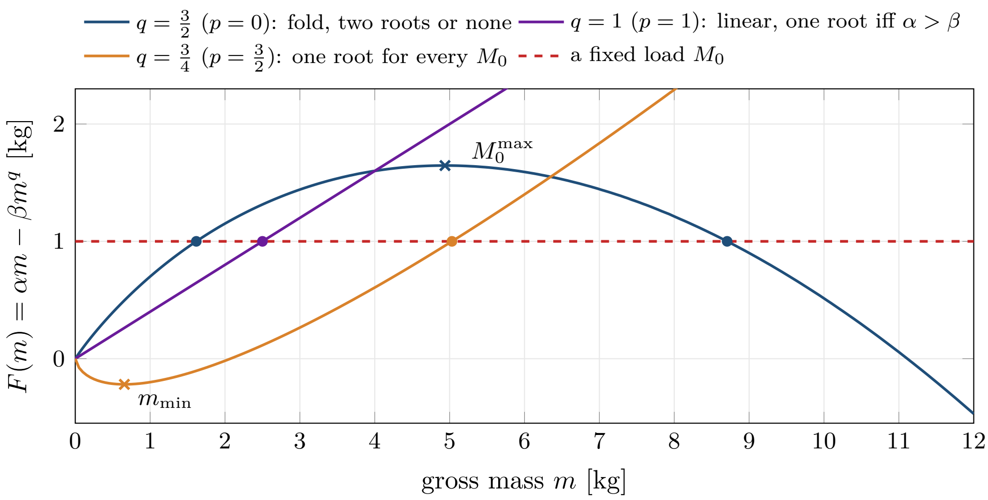


Figure 17: The shape of $F(m) = \alpha m - \beta m^q$ decides whether a fold exists (theorem 6.11), drawn with $\alpha = 1$ and illustrative β. For $q > 1$ $(p < 1)$ F is concave with an interior maximum, so a load meets it twice or not at all. For $q = 1$ it is a straight line, met once if $\alpha > \beta$. For $q < 1$ it dips below zero to a minimum at $m_{\min} = (q\beta/\alpha)^{1/(1-q)}$ and then grows without bound, so every positive load is met exactly once.

The final statement follows from $q = (3 - p)/2$: $q > 1 \iff p < 1$. $\square$

Figure 17 contrasts the three regimes.

The catastrophe is not a property of flight. It is a property of holding the rotor area fixed while the mass grows. Let the disk grow at least as fast as the mass and the fold disappears entirely, replaced by the far gentler condition eq. (42). What is then unbounded is not the mass but the rotor diameter, so the binding constraint migrates from energy to geometry, noise and safety. It does not go away. Nor does the absence of a fold mean the absence of a ceiling: at $p = 1$ exactly the longest mission that closes is still finite (proposition 7.8); only for $p > 1$ does it disappear.

7 Scaling laws and sensitivity

Corollary 6.3 gives the critical payload in closed form. Because it is a product of powers of the underlying parameters, its logarithmic derivatives are exact rational numbers, and those numbers are the design guidance.

7.1 The explicit critical payload

Proposition 7.1 (Explicit form of $M_0^{\max}$)**.** *For $q = 3/2$,*

$$M_0^{\max} = \frac{4\alpha^3}{27\beta^2} = \frac{8\,\alpha^3\,\rho\,A\,e_b^2\,\mathrm{FM}^2\,\eta^2}{27\,g^3\,t_f^2}. \tag{53}$$

Proof. From eq. (34), $\beta^2 = t_f^2 g^3/(e_b^2 \mathrm{FM}^2 \eta^2 \cdot 2\rho A)$. Hence $1/\beta^2 = 2\rho A e_b^2 \mathrm{FM}^2 \eta^2/(t_f^2 g^3)$ and $M_0^{\max} = (4\alpha^3/27) \cdot 2\rho A e_b^2 \mathrm{FM}^2 \eta^2/(t_f^2 g^3)$, which is eq. (53). □

7.2 Exact logarithmic sensitivities

Definition 7.2. For a positive quantity Y depending on a positive parameter x, the logarithmic sensitivity is $S_x := \partial \ln Y/\partial \ln x$, so that a relative change $\delta x/x$ produces $S_x\,\delta x/x$ to first order.

Theorem 7.3 (Sensitivities of the critical payload)**.** *With $M_0^{\max}$ as in eq. (53) and $\gamma_m = \lambda g/S_m$, $\alpha = 1 - f_s - \gamma_m$,*

$$S_{t_f} = -2, \qquad S_{e_b} = +2, \qquad S_{\mathrm{FM}} = +2, \qquad S_\eta = +2, \tag{54}$$

$$S_A = +1, \qquad S_D = +2, \qquad S_N = +1, \qquad S_\rho = +1, \tag{55}$$

and for the parameters entering through α,

$$S_{f_s} = -\frac{3f_s}{\alpha}, \qquad S_\lambda = -\frac{3\gamma_m}{\alpha}, \qquad S_{S_m} = +\frac{3\gamma_m}{\alpha}, \qquad S_g = -3 - \frac{3\gamma_m}{\alpha}. \tag{56}$$

Proof. Take logarithms of eq. (53):

$$\ln M_0^{\max} = \ln \tfrac{8}{27} + 3\ln\alpha + \ln\rho + \ln A + 2\ln e_b + 2\ln \mathrm{FM} + 2\ln\eta - 3\ln g - 2\ln t_f.$$

For any parameter x not appearing in α, differentiation is immediate and gives eqs. (54) and (55); the entries for D and N follow from $A = N\pi D^2/4$, so $\partial \ln A/\partial \ln D = 2$ and $\partial \ln A/\partial \ln N = 1$.

For f_s: $\partial\alpha/\partial f_s = -1$, so $\partial \ln M_0^{\max}/\partial \ln f_s = 3f_s\,\partial \ln\alpha/\partial f_s = -3f_s/\alpha$.

For λ: $\alpha = 1 - f_s - \lambda g/S_m$, so $\partial\alpha/\partial\lambda = -g/S_m$ and $\partial \ln M_0^{\max}/\partial \ln\lambda = 3\lambda(-g/S_m)/\alpha = -3\gamma_m/\alpha$. The result for S_m follows identically with the opposite sign since $\gamma_m \propto S_m^{-1}$.

For g: gravity appears both explicitly, as g^{-3}, and inside α through $\gamma_m = \lambda g/S_m$. Hence

$$S_g = -3 + 3\frac{g}{\alpha}\frac{\partial\alpha}{\partial g} = -3 + \frac{3g}{\alpha}\left(-\frac{\lambda}{S_m}\right) = -3 - \frac{3\gamma_m}{\alpha}.$$ □

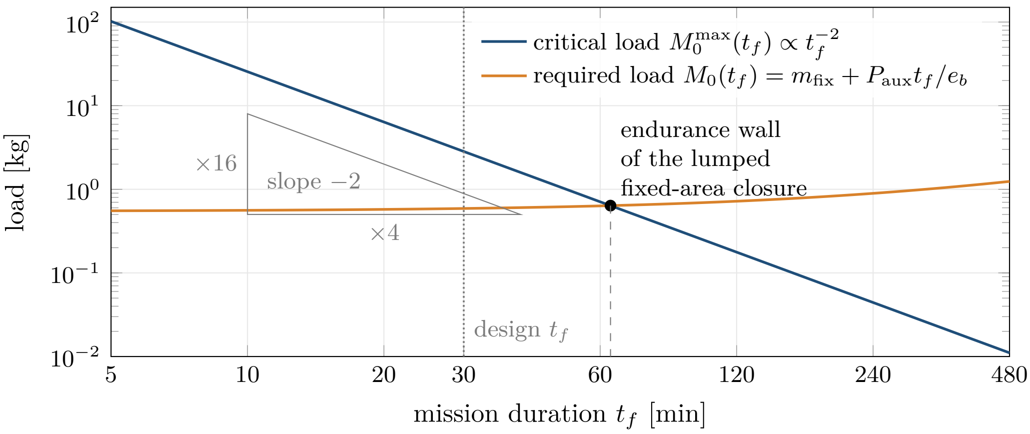


Figure 18: The inverse-square law of corollary 7.4 for the lumped fixed-area model of the design of section 17. The critical load $M_0^{\max}$ falls with slope -2 on logarithmic axes. The load the mission itself demands rises, because the battery that runs the auxiliary load, $P_{\text{aux}} t_f / e_b$, is part of M_0. The two cross at the endurance wall of this closure, 66 min, against the design's 30 min session; beyond the crossing no mass closes.

Corollary 7.4 (The inverse-square endurance law)**.** *At fixed technology and geometry, $M_0^{\max} \propto t_f^{-2}$. Multiplying the required endurance by k divides the supportable fixed load by k^2.*

Figure 18 shows the law for the lumped model of the design of section 17, and what it implies once the auxiliary load is counted.

Example 7.5 (Verification against the engine)**.** For the parameter set of section 17 the engine computes the sensitivities numerically by central differences in ln-space and returns $S_{t_f} = -2.000$, $S_{e_b} = S_{\text{FM}} = S_\eta = +2.000$, $S_D = +2.000$, $S_N = +1.000$, $S_\rho = +1.000$, $S_{f_s} = -0.943$, $S_{S_m} = +0.770$, $S_\lambda = -0.770$ and $S_g = -3.770$. With $\gamma_m = 2 \times 9.80665/120 = 0.16344$ and $\alpha = 0.63656$, eq. (56) predicts $S_{f_s} = -3(0.20)/0.63656 = -0.9426$, $S_{S_m} = 3(0.16344)/0.63656 = 0.7703$ and $S_g = -3 - 0.7703 = -3.7703$. The agreement to the printed precision is a verification of both the closed form and its implementation. See section 16.

Remark 7.6 (How to read theorem 7.3)*.* Three groups of parameters behave quite differently.

The *squared* levers, e_b, FM, η and t_f, are the ones with real authority, and their sign structure is unforgiving: a 20% better cell buys 44% more payload, once; a doubling of endurance costs three quarters of it. The figure of merit and the electrical efficiency are bounded above by unity, so at FM $= 0.73$ and $\eta = 0.78$ there remains at most a factor $(1/0.73)^2(1/0.78)^2 \approx 3.1$ in that direction, ever.

The *linear* levers are disk area and air density. Disk area is the only one of the four that the designer controls freely, and because D enters squared it is in practice the strongest available lever.

The *cubic-in-α* levers, f_s, λ, S_m, have modest sensitivity as long as α is comfortably positive, and become violent as $\alpha \to 0$. They are not where the design is won or lost unless one has been careless.

Figure 19 collects the sensitivities as the engine computes them.

7.3 The exponent, the constants, and where the wall sits

A claim one meets in discussions of this problem is that "the exponents kill you, not the constants". Theorem 7.3 lets us be precise about the sense in which that is and is not true.

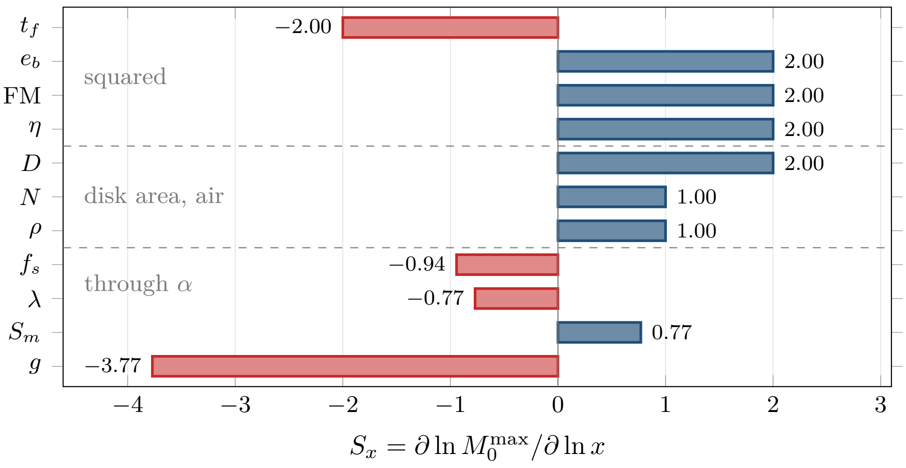


Figure 19: Logarithmic sensitivities of $M_0^{\max}$ at the engine's default parameters, computed by central differences in ln-space. The top group are the squared levers of eq. (54); the middle group are disk area, entering as D^2 and N, and air density, eq. (55); the bottom group act through α and are eq. (56), here with $\gamma_m/\alpha = 0.257$. Every bar agrees with theorem 7.3 to three decimals. A bar of length two means a 10 % change in the parameter moves the critical load by about 20 %.

Observation 7.7. The exponent q determines whether a wall exists at all (theorem 6.11: it exists iff $q > 1$) and the asymptotic behaviour of the closure. The constants α and β determine where the wall sits, through eq. (47), and their influence is not weak: $M_0^{\max}$ is cubic in α and inverse-quadratic in β. The correct statement is that the exponent creates the wall and the constants locate it.

For finite engineering at finite scales the location is what one actually cares about. Equation (53) says that location moves by a factor of two for a 41% improvement in cell energy density, or a 41% improvement in figure of merit, or a doubling of disk area, or a 29% reduction in endurance.

7.4 The endurance wall across the exponent

Combining theorem 6.11 with proposition 5.10 gives the full picture, but only once two statements that are easily conflated are separated: whether the closure has a fold, and whether there is a longest mission it can close. They are different, and they part company at $p = 1$.

Write the closure with its dependence on duration explicit. Under eq. (37) the propulsive power is $c_P m^q$ with $c_P = c_0 m_0^{p/2}$ and $c_0 = g^{3/2}/(\mathrm{FM}\,\eta\sqrt{2\rho A_0})$, so a mass m closes a mission of duration t if and only if $\alpha m - (t/e_b)(c_P m^q + P_{\mathrm{aux}}) = m_{\mathrm{fix}}$, that is if and only if $t = \Theta_p(m)$ with

$$\Theta_p(m) := \frac{e_b\,(\alpha m - m_{\mathrm{fix}})}{c_P\, m^q + P_{\mathrm{aux}}}. \tag{57}$$

Define the endurance wall at fixed m_{fix}, P_{aux} and anchor (m_0, A_0) as $t_f^{\max}(p) := \sup_{m>0} \Theta_p(m)$.

Proposition 7.8 (The endurance wall)**.** *Let $\alpha > 0$ and $m_{\mathrm{fix}} > 0$.*

(i) If $p < 1$, Θ_p attains its maximum at a finite mass $\hat{m}(p) > m_{\mathrm{fix}}/\alpha$, and $t_f^{\max}(p) = \Theta_p(\hat{m})$ is finite and attained.

(ii) If $p = 1$, Θ_1 is strictly increasing and

$$t_f^{\max}(1) = t_{\mathrm{lim}} = \frac{\alpha\, e_b}{c_P}, \tag{58}$$

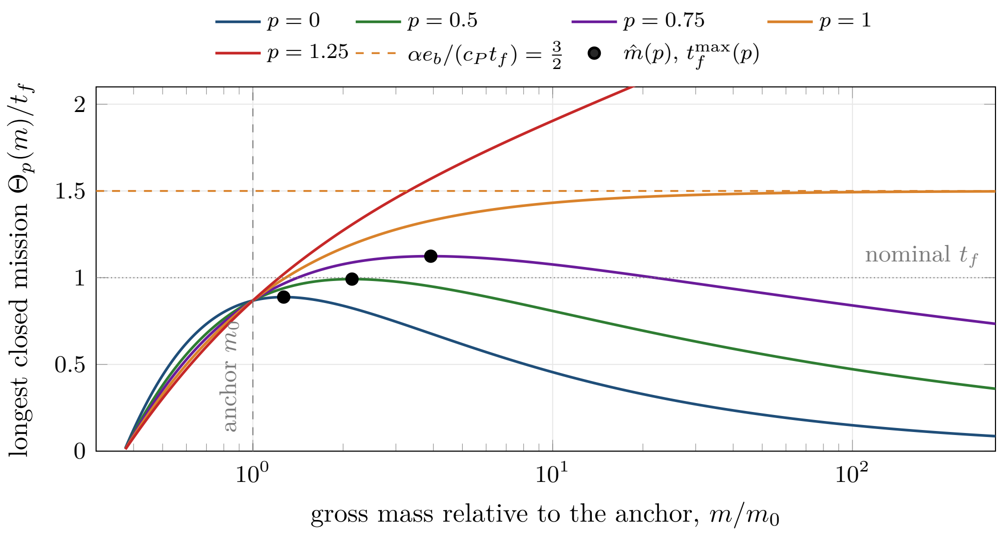


Figure 20: The longest mission a mass m can close, $\Theta_p(m)$ of eq. (57), relative to the nominal t_f, at the engine's default parameters with the anchor m_0 at the fold mass of the fixed-area closure. For $p<1$ each curve has an interior maximum $(\hat m, t_f^{\max})$ (dots), and these lie to the right of the anchor. At $p=1$ the curve rises towards $\alpha e_b/c_P=\frac{3}{2}t_f$ (dashed) without reaching it; at $p=1.25$ it grows without bound. No fold and no ceiling are different statements: $p=1$ has the first property and not the second.

which is finite and not attained: for every $t_f<t_{\lim}$ the unique mass is finite, and it diverges as $t_f\uparrow t_{\lim}$. Here e_b is net of the landing reserve, as in model 5.5.

(iii) If $p>1$, $\Theta_p(m)\to\infty$ as $m\to\infty$ and there is no wall.

(iv) Where $\hat m$ is differentiable in p,

$$\frac{\mathrm{d}\ln t_f^{\max}}{\mathrm{d}p}=\frac{\vartheta}{2}\ln\frac{\hat m}{m_0},\qquad \vartheta=\frac{c_P\hat m^q}{c_P\hat m^q+P_{\mathrm{aux}}}\in(0,1],\tag{59}$$

so the wall rises with p if and only if the mass at the wall exceeds the anchor mass. It is not monotone in p for every anchor.

Proof. Θ_p is continuous, negative for $m<m_{\mathrm{fix}}/\alpha$ and positive beyond. (i) For $q>1$ the numerator grows like m and the denominator like m^q, so $\Theta_p(m)\to 0$ as $m\to\infty$; a continuous function that is positive on $(m_{\mathrm{fix}}/\alpha,\infty)$ and tends to 0 at both ends of that interval attains its supremum. (ii) At $q=1$, $\Theta_1'(m)$ has the sign of $\alpha(c_P m+P_{\mathrm{aux}})-c_P(\alpha m-m_{\mathrm{fix}})=\alpha P_{\mathrm{aux}}+c_P m_{\mathrm{fix}}>0$, and $\Theta_1(m)\to\alpha e_b/c_P$. The mass closing duration t is $m=(m_{\mathrm{fix}}+P_{\mathrm{aux}}t/e_b)/(\alpha-c_P t/e_b)$, positive and finite exactly when $t<t_{\lim}$; with $P_{\mathrm{aux}}=0$ this is eq. (42). (iii) For $q<1$ the numerator outgrows the denominator. (iv) By the envelope theorem, $\partial_m\Theta_p(\hat m)=0$ gives $\mathrm{d}\ln t_f^{\max}/\mathrm{d}p=\partial_p\ln\Theta_p(\hat m)=-\partial_p\ln(c_P\hat m^q+P_{\mathrm{aux}})$, and with $\partial_p\ln c_P=\frac{1}{2}\ln m_0$ and $\partial_p q=-\frac{1}{2}$ this is $-\vartheta(\frac{1}{2}\ln m_0-\frac{1}{2}\ln\hat m)$. □

Remark 7.9. With $P_{\mathrm{aux}}=0$ the maximiser is explicit: $\hat m=q\,m_{\mathrm{fix}}/[\alpha(q-1)]$, which tends to infinity as $q\downarrow 1$, reconciling (i) with (ii). If the anchor is the fold mass of the fixed-area closure at the nominal duration t_f, then $\sqrt{m_0}=2\alpha/(3\beta)$ and eq. (58) gives exactly $t_{\lim}=\frac{3}{2}t_f$, independently of P_{aux}.

Figure 20 draws Θ_p for five exponents. The maxima of (i), the unattained ceiling of (ii) and the unbounded growth of (iii) are all visible, and the maxima lie to the right of the anchor, which by (iv) is why the wall rises with p here.

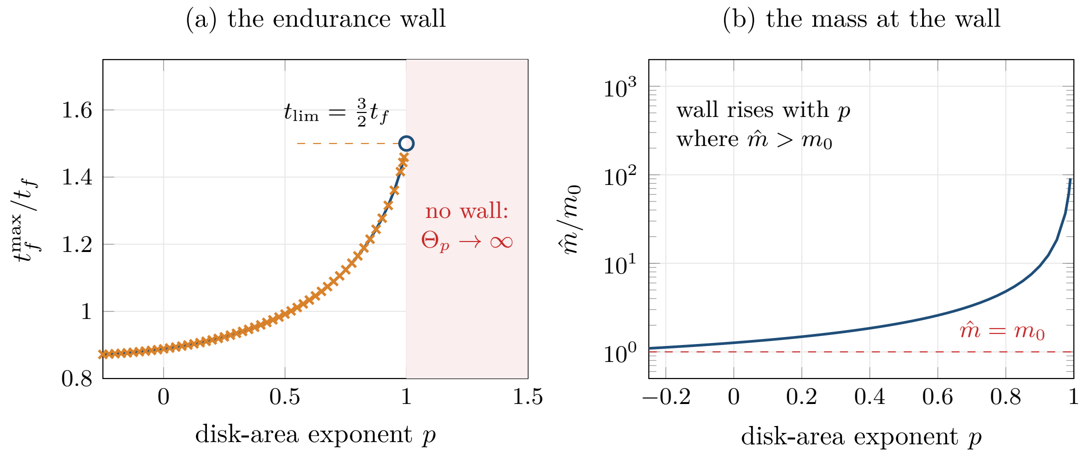


Figure 21: The endurance wall of proposition 7.8 on a fine grid of exponents, at the parameters and anchor of table 1. (a) $t_f^{\max}/t_f$ from direct maximisation of eq. (57) (line) and from the bisection of `sweep.endurance_wall` (crosses), which agree wherever the bisection runs; within about 0.01 of $p = 1$ the bisection overflows in the fold mass $(\alpha/q\beta)^{1/(q-1)}$ and only the direct value is shown. The circle at $p = 1$ is the unattained supremum $\frac{3}{2}t_f$ of eq. (58); for $p > 1$ there is no wall. (b) The mass at the wall exceeds the anchor for every $p < 1$ and diverges as $p \uparrow 1$, so by eq. (59) the wall rises monotonically on this family.

Table 1: Endurance wall against the disk-area exponent at the engine's default parameters ($t_f = 3600\,\mathrm{s}$), anchored at the fold mass $m_0 = 2.83\,\mathrm{kg}$ of the fixed-area closure. Computed by bisection on t_f with `sweep.endurance_wall` and checked against direct maximisation of eq. (57). The last column is $\hat{m}/m_0$; it exceeds one throughout, so by eq. (59) the wall rises with p for this anchor. At $p = 1$ there is no fold but there is still a ceiling, the unattained supremum eq. (58).

p	$q = (3-p)/2$	fold exists	$t_f^{\max}$	$\hat{m}/m_0$
−0.25	1.625	yes	3138 s	1.1
0.00	1.500	yes	3198 s	1.3
0.25	1.375	yes	3327 s	1.6
0.50	1.250	yes	3571 s	2.1
0.75	1.125	yes	4046 s	3.9
1.00	1.000	no	5400 s	unbounded
1.25	0.875	no	unbounded	n/a
1.50	0.750	no	unbounded	n/a

Table 1 records the computed values and fig. 21 draws them on a finer grid. The default parameters do not close at fixed area at the nominal hour, so the common anchor is the fold mass, and the $p = 1$ entry is $\frac{3}{2}t_f$ as the remark predicts.

8 First-principles component models

Models 5.3 and 5.4 introduced two constants, S_m and f_s, that carry a great deal of weight and are not laws. This section replaces both with models derived from geometry and materials, at the cost of losing the closed form. The engine keeps both layers and reports the implied values of S_m, f_s and FM so that the lumped assumptions can be audited against the detailed model.

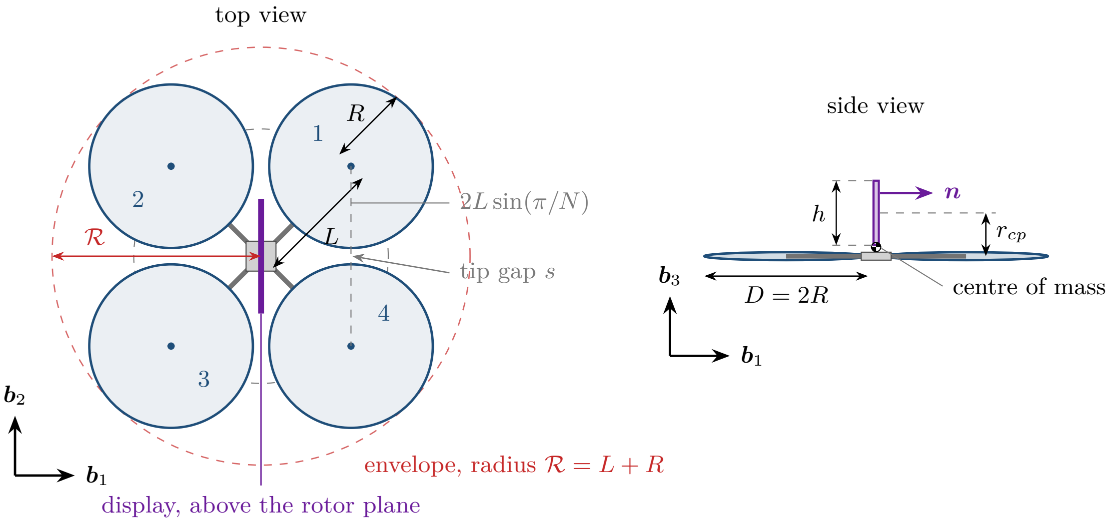


Figure 22: The design of section 17 drawn to scale. Top view: the hubs lie on a circle of radius L fixed by eq. (60), so that adjacent disks, whose centres are $2L\sin(\pi/N)$ apart, clear each other by the tip gap $s = 2L\sin(\pi/N) - 2R$. The dashed circle of radius $\mathcal{R} = L + R$ is the envelope swept by the blades; its radius is the reach eq. (61). Side view: the display is carried a distance r_{cp} above the rotor plane, and raises the centre of mass only a little because it is light compared with the rest of the machine.

8.1 Arms: a cantilevered thin-walled tube

Model 8.1 (Rotor arm)**.** Each of the N arms is a thin-walled circular tube of mean radius r, wall thickness $t \ll r$, length L, elastic modulus E, density ρ_m and allowable stress σ_{all}, built in at the hub and carrying the rotor as a point load at the tip.

Length. Adjacent rotor disks must not overlap. With N hubs on a circle of radius L, the centre-to-centre spacing of adjacent hubs is $2L\sin(\pi/N)$, so non-overlap requires $2L\sin(\pi/N) \geq 2R$, i.e.

$$L = \frac{k_{\text{gap}}\, R}{\sin(\pi/N)}, \qquad k_{\text{gap}} \geq 1, \tag{60}$$

with k_{gap} a clearance factor. The tip-to-tip span is $2(L + R)$ and the *reach*, the distance from the vehicle centre to the nearest blade tip, is

$$\mathcal{R} := L + R. \tag{61}$$

Equation (61) is trivial and turns out to be one of the most consequential quantities in the paper; see section 17. Figure 22 draws the design to scale.

Section properties. For a thin-walled tube the second moment of area about a diameter is

$$I = \pi r^3 t + O(t^2), \qquad \text{mass per unit length } \mu = 2\pi r t \rho_m. \tag{62}$$

Strength. Under the limit load $F = \mathrm{SF}\,\lambda m g/N$ at the tip, the root bending moment is $M = FL$ and the extreme-fibre stress is $\sigma = Mr/I$. Requiring $\sigma \leq \sigma_{\text{all}}$ gives

$$t \geq t_\sigma = \frac{FL}{\pi r^2 \sigma_{\text{all}}}. \tag{63}$$

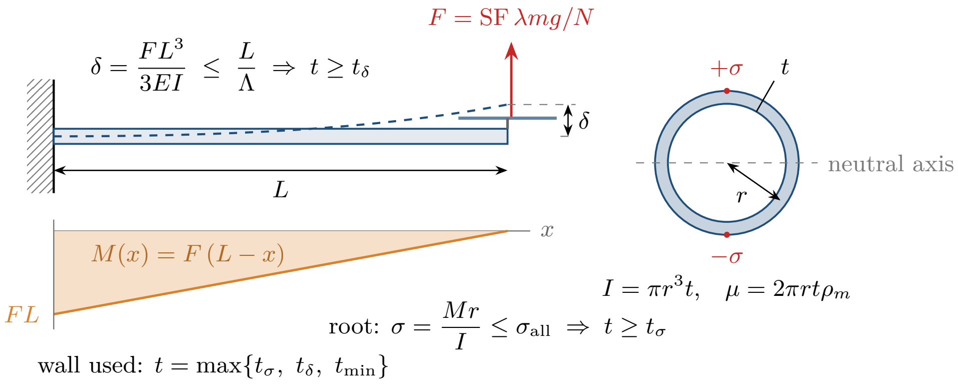


Figure 23: The arm as a cantilevered thin-walled tube. The limit load F at the tip produces a bending moment $M(x) = F(L - x)$ that grows linearly toward the root, where the extreme fibres of the section carry $\sigma = Mr/I$; that bounds the wall from below through eq. (63). The tip deflection, exaggerated here, bounds it through eq. (64). For a thin wall both I and the mass per unit length are linear in t, so every requirement on the arm is a lower bound on a single number.

Stiffness. Euler-Bernoulli theory gives the tip deflection of a cantilever under a tip load as $\delta = FL^3/(3EI)$. Imposing $\delta \le L/\Lambda$ for a slenderness limit Λ (we use $\Lambda = 100$) gives

$$t \ge t_\delta = \frac{\Lambda F L^2}{3\pi E r^3}. \tag{64}$$

Figure 23 collects the load case, the moment distribution and the section.

Dynamics. The fundamental bending frequency of a uniform cantilever of mass per unit length μ is

$$f_1 = \frac{1.875^2}{2\pi}\sqrt{\frac{EI}{\mu_{\text{eff}} L^4}}, \qquad \mu_{\text{eff}} = \mu + \frac{3m_{\text{tip}}}{L}, \tag{65}$$

where 1.875 is the first root of $\cos\zeta\cosh\zeta + 1 = 0$ and the correction to μ is the Rayleigh lumped-mass approximation for a tip mass m_{tip} (motor plus propeller). The wall thickness actually used is

$$t = \max\{t_\sigma,\ t_\delta,\ t_{\min}\}, \tag{66}$$

and the engine records which of the three governs, because the answer is informative: for the designs of section 17 it is stiffness or minimum gauge, never strength, with a working stress of 44 MPa against an allowable of 350 MPa. Figure 24 shows why.

Remark 8.2 (Consistency with assumption 2.2). Equation (65) is what makes the rigid-body assumption checkable. With $f_1 \approx 33$ Hz and an attitude loop at $\sqrt{K_R}/2\pi \approx 1.0$ Hz (section 12) the separation is a factor of 32, comfortably in the regime where treating the airframe as rigid does not corrupt the control design.

Frame total. Guards are rings around each disk, so their mass scales with circumference, not area:

$$m_{\text{guard}} = k_g\, N\, 2\pi R. \tag{67}$$

The frame is then

$$m_{\text{str}} = N\mu L\,(1 + k_{\text{hub}}) + m_{\text{guard}} + m_{\text{boom}} + m_{\text{fixed}}, \tag{68}$$

with k_{hub} accounting for the centre plate, motor mounts and fasteners, m_{boom} the optional display cantilever of section 17, and m_{fixed} landing gear and wiring.

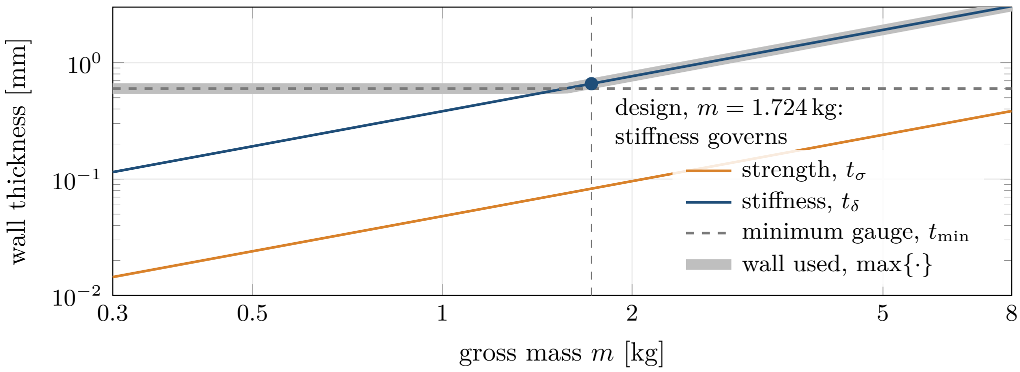


Figure 24: The three lower bounds of eq. (66) against gross mass for the design's arm tube, $r = 8.0$ mm. Strength and stiffness are both linear in the limit load $F \propto m$, so on logarithmic axes they are parallel lines of slope one, and the stiffness bound lies a factor of about eight above the strength bound: an arm stiff enough is automatically strong enough. Below about 1.6 kg the minimum gauge governs; at the design, stiffness does.

8.2 Motors: sized by torque, with a scale effect

The rotor demands a torque $Q_{\max} = P_{\mathrm{sh}}^{\max}/\Omega_{\max}$ at the sizing condition $T_{\max} = \lambda mg/N$ per rotor. A motor's continuous torque is set by the shear stress its airgap can sustain over its rotor surface, $Q \sim \tau_{\mathrm{airgap}}\,(\pi D_m L_m)\,(D_m/2)$, which for geometrically similar machines gives $Q \propto D_m^3 \propto m_{\mathrm{mot}}$, i.e. constant torque density. Real small outrunners do better than that as they grow, because the ratio of cooling surface to loss volume improves and because the winding fill factor rises. We therefore write

Model 8.3 (Motor mass)**.**

$$Q = k_\tau \left(\frac{m_{\mathrm{mot}}}{m_{\mathrm{ref}}}\right)^e m_{\mathrm{mot}} \quad\Longleftrightarrow\quad m_{\mathrm{mot}} = \left(\frac{Q\, m_{\mathrm{ref}}^e}{k_\tau}\right)^{\frac{1}{1+e}}, \tag{69}$$

with $m_{\mathrm{ref}} = 0.1$ kg, $e \approx 0.30$ and k_τ calibrated in section 16.

Because the motor is sized by the burst condition $T_{\max}$, the peak rather than continuous torque is the right basis. Equation (69) is sublinear in Q, which matters: at $e = 0.3$ a tenfold increase in torque costs a factor $10^{1/1.3} \approx 5.9$ in mass, not ten (fig. 25).

Speed controllers are sized by power, $m_{\mathrm{esc}} = P_{\mathrm{el}}^{\max}/k_P$, and propellers by an empirical areal law $m_{\mathrm{prop}} = k_p D^{2.6}(N_b/2)$ fitted to carbon propellers.

Remark 8.4 (The electrical chain that mass does not see)*.* Equation (69) determines the size of the motor but not whether it can be driven. A rotor turning at Ω requires a back-EMF Ω/K_V below the pack voltage with margin for the current-resistance drop, and the motor constant satisfies $K_t\,[\mathrm{N\,m/A}] = 9.5493/K_V\ [\mathrm{rpm/V}]$. Sizing K_V so that maximum rotor speed is reached at the *minimum* pack voltage, not the nominal or full one, is what keeps the machine flyable at the bottom of a discharge. This constraint is decoupled from the mass closure and is treated separately in the engine; it is what forces the low-K_V, custom-wound motors of section 17.

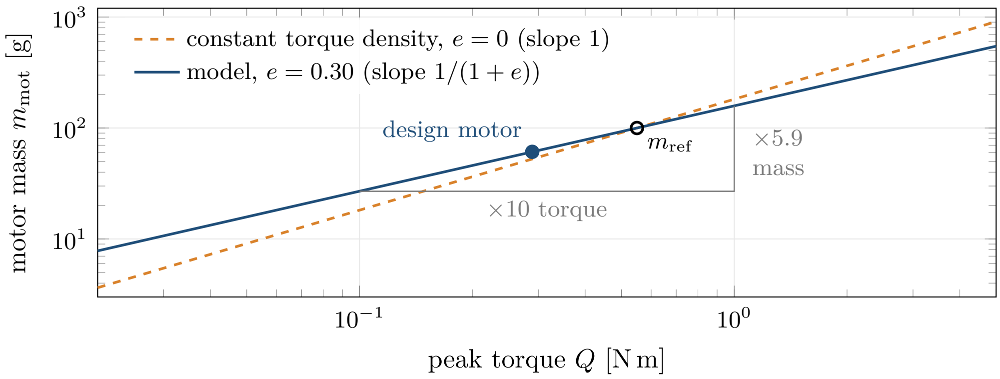


Figure 25: Motor mass against the peak torque it must hold, eq. (69). Constant torque density, $e = 0$, would make mass proportional to torque; the scale effect bends the line to slope $1/(1+e)$ on logarithmic axes, and the two agree at the reference motor. The step shows a tenfold torque costing $10^{1/1.3} \approx 5.9$ times the mass. The filled point is the design's motor, sized by the torque at $T_{\max}$; the calibration of k_τ against catalogue motors is in section 16.

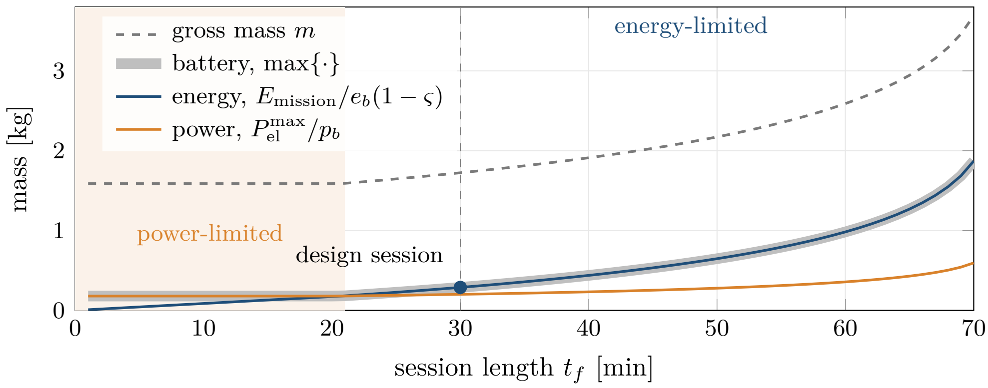


Figure 26: The two requirements of eq. (70) along the closed design as the session length varies, every other parameter fixed. The power term follows the peak electrical power and barely moves. The energy term grows with t_f, and faster than linearly, because through the closure a heavier battery needs a heavier machine that needs more power. Below about 21 min the pack is sized by power, and endurance up to that point costs nothing: the gross mass is flat. The design session of 30 min is energy-limited.

8.3 Battery: two constraints, not one

Model 5.5 sizes the pack by energy. A pack must also deliver power. Writing p_b for the usable specific power and ς for the state of charge held in reserve,

$$m_{\text{bat}} = \max\left\{ \underbrace{\frac{E_{\text{mission}}}{e_b\,(1-\varsigma)}}_{\text{energy}}, \underbrace{\frac{P_{\text{el}}^{\max}}{p_b}}_{\text{power}} \right\}. \tag{70}$$

Which term binds depends on the mission: for short sessions the power term dominates and endurance is effectively free, while for long sessions the energy term takes over. The engine reports which is active, because a design that is power-limited responds to entirely different levers than one that is energy-limited. Figure 26 traces both terms along the closed design.

High-energy lithium-ion cells are not high-power cells. Sizing against the burst condition $T_{\max}$ requires the burst rating; the continuous hover draw must then be checked separately against the continuous rating. Both checks are implemented.

8.4 Cost of the first-principles layer

Replacing models 5.3 and 5.4 by models 8.1 and 8.3 removes the closed form: $m_{\text{str}}(m)$ and $m_{\text{prop}}(m)$ are no longer proportional to m, so eq. (36) is replaced by the implicit equation

$$\mathcal{F}(m) := m - \Big[m_{\text{fix}} + m_{\text{str}}(m) + m_{\text{prop}}(m) + m_{\text{bat}}(m)\Big] = 0, \tag{71}$$

solved numerically. For the designs studied $\mathcal{F}$ has the qualitative shape of section 6: negative for small m, positive on the branch where a design exists, negative again beyond the fold. That shape is observed, not proved. Lemma 6.1 is a statement about the lumped residual; it does not transfer to $\mathcal{F}$, which contains clipped quantities, maxima (eq. (70)) and empirical fits. Near the fold the positive stretch can also be very narrow, and section 15 describes how the engine searches for it and what it reports when it does not find it. A failed search is not a proof that no mass closes.

9 Acoustics

For a machine that hovers beside a person's head, the acoustic model is not a secondary consideration; section 17 finds it to be the binding constraint. It is also the place where the most error was found, so it is developed carefully.

9.1 Why the sixth power

Model 9.1 (Broadband loading noise)**.** At the scales and Reynolds numbers considered, rotor noise is dominated by broadband loading noise: unsteady lift on the blade sections, arising from inflow turbulence and boundary-layer vortex shedding, radiating as a distribution of acoustic dipoles.

Curle's extension of Lighthill's acoustic analogy to flows bounded by solid surfaces [15, 6] shows that the surface term is dipole in character, and dimensional analysis of the dipole source strength gives the acoustic power radiated by a body of characteristic area A_b in a flow of characteristic velocity U:

$$W_{\text{ac}} \sim \frac{\rho\, A_b\, U^6}{c_0^3}, \tag{72}$$

with c_0 the speed of sound. This is the classical U^6 dipole law, against U^8 for the quadrupole (free jet) case; see Ffowcs Williams and Hawkings [23] for the rotating-surface generalisation, and Brooks, Pope and Marcolini [3] for the airfoil self-noise mechanisms this lumps together.

Taking $10\log_{10}$ of eq. (72) with $U = V_{\text{tip}}$ gives a $60\log_{10} V_{\text{tip}}$ dependence. Loading enters through the sectional lift, for which we take an empirical $10\log_{10} T$, and N statistically independent rotors add incoherently, contributing $10\log_{10} N$. Hence

Model 9.2 (Source model)**.**

$$L_p(1\,\text{m}) = C + 60\log_{10}\frac{V_{\text{tip}}}{V_{\text{tip}}^{\text{ref}}} + 10\log_{10}\frac{T_{\text{rotor}}}{T^{\text{ref}}} + 10\log_{10} N, \tag{73}$$

with $V_{\text{tip}}^{\text{ref}} = 100\,\text{m/s}$, $T^{\text{ref}} = 5\,\text{N}$ and C an anchor calibrated against measured aircraft in section 16.

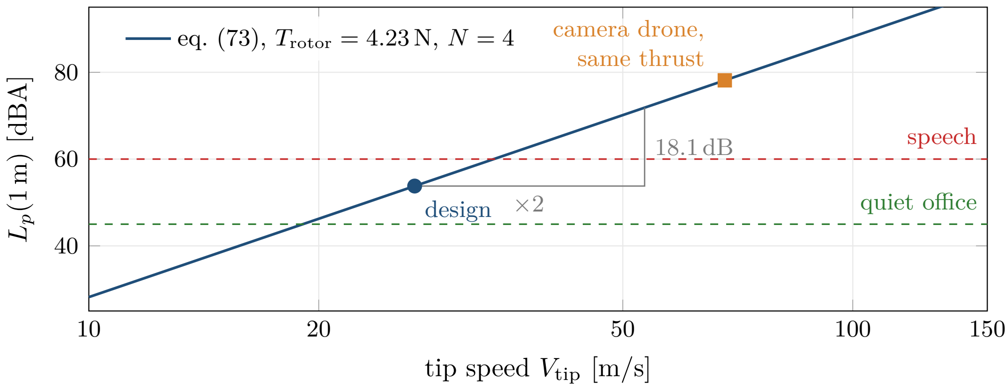


Figure 27: The source model eq. (73) at the design's thrust per rotor. On a logarithmic tip-speed axis the level is a straight line of 60 dB per decade, so halving the tip speed buys 18.1 dB wherever one starts. The design turns at 26.7 m/s; carrying the same thrust at a camera drone's tip speed would cost about 24 dB, the difference between a quiet office and a raised voice.

Observation 9.3 (The design lever). Halving tip speed reduces the level by $60 \log_{10} 2 = 18.1\,\text{dB}$. No other parameter in the model has comparable authority. Together with corollary 4.6, which says that raising solidity lowers the tip speed needed to carry a given thrust, this determines the entire shape of the machine: large, fat-bladed, slowly turning rotors.

Figure 27 draws eq. (73) at the design's thrust.

9.2 Sound power against sound pressure

Definition 9.4. The sound power level L_W of a source is $L_W = 10 \log_{10}(W/W_0)$, $W_0 = 1 \times 10^{-12}\,\text{W}$, a property of the source alone. The sound pressure level L_p at a point is $L_p = 20 \log_{10}(p_{\text{rms}}/p_0)$, $p_0 = 20\,\mu\text{Pa}$, and depends on the environment.

Proposition 9.5 (Free-field conversion)**.** *For a point source of directivity factor Q radiating into free space above a rigid plane, the mean-square pressure at distance r satisfies*

$$L_p(r) = L_W + 10 \log_{10}\left(\frac{Q}{4\pi r^2}\right), \tag{74}$$

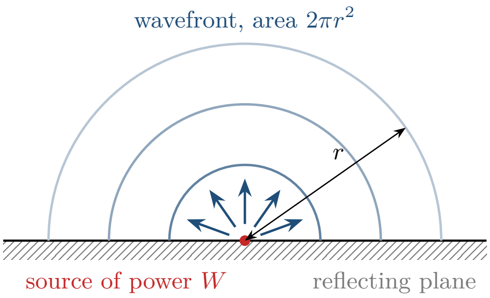
so for a source above a reflecting plane, $Q = 2$, and at $r = 1\,\text{m}$,

$$L_p(1\,\text{m}) = L_W - 10 \log_{10}(2\pi) = L_W - 7.98\,\text{dB}. \tag{75}$$

Proof. Acoustic intensity at distance r for a source of power W radiating with directivity Q is $I = QW/(4\pi r^2)$. In the far field of a plane wave $I = p_{\text{rms}}^2/(\rho c_0)$, and the reference quantities are chosen so that $10 \log_{10}(I/I_0) = L_p$ with $I_0 = p_0^2/(\rho c_0)$ to within 0.1 dB at standard conditions. Taking logarithms gives eq. (74); substituting $Q = 2$, $r = 1$ gives eq. (75). □

Figure 28 is the geometry behind $Q = 2$.

Remark 9.6 (A trap worth naming). European regulation 2019/945 requires manufacturers of unmanned aircraft to declare a *sound power* level L_{WA}. It is routinely misread as a pressure at one metre. By proposition 9.5 that misreading is worth 8 dB, and in the wrong direction: it makes published aircraft sound louder than they are, and so makes a design calibrated against the misreading look quieter than it is. Section 16 records the consequences.



$$I = \frac{W}{2\pi r^2} = \frac{QW}{4\pi r^2}, \quad Q = 2$$

$$L_p(r) = L_W + 10\log_{10}\frac{Q}{4\pi r^2}$$

$$\begin{aligned} L_p(1\,\mathrm{m}) &= L_W - 10\log_{10} 2\pi \\ &= L_W - 7.98\,\mathrm{dB} \end{aligned}$$

Figure 28: Sound power against sound pressure, proposition 9.5. A source on a reflecting plane spreads its power W over hemispherical wavefronts of area $2\pi r^2$, which is the directivity $Q = 2$ of eq. (74). At one metre that area is $2\pi\,\mathrm{m}^2$, and $10\log_{10} 2\pi = 7.98\,\mathrm{dB}$ is the whole difference between a declared sound power level and the pressure level a listener at one metre hears.

9.3 The room

Free field is the wrong model indoors. The classical treatment (Sabine; see Beranek [2] or Kuttruff [11]) writes the steady-state level as the sum of a direct field, which falls with distance, and a diffuse reverberant field, which does not.

Definition 9.7 (Room constant)**.** For a room of total interior surface area S and average absorption coefficient $\bar{\alpha}$,

$$\mathcal{R}_{\mathrm{room}} := \frac{S\,\bar{\alpha}}{1-\bar{\alpha}}. \tag{76}$$

Proposition 9.8 (Steady-state level in a room)**.**

$$L_p(r) = L_W + 10\log_{10}\left(\frac{Q}{4\pi r^2} + \frac{4}{\mathcal{R}_{\mathrm{room}}}\right). \tag{77}$$

The two contributions are equal at the critical distance

$$r_c = \sqrt{\frac{Q\,\mathcal{R}_{\mathrm{room}}}{16\pi}}. \tag{78}$$

Sketch. The direct term is eq. (74). For the reverberant term, in the diffuse field approximation the energy density is uniform and in steady state the power absorbed equals the power injected: $W = \frac{1}{4}\bar{\alpha}S\,c_0\,E$ with E the energy density, whence $E = 4W/(c_0 S\bar{\alpha})$; correcting for the fact that the first reflection has already occurred replaces $S\bar{\alpha}$ by $\mathcal{R}_{\mathrm{room}}$, and $p^2_{\mathrm{rms}} = \rho c_0^2 E$ gives the $4/\mathcal{R}_{\mathrm{room}}$ term. Setting the two terms equal and solving for r gives eq. (78). □

Proposition 9.8 has a consequence that no free-field figure can express: beyond r_c, moving away from the machine buys nothing. The level is then set by how absorbent the room is, not by how far away one stands. A flying display does not make a quiet place near itself and a loud place far away; it raises the level of the entire room to within a few decibels of the level at the user's own ear.

For an ordinary furnished room, $\bar{\alpha} \approx 0.20$ and r_c is under a metre, which is smaller than the standoff at which such a machine is useful. The user therefore sits close to the boundary between the two regimes and receives both contributions (fig. 29).

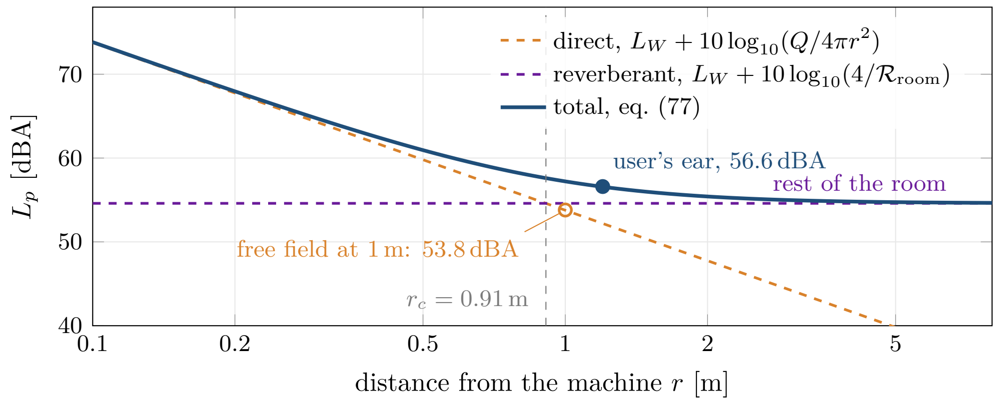


Figure 29: Equation (77) for the design in the reference $5 \times 4 \times 2.4$ m room with $\bar{\alpha} = 0.20$. The direct field falls by 6 dB per doubling of distance, the reverberant field does not fall at all, and the two are equal at the critical distance $r_c = 0.91$ m of eq. (78). The user at 1.20 m is just past it and hears 56.6 dBA, 2.8 dB above the free-field figure at one metre; the rest of the room settles at the reverberant level of about 54.8 dBA.

9.4 Weighting and tonal content

The A-weighting of IEC 61672 is

$$R_A(f) = \frac{12194^2 f^4}{(f^2 + 20.6^2)\sqrt{(f^2 + 107.7^2)(f^2 + 737.9^2)}(f^2 + 12194^2)},$$
$$A(f) = 20\log_{10} R_A(f) + 2.00. \tag{79}$$

The blade passage frequency is

$$f_{\text{BPF}} = \frac{N_b\,\Omega}{2\pi} = \frac{N_b\,\text{rpm}}{60}. \tag{80}$$

Remark 9.9 (What the broadband number omits). Equation (73) predicts a broadband level. What makes a rotor *annoying* rather than merely loud is tonal content at f_{BPF} and its harmonics. For the designs of section 17, $f_{\text{BPF}} \approx 35$ Hz, which eq. (79) discounts by about 38 dB (fig. 30). That is a genuine perceptual benefit of turning slowly and it is also a warning: the acoustic energy has not gone anywhere. It has moved to low frequency, where rooms absorb poorly, where Sabine's diffuse-field assumption fails because the modal density is low, and where humans report a throb rather than a hiss. Furthermore several rotors at slightly different speeds beat at the difference frequency, which is perceptually worse than a steady tone. None of this is in the A-weighted broadband figure, and it is the most likely way for a built prototype to be unacceptable at a level the model calls acceptable.

The Schroeder frequency, above which a room behaves diffusely,

$$f_S \approx 2000\sqrt{\frac{T_{60}}{V}}, \tag{81}$$

is of order 200 Hz for a domestic room, which is precisely where the first several blade harmonics lie. Proposition 9.8 is therefore being applied slightly outside its domain of validity for exactly the components that matter most perceptually. This is stated as a limitation rather than repaired.

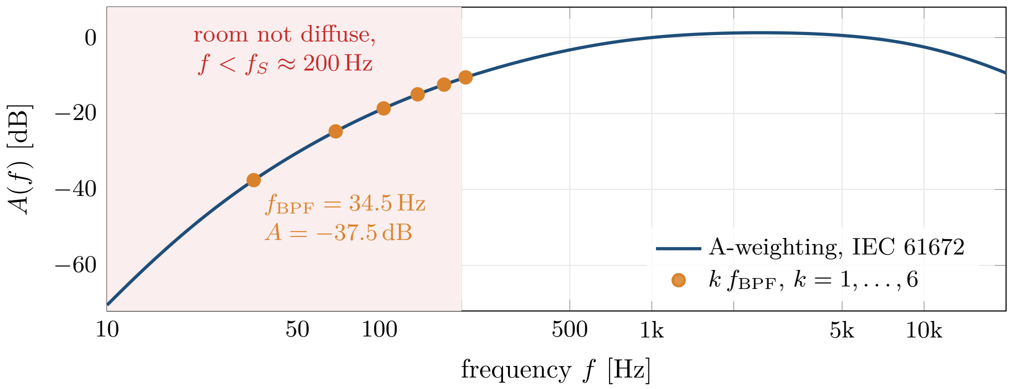


Figure 30: The A-weighting eq. (79) with the design's blade passage frequency and its first five harmonics. The fundamental at 34.5 Hz is weighted down by 37.5 dB, and every harmonic up to the sixth lies in the shaded band below the Schroeder frequency eq. (81), where the diffuse-field model of proposition 9.8 does not hold. The A-weighted broadband figure is therefore least informative about exactly the tonal components a listener notices.

10 Near-field aerodynamic corrections

Model 3.1 treats an isolated rotor in an unbounded fluid at rest. A machine hovering at head height in a room violates all three conditions: there is a floor below, a ceiling above, other rotors alongside, and after a short time the air is not at rest but circulating. This section quantifies the first three and measures the fourth.

10.1 Ground effect

When the wake is constrained by a surface a distance z below the rotor, the streamtube cannot contract freely, the induced velocity at the disk falls, and the rotor produces more thrust for the same power.

Model 10.1 (Cheeseman-Bennett)**.** Modelling the ground by an image source, Cheeseman and Bennett [4] obtain, at constant power,

$$\left.\frac{T_{\text{IGE}}}{T_{\text{OGE}}}\right|_P = \frac{1}{1 - \left(\dfrac{R}{4z}\right)^2}, \tag{82}$$

valid for $z/R \gtrsim 0.5$; equivalently, at constant thrust the induced power is reduced by the same factor.

The dependence is inverse-square in z/R, which decides the magnitude immediately.

Example 10.2. For $R = 0.246$ m and a rotor plane $z = 1.37$ m above the floor, the adopted layout of section 17, $(R/4z)^2 \approx 2 \times 10^{-3}$ and the effect is 0.2%. Ground effect is negligible for a small rotor at chest height, and would only matter within about one radius of a surface.

10.2 Ceiling effect

A ceiling above the rotor restricts the *inflow* rather than the wake. The low-pressure region above the disk is bounded, the rotor is drawn toward the surface, and thrust rises at constant power. The functional form is taken to be the same as eq. (82) with a separate coefficient,

$$\left.\frac{T_{\text{ICE}}}{T_{\text{OGE}}}\right|_P = \frac{1}{1 - k_c\left(\dfrac{R}{4z_c}\right)^2}, \tag{83}$$

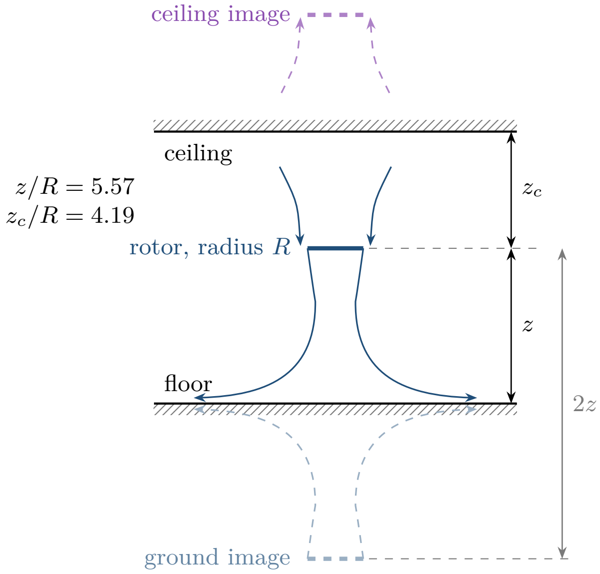


Figure 31: The image construction behind eqs. (82) and (83), drawn to scale for the design's rotor in the reference room. A mirror rotor below the floor, with its wake mirrored, makes the floor a plane of symmetry across which no air passes, which is the boundary condition the floor imposes; a mirror above the ceiling does the same for the inflow. The images sit $2z$ and $2z_c$ away, many rotor radii, and the corrections fall as the square of $R/4z$, which is why both are small here.

with z_c the distance to the ceiling and $k_c > 1$ because at equal spacing ceiling effect is the stronger of the two. Figure 31 shows the image construction for both surfaces.

Remark 10.3 (Honesty about k_c). Published values for ceiling proximity scatter considerably across geometries and measurement techniques [21, 5]. k_c is the least certain constant in this paper. It is exposed as a user parameter and the engine reports the resulting factor separately so that its influence can be bounded rather than believed. For the design here it reduces induced power by 0.7 % at $k_c = 2.0$, and the entire range $k_c \in [0.5, 6]$ moves the design mass by 13 g (fig. 32).

10.3 Rotor-to-rotor interference

Adjacent rotors each operate partly in the other's induced flow. The penalty is negligible when the tip-to-tip gap exceeds roughly a quarter diameter and grows as the disks approach.

Model 10.4 (Interference factor)**.** With s the tip-to-tip gap and D the diameter,

$$\kappa_{\text{int}}(s/D) = \begin{cases} 1, & s/D \geq 0.25, \\ 1 + \varepsilon\left(1 - \dfrac{s/D}{0.25}\right)^2, & 0 \leq s/D < 0.25, \end{cases} \qquad \varepsilon = 0.07, \tag{84}$$

applied as a multiplier on induced power. The quadratic interpolation is chosen so that the penalty and its derivative vanish at the matching point, consistent with an induced-flow interaction that decays smoothly.

For a quadrotor with $k_{\text{gap}} = 1.10$ in eq. (60), the adjacent hub spacing is $2L\sin(\pi/4) = 2.2R$ and the tip-to-tip gap is $0.2R = 0.1D$, giving $\kappa_{\text{int}} = 1.025$ (fig. 33).

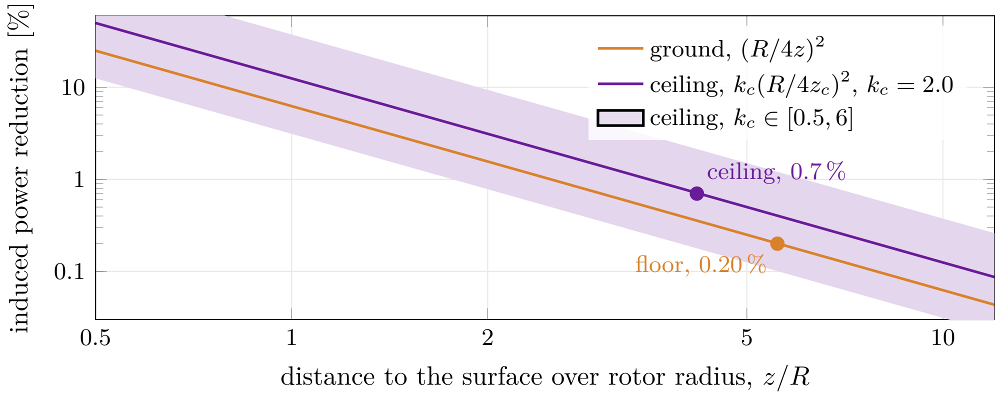


Figure 32: Fractional reduction of induced power from eqs. (82) and (83), $1 - 1/k = k_s(R/4z)^2$ with $k_s = 1$ for the floor and $k_s = k_c$ for the ceiling, against distance over rotor radius. On logarithmic axes both are lines of slope -2; the band spans the published range $k_c \in [0.5, 6]$. At the design's two distances the floor is worth 0.20 % and the ceiling 0.7 %, and even the upper edge of the band stays below 3 %.

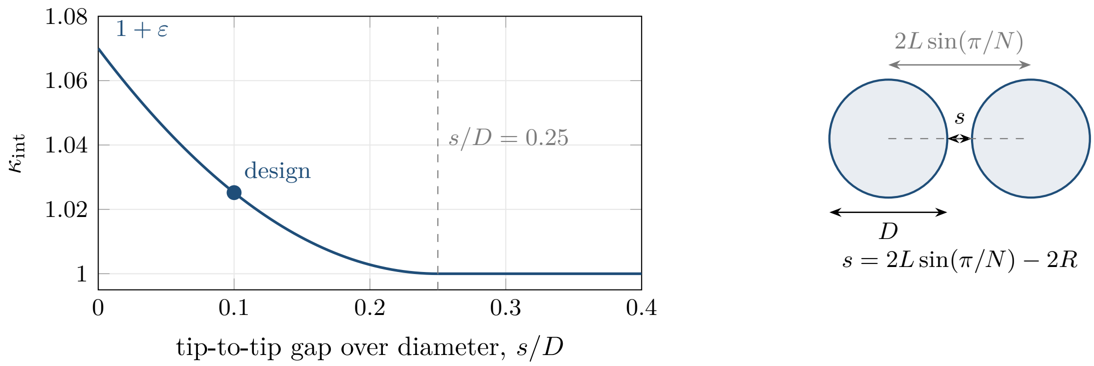


Figure 33: The interference factor eq. (84) against tip gap over diameter, with the geometry that defines the gap. The penalty and its slope vanish together at $s/D = 0.25$, so the model joins the isolated-rotor case smoothly. The design, with $k_{\text{gap}} = 1.100$ in eq. (60), sits at $s/D = 0.100$ and pays $\kappa_{\text{int}} = 1.0252$.

10.4 Net effect and why it is worth stating

The three corrections enter the induced power as

$$P_{\text{ind}} = \frac{\kappa\,\kappa_{\text{int}}}{k_{\text{ground}}\,k_{\text{ceiling}}}\,T v_i, \tag{85}$$

and they partly cancel. Table 2 gives the measured influence on a converged design.

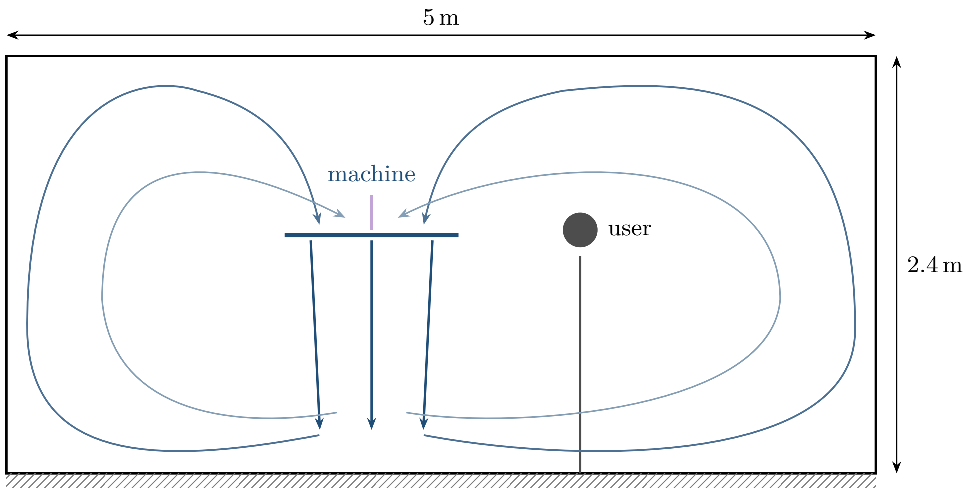
The corrections were worth adding not because they changed the design but because they bounded their own importance. A caveat that says "ground and ceiling effect are not modelled" is unfalsifiable; a table that says they are worth a quarter of a decibel is a result. The value of a model is often in the size of the term it lets you discard.

10.5 Recirculation: the correction that is not made

Model 3.1 assumes the fluid far upstream is at rest. In a closed room this is false, and the timescale over which it becomes false is computable.

Table 2: Effect of the near-field corrections on the converged design of section 17; the four rows differ only in the corrections applied. κ_{tot} is the induced power factor actually used, eq. (85). The whole span of the four combinations is 0.22 dB and 22 g.

Model	κ_{tot}	Gross mass	Hover power	$L_p(1\,\text{m})$
Both corrections	1.168	1.724 kg	106.0 W	53.78 dB
No surface effects	1.179	1.730 kg	107.2 W	53.84 dB
No interference	1.139	1.708 kg	103.0 W	53.61 dB
Neither	1.150	1.714 kg	104.1 W	53.67 dB



$\dot{V} = Av_i \approx 2.3\,\text{m}^3/\text{s}$, $t_{\text{ex}} = V_{\text{room}}/\dot{V} \approx 21\,\text{s}$, mean return velocity $\dot{V}/(\text{floor area}) \approx 0.1\,\text{m/s}$

Figure 34: Recirculation in the reference room, schematic. The wake strikes the floor, spreads, rises along the walls and returns across the ceiling into the rotor inflow. At $\dot{V} = Av_i$ a volume of air equal to the room's passes through the disks every t_{ex} of eq. (86), so within the first minute of a session the machine is drawing in air it has already accelerated, and the quiescent inflow of model 3.1 no longer holds.

Definition 10.5 (Room exchange time)**.** The volumetric flow through the disks is $\dot{V} = Av_i$. For a room of volume V_{room},

$$t_{\text{ex}} := \frac{V_{\text{room}}}{Av_i} \tag{86}$$

is the time for a volume of air equal to the room's to pass through the rotor disks once.

Example 10.6. For the design of section 17, $A = 0.760\,\text{m}^2$ and $v_i = 3.0\,\text{m/s}$, so in a $5\times4\times2.4\,\text{m}$ room $\dot{V} = 2.28\,\text{m}^3/\text{s}$, $V_{\text{room}} = 48\,\text{m}^3$ and $t_{\text{ex}} = 21\,\text{s}$. The mean return velocity distributed over the floor plan is $\dot{V}/(5 \times 4) = 0.11\,\text{m/s}$.

Figure 34 shows the circulation this sets up in the room.

Within one minute of a thirty-minute session the machine is flying in air it has already accelerated. The quiescent-inflow assumption underlying every result of section 3 has quietly ceased to hold. Correcting this is not a matter of a coefficient: it requires solving the room flow, which is a computational fluid dynamics problem with a moving boundary. It is the largest outstanding gap in the model and is recorded as such in section 18. A secondary consequence is a persistent draught of order $0.1\,\mathrm{m/s}$ *throughout the room, far below the* $6.0\,\mathrm{m/s}$ *wake velocity but present continuously and over the whole floor area.*

Part II
The Differential Layer: Flight

11 Flight dynamics

Once the machine exists its mass is fixed (assumption 2.3) and the problem becomes one of differential equations. This section derives them.

11.1 Configuration and kinematics

The configuration space is $\mathrm{SE}(3) = \mathbb{R}^3 \rtimes \mathrm{SO}(3)$, with state $(\boldsymbol{r}, R)$ and velocities $(\boldsymbol{v}, \boldsymbol{\omega})$.

Proposition 11.1 (Attitude kinematics)**.**

$$\dot{R} = R\,\widehat{\boldsymbol{\omega}}. \tag{87}$$

In terms of the unit quaternion $q = (q_0, \boldsymbol{q}_v)$ *representing* R*,*

$$\dot{q} = \tfrac{1}{2}\, q \otimes (0, \boldsymbol{\omega}) = \tfrac{1}{2} \begin{bmatrix} -\boldsymbol{q}_v^{\mathsf{T}} \\ q_0 I + \widehat{\boldsymbol{q}_v} \end{bmatrix} \boldsymbol{\omega} =: \tfrac{1}{2}\, \Omega(\boldsymbol{\omega})\, q. \tag{88}$$

Proof. Let $\boldsymbol{u}$ be a vector fixed in $\mathcal{B}$, so its inertial coordinates are $R\boldsymbol{u}$ and its inertial time derivative is $\boldsymbol{\omega}_{\mathcal{I}} \times R\boldsymbol{u}$ where $\boldsymbol{\omega}_{\mathcal{I}} = R\boldsymbol{\omega}$. Then $\dot{R}\boldsymbol{u} = \widehat{(R\boldsymbol{\omega})}R\boldsymbol{u}$ for all $\boldsymbol{u}$, so $\dot{R} = \widehat{(R\boldsymbol{\omega})}R = R\widehat{\boldsymbol{\omega}}R^{\mathsf{T}}R = R\widehat{\boldsymbol{\omega}}$ using eq. (4). Equation (88) follows by differentiating eq. (5) and using quaternion multiplication; see [17, §1.4]. □

Remark 11.2. Equation (88) preserves $\|q\|$ exactly, since $q^{\mathsf{T}}\dot{q} = \frac{1}{2}q^{\mathsf{T}}\Omega q = 0$ by antisymmetry. Numerically, integration error accumulates and the engine renormalises after each step; see section 15.

11.2 Newton-Euler equations

Theorem 11.3 (Equations of motion)**.** *For a rigid body of mass* m *and body-frame inertia tensor* J *subject to a total external force* $\boldsymbol{F}$ *expressed in* $\mathcal{I}$ *and a total external moment* $\boldsymbol{\tau}$ *about the centre of mass expressed in* $\mathcal{B}$*,*

$$\dot{\boldsymbol{r}} = \boldsymbol{v}, \tag{89}$$

$$m\,\dot{\boldsymbol{v}} = \boldsymbol{F}, \tag{90}$$

$$J\,\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times J\boldsymbol{\omega} = \boldsymbol{\tau}. \tag{91}$$

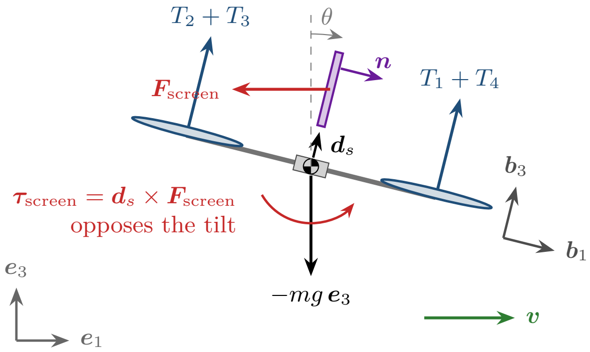


Figure 35: Free-body diagram in forward flight, side view with $\boldsymbol{b}_2$ into the page. The rotor thrusts act along $\boldsymbol{b}_3$, tilted by θ toward the motion, so that $TR\boldsymbol{e}_3 - mg\boldsymbol{e}_3$ has a horizontal component that accelerates the machine; the weight acts at the centre of mass; the screen drag acts at the offset $\boldsymbol{d}_s$ and produces the moment $\boldsymbol{d}_s \times \boldsymbol{F}_{\text{screen}}$ of eq. (100). With the display above the centre of mass that moment turns the airframe back against its tilt.

Proof. Equations (89) and (90) are Newton's second law in an inertial frame. For eq. (91), the angular momentum about the centre of mass expressed in $\mathcal{I}$ is $\boldsymbol{h}_{\mathcal{I}} = RJ\boldsymbol{\omega}$ and $\dot{\boldsymbol{h}}_{\mathcal{I}} = \boldsymbol{\tau}_{\mathcal{I}} = R\boldsymbol{\tau}$. Differentiating using eq. (87),

$$\dot{\boldsymbol{h}}_{\mathcal{I}} = \dot{R}J\boldsymbol{\omega} + RJ\dot{\boldsymbol{\omega}} = R\hat{\boldsymbol{\omega}}J\boldsymbol{\omega} + RJ\dot{\boldsymbol{\omega}} = R(\boldsymbol{\omega} \times J\boldsymbol{\omega} + J\dot{\boldsymbol{\omega}}).$$

Equating and cancelling R gives eq. (91). □

The forces are

$$\boldsymbol{F} = T\,R\boldsymbol{e}_3 - mg\,\boldsymbol{e}_3 + \boldsymbol{F}_{\text{aero}} + \boldsymbol{F}_{\text{dist}}, \tag{92}$$

with T the total rotor thrust along the body $\boldsymbol{b}_3$ axis. Figure 35 shows the forces in forward flight.

11.3 Rotor wrench and the allocation map

Model 11.4 (Quasi-steady rotor)**.** Rotor i turning at Ω_i produces thrust and reaction torque

$$T_i = k_T\Omega_i^2, \qquad Q_i = k_Q\Omega_i^2, \tag{93}$$

with k_T, k_Q determined by sections 3 and 4 rather than fitted: $k_T = T/\Omega^2$ and $k_Q = P_{\text{sh}}/\Omega^3$ evaluated at the design point.

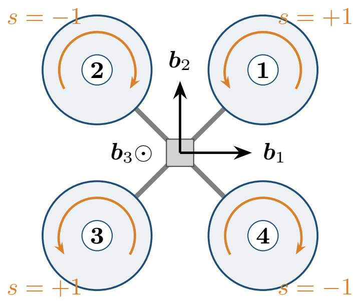
Let rotor i sit at body position $\boldsymbol{d}_i = (x_i, y_i, 0)$ and have spin sense $s_i \in \{+1, -1\}$. Its contribution to the body moment is $\boldsymbol{d}_i \times T_i\boldsymbol{e}_3$ from thrust and $-s_iQ_i\boldsymbol{e}_3$ from reaction torque, so

$$\boldsymbol{\tau} = \sum_{i=1}^{N} \begin{bmatrix} y_i\,k_T \\ -x_i\,k_T \\ -s_i\,k_Q \end{bmatrix} \Omega_i^2, \qquad T = \sum_{i=1}^{N} k_T\Omega_i^2. \tag{94}$$

Definition 11.5 (Allocation matrix)**.** Collecting eq. (94) with $\boldsymbol{u} = (\Omega_1^2, \dots, \Omega_N^2)$,

$$\begin{bmatrix} T \\ \boldsymbol{\tau} \end{bmatrix} = \boldsymbol{M}\,\boldsymbol{u}, \qquad \boldsymbol{M} = \begin{bmatrix} k_T & \cdots & k_T \\ y_1k_T & \cdots & y_Nk_T \\ -x_1k_T & \cdots & -x_Nk_T \\ -s_1k_Q & \cdots & -s_Nk_Q \end{bmatrix} \in \mathbb{R}^{4\times N}. \tag{95}$$

Figure 36 writes $\boldsymbol{M}$ out for the layout used throughout.



$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \underbrace{\begin{bmatrix} k_T & k_T & k_T & k_T \\ \ell k_T & \ell k_T & -\ell k_T & -\ell k_T \\ -\ell k_T & \ell k_T & \ell k_T & -\ell k_T \\ -k_Q & k_Q & -k_Q & k_Q \end{bmatrix}}_{\boldsymbol{M}} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

rows: thrust; roll $y_i k_T$; pitch $-x_i k_T$; yaw $-s_i k_Q$
rotor i at $(x_i, y_i) = L(\cos\psi_i, \sin\psi_i)$, $\psi_i = \frac{\pi}{4} + \frac{(i-1)\pi}{2}$, $\ell = L/\sqrt{2}$

Figure 36: The allocation map eq. (95) for the $\times$-layout quadrotor used throughout. Rotor i sits at angle $\psi_i = \pi/4 + (i-1)\pi/2$ from $\boldsymbol{b}_1$, with alternating spin: $s_i = +1$ turns counterclockwise seen from above and reacts on the body with $-k_Q\Omega_i^2$ about $\boldsymbol{b}_3$. Each column of $\boldsymbol{M}$ is one rotor's contribution to thrust, roll, pitch and yaw. The sign patterns of the three moment rows, $(+,+,-,-)$, $(-,+,+,-)$ and $(-,+,-,+)$, are orthogonal to each other and to the thrust row: that is why rank $\boldsymbol{M} = 4$ (proposition 11.6), and why a balanced change of rotor speeds can move one moment and nothing else.

Proposition 11.6 (Controllability of the wrench)**.** *The set of achievable wrenches is $\boldsymbol{M}\mathbb{R}^N_{\geq 0}$. Full four-axis authority requires* rank $\boldsymbol{M} = 4$*, which for hubs equispaced on a circle of radius L with alternating spin senses holds whenever $N \geq 4$ is even. For odd N, $\sum_i s_i \neq 0$ and the hover trim $\Omega_i \equiv \Omega_h$ produces a nonzero yaw moment, so the vehicle cannot trim without unequal thrusts.*

Proof. The rank statement is a direct computation on eq. (95): the first row is constant, rows two and three span the $(y_i, -x_i)$ pattern which has rank two for $N \geq 3$ equispaced hubs, and row four is independent of the others exactly when the s_i are not all equal. For odd N with alternating signs, $\sum_i s_i = \pm 1$, so at equal Ω_i the yaw component of $\boldsymbol{M}\boldsymbol{u}$ is $\mp k_Q \Omega_h^2 \neq 0$. □

Remark 11.7. Proposition 11.6 is why the engine restricts N to even values. A tricopter resolves the trim problem with a tilting motor, which is a different actuation model and is outside the scope of eq. (95).

11.4 Motor dynamics

Rotor speed does not follow command instantaneously. The simplest adequate model is first order,

$$\tau_m\,\dot{\Omega}_i = \Omega_i^{\text{cmd}} - \Omega_i, \tag{96}$$

with τ_m of order 20 to 60 ms. The underlying electromechanical model is

$$L\dot{\imath}_i = V_i - R_m \imath_i - k_e\Omega_i, \qquad J_r\dot{\Omega}_i = k_\tau \imath_i - k_Q\Omega_i^2, \tag{97}$$

whose electrical time constant L/R_m is typically two orders of magnitude faster than the mechanical one, so that singular perturbation reduces eq. (97) to a first-order system in Ω_i; eq. (96) is its linearisation about the trim point. We use eq. (96) and record the simplification.

11.5 Battery state

$$\dot{E} = -P, \qquad P = \frac{1}{\eta}\sum_{i=1}^{N} k_P\,\Omega_i^3 + P_{\text{aux}}, \qquad k_P = \left.\frac{P_{\text{sh}}}{\Omega^3}\right|_{\text{design}}, \tag{98}$$

or in normalised form $\dot{s} = -P/E_{\max}$ with $s = E/E_{\max}$ the state of charge. The cubic dependence follows from $P_{\text{sh}} = Q\Omega$ with $Q \propto \Omega^2$. Equation (98) is the coupling that makes control effort cost endurance, and it is what section 13 closes.

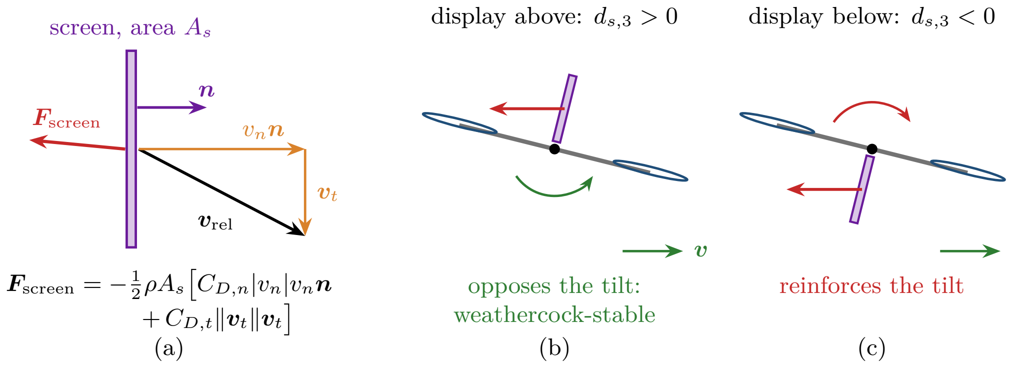


Figure 37: (a) The flat-plate model eq. (99). The plate's velocity relative to the air splits into a normal part $v_n\boldsymbol{n}$, resisted with $C_{D,n}$ over the whole area, and a tangential part $\boldsymbol{v}_t$, resisted only by skin friction, so the drag points almost along $-\boldsymbol{n}$ whatever the direction of motion. (b), (c) The moment of that drag about the centre of mass in forward flight, airframe tilted toward the motion. With the display above, the moment turns the airframe back against its tilt; with the display below, further into it. The magnitude is the same; the sign is not.

Equation (98) is calibrated at hover and has no inflow term. At fixed rotor speed it returns the same power in hover and in axial translation, so it does not contain the climb correction of eq. (16), which the component model of section 3 does. It is a near-hover approximation, and so are the mission energies computed with it: they are appropriate for following a person indoors at walking speed and vertical rates of a few tenths of a metre per second, and they are not a model of climbs, descents or aggressive manoeuvres. Adding a climb term to the power alone, while thrust keeps its hover calibration, would make the two inconsistent; a consistent treatment couples an axial and oblique inflow model to both.

11.6 The screen as an aerodynamic surface

This is what distinguishes the vehicle from a camera drone. The display is a large flat plate, rigidly attached through a gimbal, and therefore both generates orientation-dependent drag and applies that drag at a point offset from the centre of mass.

Model 11.8 (Flat plate with orientation)**.** Let $\boldsymbol{n}$ be the outward normal of the screen in inertial coordinates and $\boldsymbol{v}_{\text{rel}} = \boldsymbol{v} - \boldsymbol{v}_{\text{air}}$. Decompose $\boldsymbol{v}_{\text{rel}} = v_n\boldsymbol{n} + \boldsymbol{v}_t$ with $v_n = \boldsymbol{n}^{\mathsf{T}}\boldsymbol{v}_{\text{rel}}$. Then

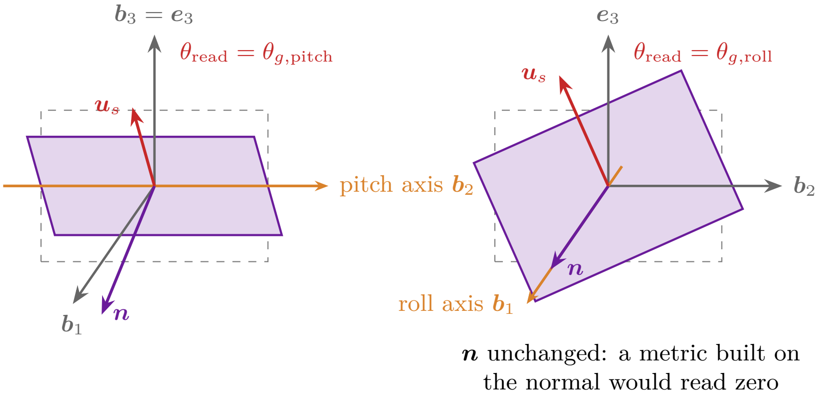
$$\boldsymbol{F}_{\text{screen}} = -\tfrac{1}{2}\rho A_s\Big[C_{D,n}\,|v_n|\,v_n\,\boldsymbol{n} + C_{D,t}\,\|\boldsymbol{v}_t\|\,\boldsymbol{v}_t\Big], \tag{99}$$

with $C_{D,n} \approx 1.17$ for a normal flat plate and $C_{D,t} \ll C_{D,n}$ skin friction.

The moment about the centre of mass follows from the offset $\boldsymbol{d}_s$ of the screen's centre of pressure in body coordinates:

$$\boldsymbol{\tau}_{\text{screen}} = R^{\mathsf{T}}\big[(R\boldsymbol{d}_s) \times \boldsymbol{F}_{\text{screen}}\big]. \tag{100}$$

Remark 11.9 (The sign of $\boldsymbol{d}_s$ matters)*.* If the display is carried *above* the rotor plane, $d_{s,3} > 0$; if it hangs below, $d_{s,3} < 0$. In forward flight the drag acts rearward, so a screen above the centre of mass produces a nose-up moment that opposes the forward tilt, a weathercock-stable arrangement, while a screen below produces a nose-down moment that reinforces it. The magnitude is identical; only the sign changes, and it changes the closed-loop behaviour measurably. Figure 37 shows the decomposition and the two signs.



(a) gimbal pitch (b) gimbal roll

Figure 38: The gimbal and the readability error eq. (102), airframe level, dashed outline the display at rest in the $\boldsymbol{b}_2\boldsymbol{b}_3$ plane. (a) Gimbal pitch about $\boldsymbol{b}_2$ tilts the normal $\boldsymbol{n}$ and the up vector $\boldsymbol{u}_s$ together, and θ_{read} equals the pitch angle. (b) Gimbal roll about $\boldsymbol{b}_1$ turns the image in its own plane: $\boldsymbol{n}$ does not move, so a metric built on the normal reads zero while the reader sees the image rotated by the roll angle. Measuring $\boldsymbol{u}_s$ against $\boldsymbol{e}_3$ catches both.

Remark 11.10 (Why the screen is nearly transparent to the rotors)*.* The screen is vertical and the rotor inflow and wake are vertical, so the plate is edge-on to the rotor flow and blocks a projected area of order (thickness × width), some $2 \times 10^{-3}\,\text{m}^2$ against a disk area of $0.76\,\text{m}^2$. Placing it above or below the rotor plane is therefore almost free aerodynamically. What it changes is geometry relative to the user, which section 17 shows to be the decisive consideration.

11.7 Gimbal

The display is carried on a gimbal whose job is to hold it upright while the airframe tilts to accelerate. Modelling the pitch and roll axes as independently actuated single-degree-of-freedom systems,

$$J_g\ddot{\theta}_{g,j} + b_g\dot{\theta}_{g,j} = \tau_{g,j} - \tau_{\text{wind},j}, \qquad |\tau_{g,j}| \le \tau_g^{\max}, \quad j \in \{\text{pitch}, \text{roll}\}. \tag{101}$$

The gimbal motor torque reacts on the airframe and is included in $\boldsymbol{\tau}$.

Definition 11.11 (Readability error)**.** Let $\boldsymbol{u}_s$ be the unit vector along the top edge of the display in inertial coordinates. The readability error is

$$\theta_{\text{read}} = \arccos\left(\boldsymbol{e}_3^{\mathsf{T}}\boldsymbol{u}_s\right), \tag{102}$$

the angle between the top of the image and vertical.

Remark 11.12*.* Equation (102) is deliberately not the angle between the screen normal and horizontal. Gimbal roll leaves the normal unchanged while rotating the image in its own plane, so a normal-based metric is blind to exactly the error a reader notices most. Measuring the up-vector captures both axes (fig. 38).

11.8 The complete system

Assembling, the state is

$$x = \left(\boldsymbol{r},\ \boldsymbol{v},\ q,\ \boldsymbol{\omega},\ \Omega_1, \ldots, \Omega_N,\ E,\ \boldsymbol{\theta}_g,\ \dot{\boldsymbol{\theta}}_g\right) \in \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{S}^3 \times \mathbb{R}^3 \times \mathbb{R}^N \times \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2, \tag{103}$$

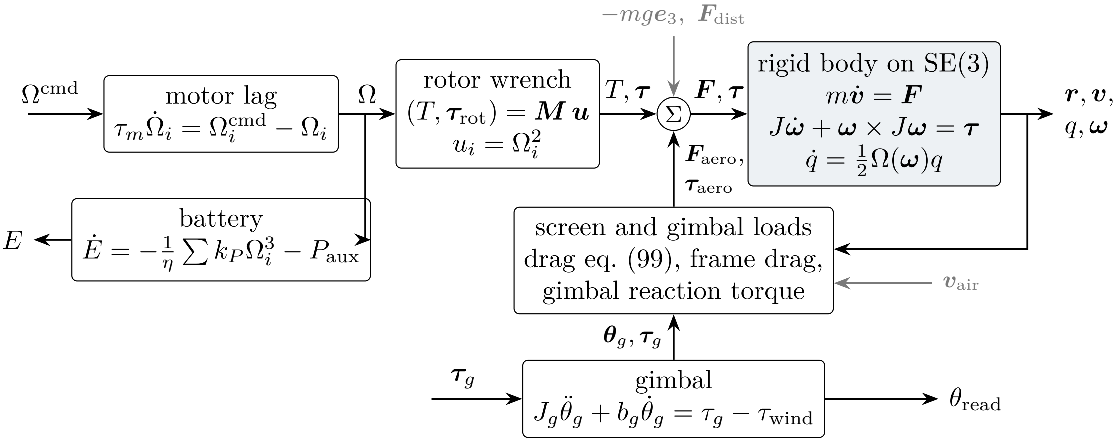


Figure 39: The plant $\dot{x} = f(x, u, w; p)$ of eq. (104). Commanded rotor speeds pass through the motor lag eq. (96) and the allocation eq. (95) into the rigid-body equations eqs. (90) and (91); velocity and attitude feed back through the screen and frame drag, and the gimbal enters through the screen's orientation and its reaction torque. The battery integrates the electrical power eq. (98) and is the one state that does not feed back into the motion, since no voltage sag is modelled: within a flight the energy coupling runs one way, and it closes only through the battery mass of section 13.

of dimension $18 + N$, and the dynamics are

$$\dot{x} = f(x, u, w;\, p), \tag{104}$$

with control $u = (\Omega_{1:N}^{\text{cmd}}, \boldsymbol{\tau}_g)$, disturbance w comprising wind and human motion, and design parameters p. Figure 39 shows how the pieces of this section connect.

Crucially, eq. (104) is not the whole problem. The parameters p are themselves constrained by the algebraic closure of section 5, giving a differential-algebraic system

$$\dot{x} = f(x, u, w; p), \qquad 0 = h(x, u, p), \tag{105}$$

whose algebraic part contains eq. (27) with the mission energy computed from a solution of the differential part. Section 13 makes this precise.

12 Control

12.1 Why a cascade

The vehicle is underactuated: four independent rotor speeds control six degrees of freedom. Horizontal motion is available only by tilting the thrust vector, so the causal chain is

$$\tau_y \;\to\; \ddot{\theta} \;\to\; \theta \;\to\; \ddot{x} \;\to\; x, \tag{106}$$

four integrations from moment to position. Near hover, linearising eqs. (90) and (91) about $R = I$, $T = mg$ gives the classical decoupling

$$\ddot{x} \approx g\theta, \quad \ddot{y} \approx -g\phi, \quad \ddot{z} \approx \frac{\delta T}{m}, \quad \ddot{\phi} = \frac{\tau_x}{J_x}, \quad \ddot{\theta} = \frac{\tau_y}{J_y}, \quad \ddot{\psi} = \frac{\tau_z}{J_z}. \tag{107}$$

The attitude loop is two integrations from its input, the position loop four. Since $\sqrt{K_R} \gg \sqrt{K_p}$ in any sensible design, time-scale separation justifies the cascade

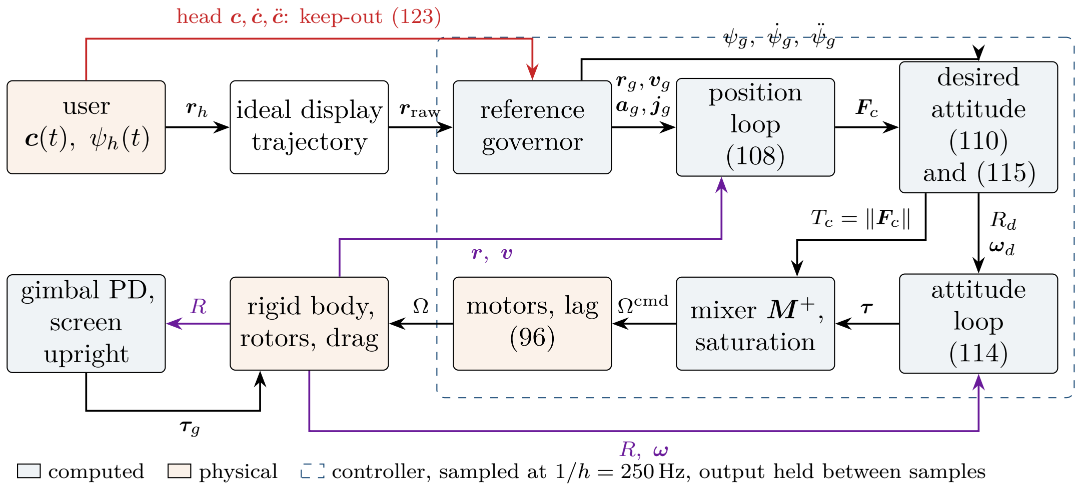


Figure 40: The cascade as the engine implements it. Blue boxes are computed at each sample of the controller, orange boxes are physical. The reference governor turns the ideal display trajectory into one the aircraft can fly and holds it outside the keep-out sphere about the user's head; the position loop turns the governed reference into a force $\boldsymbol{F}_c$; the thrust magnitude, the desired attitude R_d and its rate $\boldsymbol{\omega}_d$ follow from that force and its derivative; the attitude loop and the mixer turn them into rotor speed commands. The translational state closes the outer loop, the rotational state the inner one. The gimbal loop runs beside the cascade and keeps the screen upright while the airframe tilts.

$$\text{position} \to \text{velocity} \to \text{attitude} \to \text{body rate} \to \text{rotor speed},$$

each level treating the level below as instantaneous. This is standard practice [16], and eq. (107) is its justification. Figure 40 shows the cascade as the engine implements it, and fig. 41 the separation of its time scales that the argument needs.

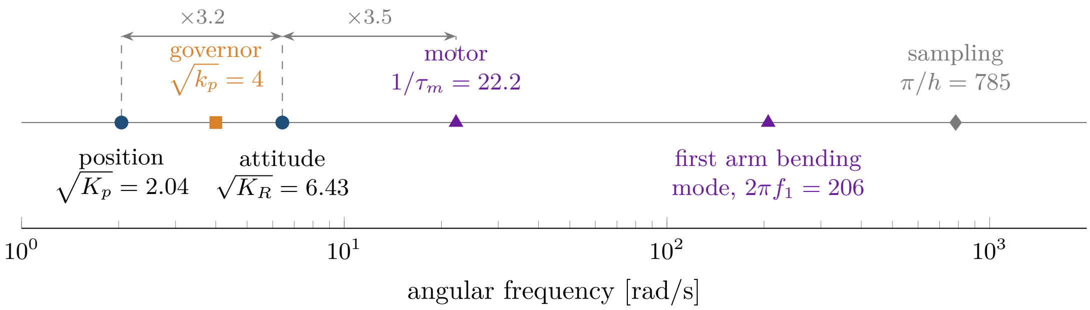


Figure 41: The frequency ladder of the design's cascade on a logarithmic axis, with the gains of model 12.7. Each loop is at least a factor of three slower than the one it relies on: position at $\sqrt{K_p}$, attitude at $\sqrt{K_R}$, the motors at $1/\tau_m$. The governor's filter at $\sqrt{k_p}$ shapes the reference and closes no loop on the vehicle. The first bending mode of the arms and the sampling rate of the controller lie one and two orders of magnitude above the fastest loop, which is the condition under which the airframe may be treated as rigid and the sampled controller as continuous.

Remark 12.1 (Flatness). The quadrotor with these dynamics is differentially flat with flat outputs (x, y, z, ψ) [18]: every state and input is an algebraic function of the flat outputs and finitely many derivatives. This is what makes feedforward from a desired trajectory possible and is why the controller below uses $\ddot{\boldsymbol{r}}_d$ directly rather than relying on feedback alone.

12.2 Position loop

Given a reference $(\boldsymbol{r}_d, \dot{\boldsymbol{r}}_d, \ddot{\boldsymbol{r}}_d)$, define errors $\boldsymbol{e}_p = \boldsymbol{r}_d - \boldsymbol{r}$, $\boldsymbol{e}_v = \dot{\boldsymbol{r}}_d - \boldsymbol{v}$, and command

$$\boldsymbol{a}_c = \ddot{\boldsymbol{r}}_d + K_p \boldsymbol{e}_p + K_d \boldsymbol{e}_v + K_i \int \boldsymbol{e}_p \, \mathrm{d}t, \qquad \boldsymbol{F}_c = m(\boldsymbol{a}_c + g\boldsymbol{e}_3). \tag{108}$$

The desired thrust magnitude and thrust direction are

$$T_c = \|\boldsymbol{F}_c\|, \qquad \boldsymbol{b}_3^d = \frac{\boldsymbol{F}_c}{\|\boldsymbol{F}_c\|}, \tag{109}$$

and with a desired yaw ψ_d the desired attitude is completed by

$$\boldsymbol{b}_2^d = \frac{\boldsymbol{b}_3^d \times \boldsymbol{b}_1^\psi}{\|\boldsymbol{b}_3^d \times \boldsymbol{b}_1^\psi\|}, \quad \boldsymbol{b}_1^d = \boldsymbol{b}_2^d \times \boldsymbol{b}_3^d, \quad R_d = [\, \boldsymbol{b}_1^d \; \boldsymbol{b}_2^d \; \boldsymbol{b}_3^d \,], \tag{110}$$

where $\boldsymbol{b}_1^\psi = (\cos\psi_d, \sin\psi_d, 0)$. Equation (110) is singular only when $\boldsymbol{b}_3^d \parallel \boldsymbol{b}_1^\psi$, i.e. at 90° of tilt, which the thrust limit precludes; the engine falls back to an alternative reference axis if it occurs.

12.3 Geometric attitude control on SO(3)

Attitude error is measured on the group rather than in coordinates, which avoids both the singularities of Euler angles and the unwinding of naive quaternion feedback.

Definition 12.2 (Attitude error)**.**

$$\boldsymbol{e}_R = \tfrac{1}{2}(R_d^\top R - R^\top R_d)^\vee, \qquad \boldsymbol{e}_\omega = \boldsymbol{\omega} - R^\top R_d\, \boldsymbol{\omega}_d, \tag{111}$$

with configuration error function $\Psi(R, R_d) = \frac{1}{2}\operatorname{tr}(I - R_d^\top R) \in [0, 2]$.

Model 12.3 (Geometric attitude law)**.**

$$\boldsymbol{\tau} = -K_R\, \boldsymbol{e}_R - K_\omega\, \boldsymbol{e}_\omega + \boldsymbol{\omega} \times J\boldsymbol{\omega} - J(\widehat{\boldsymbol{\omega}} R^\top R_d \boldsymbol{\omega}_d - R^\top R_d \dot{\boldsymbol{\omega}}_d). \tag{112}$$

Theorem 12.4 (Exponential attitude stability; Lee, Leok and McClamroch [13], Proposition 1)**.** *Consider eq.* (91) *under eq.* (112) *with positive scalar gains k_R, k_ω in place of K_R, K_ω and a smooth reference $(R_d, \boldsymbol{\omega}_d, \dot{\boldsymbol{\omega}}_d)$. The equilibrium $(\boldsymbol{e}_R, \boldsymbol{e}_\omega) = (0, 0)$ is exponentially stable, and its region of attraction contains the set*

$$\Psi(R(0), R_d(0)) < 2, \qquad \|\boldsymbol{e}_\omega(0)\|^2 < \frac{2}{\lambda_{\max}(J)}\, k_R(2 - \Psi(R(0), R_d(0))). \tag{113}$$

The excluded set is $\Psi = 2$, where $R_d^\top R$ is a rotation by π. For the unweighted Ψ used here every such rotation is a critical point: if $R_d^\top R = 2\boldsymbol{u}\boldsymbol{u}^\top - I$ for a unit vector $\boldsymbol{u}$, the matrix is symmetric and $\boldsymbol{e}_R = 0$. The critical set of Ψ is therefore the identity together with a two-dimensional family of attitudes, not three isolated points. Theorem 12.4 is a statement on the sublevel set eq. (113), and no almost-global claim is made here. Figure 42 shows the mechanism along any single axis of rotation.

The engine implements

$$\boldsymbol{\tau} = J(-K_R \boldsymbol{e}_R - K_\omega \boldsymbol{e}_\omega) + \boldsymbol{\omega} \times J\boldsymbol{\omega} - J(\widehat{\boldsymbol{\omega}} R^\top R_d \boldsymbol{\omega}_d - R^\top R_d \dot{\boldsymbol{\omega}}_d^\psi), \qquad \boldsymbol{e}_\omega = \boldsymbol{\omega} - R^\top R_d \boldsymbol{\omega}_d, \tag{114}$$

with the desired angular velocity taken from differential flatness [18]. With $\boldsymbol{j}_d$ the reference jerk and T_c the commanded thrust,

$$\boldsymbol{h}_\omega = \frac{m}{T_c}\Big(\boldsymbol{j}_d - (\boldsymbol{b}_3^d \cdot \boldsymbol{j}_d)\, \boldsymbol{b}_3^d\Big), \quad \boldsymbol{\omega}_d = \begin{bmatrix} -\boldsymbol{h}_\omega \cdot \boldsymbol{b}_2^d \\ \boldsymbol{h}_\omega \cdot \boldsymbol{b}_1^d \\ \dot{\psi}_d\, (\boldsymbol{b}_3^d \cdot \boldsymbol{e}_3) \end{bmatrix}, \quad \dot{\boldsymbol{\omega}}_d^\psi = \begin{bmatrix} 0 \\ 0 \\ \ddot{\psi}_d\, (\boldsymbol{b}_3^d \cdot \boldsymbol{e}_3) \end{bmatrix}. \tag{115}$$

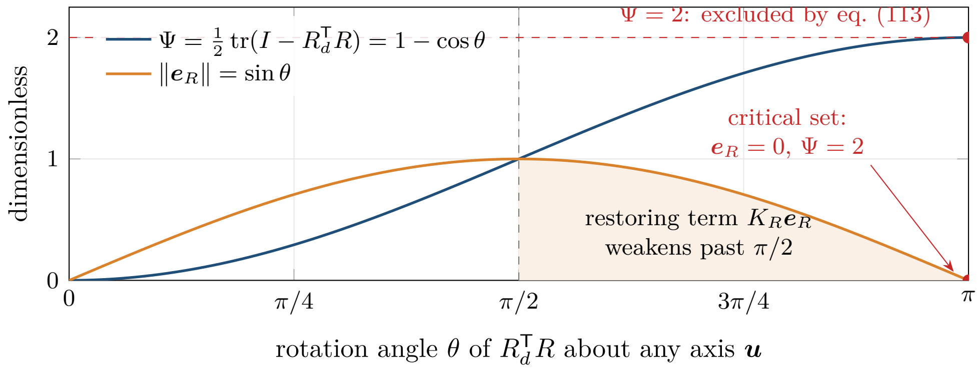


Figure 42: The attitude error for a rotation of $R_d^\mathsf{T} R$ by the angle θ about any unit axis $\boldsymbol{u}$, for which $\operatorname{tr}(R_d^\mathsf{T} R) = 1 + 2\cos\theta$ gives $\Psi = 1 - \cos\theta$ and $\boldsymbol{e}_R = \sin\theta\,\boldsymbol{u}$. The configuration error grows monotonically to its maximum 2 at $\theta = \pi$, but the restoring term, proportional to $\boldsymbol{e}_R$, peaks at $\pi/2$ and decays to zero at π (shaded). Every rotation by π is therefore a critical point of Ψ at which the proportional feedback vanishes, which is why eq. (113) excludes $\Psi = 2$ and why no continuous feedback of this kind is global on SO(3).

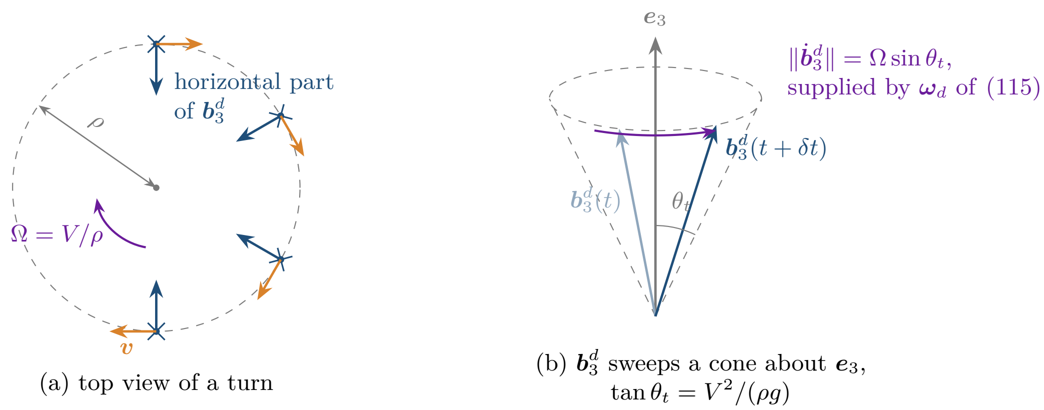


Figure 43: Why a curved path needs angular-rate feedforward. (a) On a turn of radius ρ flown at speed V the commanded force has a horizontal part pointing at the centre of the turn, so the horizontal part of $\boldsymbol{b}_3^d$ turns with the path at $\Omega = V/\rho$. (b) $\boldsymbol{b}_3^d$ therefore sweeps a cone of half-angle θ_t about $\boldsymbol{e}_3$, with $\tan\theta_t = V^2/(\rho g)$, and changes at the rate $\Omega\sin\theta_t$ even in a perfectly steady turn. With $\boldsymbol{\omega}_d = 0$ the attitude loop sees that rate as a persistent error and settles at a lag of about $K_\omega\|\boldsymbol{\omega}_d\|/K_R$; eq. (115) supplies the rate instead.

Equation (114) differs from the law of theorem 12.4 in three ways. The gains JK_R and JK_ω are matrices, which coincide with scalar gains only when J is a multiple of the identity. The tilt part of $\dot{\boldsymbol{\omega}}_d$, which needs the reference snap, is omitted. And the plant has motor lag and rotor saturation, which the theorem does not. Theorem 12.4 therefore motivates eq. (114) and does not certify it: the behaviour of the implemented loop is established by simulation for the trajectories reported, not by proof. Normalising by J makes K_R carry units of $1/\mathrm{s}^2$ and gives it a direct reading: the angular acceleration commanded per radian of attitude error.

Remark 12.5 (Why the rate feedforward matters). An earlier version of the engine set $\boldsymbol{\omega}_d = 0$. On a curved path the thrust direction rotates at the turn rate of the path (fig. 43), and a loop with no rate feedforward follows it with a standing attitude error of about $K_\omega\|\boldsymbol{\omega}_d\|/K_R$. On the turnaround mission of section 17 that version reached a peak tracking error of 0.27 m and a rotor disk passed 0.34 m from the user's eye. With eq. (115) the same mission gives 64 mm and 0.61 m.

12.4 Control authority and gain selection

Theorem 12.4 says nothing about saturation, and saturation is what actually limits this vehicle.

Definition 12.6 (Torque and angular acceleration authority)**.** Holding total thrust at mg and the other two moments at zero, the authority about body axis j is the smaller of the two one-sided maxima

$$\tau_j^{\max} = \min\{\tau_j^+, \tau_j^-\}, \qquad \tau_j^{\pm} = \max\Big\{ \pm\tau : \boldsymbol{M}\boldsymbol{u} = (mg,\ \tau\boldsymbol{e}_j),\ 0 \le u_i \le \Omega_{\max}^2 \Big\}, \tag{116}$$

a linear program in $\boldsymbol{u} = (\Omega_1^2, \ldots, \Omega_N^2)$ that the engine solves for every layout, with $\Omega_{\max}^2 = \lambda\,\Omega_h^2$ and $\Omega_h^2 = mg/(Nk_T)$. For the symmetric quadrotor with alternating spins it has the closed form (appendix B)

$$\tau_j^{\max} = \Delta \sum_{i=1}^{4} |\boldsymbol{M}_{j+1,i}|, \qquad \Delta = \min\{\Omega_h^2,\ \Omega_{\max}^2 - \Omega_h^2\}, \tag{117}$$

the differential being bounded by whichever is smaller, the room to speed two rotors up or the room to slow the other two down. The angular acceleration authority is $\alpha_j^{\max} = (J^{-1})_{jj}\,\tau_j^{\max}$, which is $\tau_j^{\max}/J_{jj}$ only when J is diagonal.

For the design of section 17, eq. (116) gives $\tau_{\text{roll}}^{\max} = \tau_{\text{pitch}}^{\max} = 3.66\,\text{N m}$ and $\tau_{\text{yaw}}^{\max} = 0.518\,\text{N m}$, i.e. $\alpha_{\text{roll/pitch/yaw}}^{\max} = 56/60/4.5\ \text{rad/s}^2$. An earlier version of eq. (116) divided the sum by N, on the reasoning that the thrust constraint "shares" the differential among the rotors; it does not, the constraint is met by pairing increments with decrements, and the error was a factor of four. It was found by independent review and is recorded in section 16. Figure 44 draws the whole attainable set, of which eq. (116) takes the axis intercepts.

Model 12.7 (Gain heuristic)**.** With $e_{\max}$ the largest attitude error the loop is expected to correct and ζ a damping ratio, the engine sets

$$K_R = \min\left\{\frac{\alpha^{\max}}{2\,e_{\max}},\ \left(\frac{0.3}{\tau_m}\right)^2\right\}, \qquad \alpha^{\max} := \min\{\alpha_{\text{roll}}^{\max}, \alpha_{\text{pitch}}^{\max}\}, \qquad K_\omega = 2\zeta\sqrt{K_R}, \tag{118}$$

and, one level up, $K_p = \min\{a_{\max}/e_{p,\max},\ K_R/9\}$, $K_d = 2\zeta\sqrt{K_p}$, with $a_{\max}$ the acceleration cap of the reference governor. The second term keeps the position bandwidth $\sqrt{K_p}$ at least a factor three below the attitude bandwidth $\sqrt{K_R}$, which is the time-scale separation the cascade of eq. (106) rests on; a machine whose attitude authority has been cut, by a display on a boom, is destabilised by position gains that were fine without the boom.

The first bound is the linearised saturation argument: with $\boldsymbol{e}_\omega$ small the commanded angular acceleration is $K_R\|\boldsymbol{e}_R\| \le K_R e_{\max}$, and half the single-axis authority is reserved for the rate term, for simultaneous demands on the other axes and for yaw. The second bound keeps the attitude bandwidth $\sqrt{K_R}$ below $0.3/\tau_m$, so that the first-order motor lag of eq. (96) contributes less than about 17° of phase at crossover. K_ω follows from the linearised closed loop $\ddot{e} + K_\omega\dot{e} + K_R e = 0$. None of this is a saturation certificate: it neglects the rate feedback term, simultaneous torque demands, the yaw axis and the actuator dynamics, and section 17 verifies by simulation that the resulting loop stays out of saturation on the missions flown. For the design the two bounds are $65/\text{s}^2$ and $44.4/\text{s}^2$, the lag bound binds, and the value used is $K_R = 41.4/\text{s}^2$, $K_\omega = 12.9/\text{s}$; fig. 41 places these values on one frequency axis.

Remark 12.8 (Yaw is the weak axis). By eq. (95) the yaw column involves k_Q rather than $k_T L$, and $k_Q/k_T \sim C_{d_0}$-scale quantities. A rotor optimised for quietness by observation 9.3 turns slowly and has small reaction torque, so $\alpha_{\text{yaw}}^{\max}$ is an order of magnitude below the roll and pitch values: $4.5\,\text{rad/s}^2$ against $56\,\text{rad/s}^2$ for the design. A quiet hovering display therefore

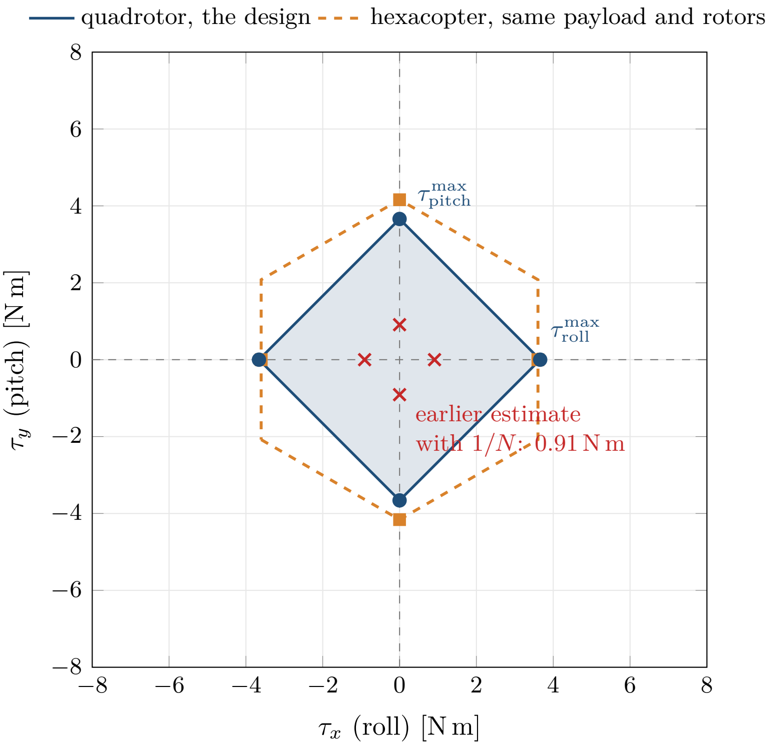


Figure 44: Attainable torque sets in the roll and pitch plane with thrust held at mg and the yaw moment at zero, computed from the engine through their support function: one linear program of the form eq. (116) for each of 240 directions. The quadrotor's set is a square turned by 45° whose axis intercepts are $\tau_{\text{roll}}^{\max} = \tau_{\text{pitch}}^{\max} = 3.66\,\text{N m}$, the balanced differential of eq. (117); the crosses mark the earlier estimate with the factor $1/N$, a quarter of it. The hexacopter of table 12, re-optimised under the same limits (gross 1.935 kg), has a hexagonal set that reaches further in pitch than in roll: its roll differential also produces a yaw moment, which the zero-yaw constraint must cancel, while its pitch differential does not. Its authority in the sense of eq. (116) is the smaller intercept, roll.

cannot turn to face its user quickly. The engine responds by governing the heading reference as it governs position, with a rate cap of 1.5 rad/s and an acceleration cap of half the yaw authority; the architectural response, a yaw axis on the gimbal where a 30 g motor solves what the airframe cannot, is not modelled.

12.5 Control allocation under saturation

Given a desired wrench $\boldsymbol{\chi}_d = (T_c, \boldsymbol{\tau})$ the allocation problem is

$$\text{find } \boldsymbol{u} \in [0, \Omega_{\max}^2]^N \quad \text{with} \quad \boldsymbol{M}\boldsymbol{u} = \boldsymbol{\chi}_d. \tag{119}$$

When $\boldsymbol{\chi}_d$ lies outside $\boldsymbol{M}[0, \Omega_{\max}^2]^N$ no exact solution exists and a priority must be chosen. We solve the unconstrained least-norm problem $\boldsymbol{u} = \boldsymbol{M}^+\boldsymbol{\chi}_d$ and, on violation, iterate

$$\boldsymbol{u} \leftarrow \Pi_{[0,\Omega_{\max}^2]}(\boldsymbol{u}), \qquad \boldsymbol{u} \leftarrow \boldsymbol{u} + \boldsymbol{M}^+(\boldsymbol{\chi}_d - \boldsymbol{M}\boldsymbol{u}) \odot \boldsymbol{\nu}, \tag{120}$$

with weights $\boldsymbol{\nu}$ that de-prioritise the thrust channel. The engineering content is the priority: *attitude is preserved and thrust is sacrificed*, because losing attitude loses the vehicle whereas losing thrust loses altitude, which is recoverable.

12.6 The reference governor

Tracking a person's face rigidly is the wrong specification, and this is a statement about kinematics rather than taste.

Proposition 12.9 (The cost of a rigid standoff during a turn). *Let the user rotate their heading by π in time Δt while the display is required to remain at standoff d ahead of them. Then the display must traverse a semicircle of radius d, and its mean speed and peak centripetal acceleration satisfy*

$$\bar{v} = \frac{\pi d}{\Delta t}, \qquad a_c = \frac{\bar{v}^2}{d} = \frac{\pi^2 d}{\Delta t^2}. \tag{121}$$

For $d = 0.7\,\mathrm{m}$ and $\Delta t = 1\,\mathrm{s}$ this is $2.2\,\mathrm{m/s}$ and $6.9\,\mathrm{m/s^2}$, and the simulated demand including the translation is over $10\,\mathrm{m/s^2}$. The engine confirms that increasing the thrust margin makes the outcome *worse*, because the controller then tilts harder and the gimbal cannot keep up.

Model 12.10 (Rate-limited reference with a moving keep-out). Let $(\boldsymbol{r}_{\mathrm{raw}}, \boldsymbol{v}_{\mathrm{raw}}, \boldsymbol{a}_{\mathrm{raw}}, \boldsymbol{j}_{\mathrm{raw}})$ be the ideal display trajectory and its derivatives, and let $\boldsymbol{c}(t)$ be the user's head with velocity $\dot{\boldsymbol{c}}$ and acceleration $\ddot{\boldsymbol{c}}$. The governed reference is the state $(\boldsymbol{r}_g, \boldsymbol{v}_g)$ of a second-order filter with acceleration feedforward,

$$\boldsymbol{a}^\star = \boldsymbol{a}_{\mathrm{raw}} + k_p(\boldsymbol{r}_{\mathrm{raw}} - \boldsymbol{r}_g) + k_d(\boldsymbol{v}_{\mathrm{raw}} - \boldsymbol{v}_g), \tag{122}$$

whose acceleration is projected, by alternating projections, onto the intersection of three convex sets: the acceleration ball $\|\boldsymbol{a}\| \le a_{\max}$, the next-step speed ball $\|\boldsymbol{v}_g + \boldsymbol{a}\ \mathrm{d}t\| \le v_{\max}$, and the keep-out half-space $\boldsymbol{n}^\mathsf{T}\boldsymbol{a} \ge \ell$. With $\varrho = d_{\mathrm{safe}} + \mathcal{R} + \varepsilon$ the keep-out radius, $\boldsymbol{n} = (\boldsymbol{r}_g - \boldsymbol{c})/\|\boldsymbol{r}_g - \boldsymbol{c}\|$, $h = \|\boldsymbol{r}_g - \boldsymbol{c}\| - \varrho$, $\dot{h} = \boldsymbol{n}^\mathsf{T}(\boldsymbol{v}_g - \dot{\boldsymbol{c}})$ and $\|\boldsymbol{v}_t\|^2 = \|\boldsymbol{v}_g - \dot{\boldsymbol{c}}\|^2 - \dot{h}^2$,

$$\ell = \max\left\{\boldsymbol{n}^\mathsf{T}\ddot{\boldsymbol{c}} - \frac{\|\boldsymbol{v}_t\|^2}{\|\boldsymbol{r}_g - \boldsymbol{c}\|} - 2\gamma\,\dot{h} - \gamma^2 h,\ \boldsymbol{n}^\mathsf{T}\ddot{\boldsymbol{c}} - \frac{2\,(h + \dot{h}\ \mathrm{d}t)}{\mathrm{d}t^2}\right\}, \qquad \gamma = 4. \tag{123}$$

The state then advances by $\boldsymbol{r}_g \leftarrow \boldsymbol{r}_g + \boldsymbol{v}_g\ \mathrm{d}t + \frac{1}{2}\boldsymbol{a}\ \mathrm{d}t^2$, $\boldsymbol{v}_g \leftarrow \boldsymbol{v}_g + \boldsymbol{a}\ \mathrm{d}t$. There is no position projection. If the three sets have empty intersection the step is flagged infeasible, the acceleration is scaled back to $a_{\max}$, and the violation is reported. The heading reference is governed separately by the same filter in one dimension with a rate cap and an acceleration cap.

Proposition 12.11 (What the keep-out guarantees). *(i) The caps alone do not keep the reference out of the user. On a curved raw path the acceleration cap makes the governed reference cut the corner, and since the raw path lies at distance d from the user by construction, the governed reference can approach the user arbitrarily closely as $a_{\max}$ decreases.*

(ii) In continuous time, and while the first term of eq. (123) *is the active constraint, the set $\{h \ge 0,\ \dot{h} + \gamma h \ge 0\}$ is forward invariant: a reference that starts outside the keep-out sphere with an admissible closing rate never enters it, whatever the user does.*

(iii) The second term of eq. (123) *is the one-step condition $h_{k+1} \ge 0$ under constant $\ddot{\boldsymbol{c}}$ over the step, linearised in the tangent plane. It is what the discrete update enforces.*

(iv) The vehicle, not the reference, is what must stay clear. If $\sup_t \|\boldsymbol{r}(t) - \boldsymbol{r}_g(t)\| \le \varepsilon$ on a mission, then on that mission the vehicle's centre stays at least $d_{\mathrm{safe}} + \mathcal{R}$ from the head, and hence every blade tip at least d_{safe}. The hypothesis is checked per run and reported; it is not proved.

Proof. (i) In the limit of small $a_{\max}$ the filter output is a convex combination of past raw positions, and a chord of a circular arc lies inside the arc, so the reference lies inside the

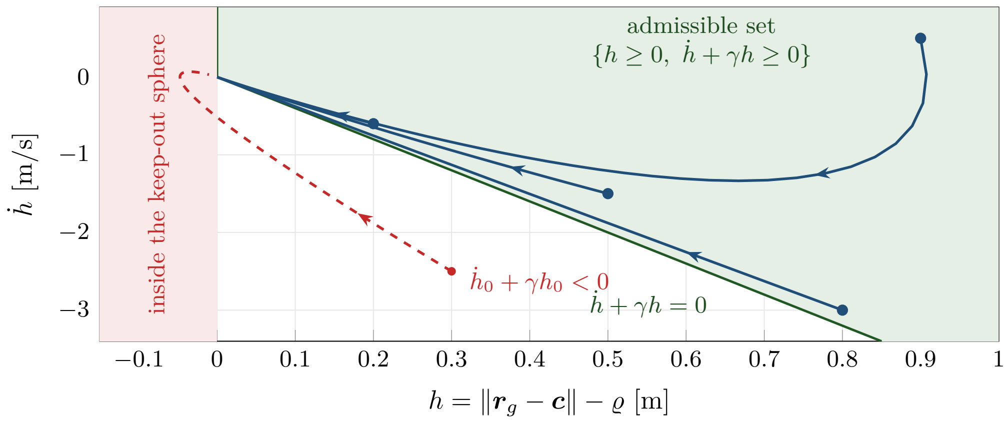


Figure 45: The phase plane of the keep-out barrier $h = \|\boldsymbol{r}_g - \boldsymbol{c}\| - \varrho$ with $\gamma = 4$. The shaded set $\{h \geq 0,\ \dot{h} + \gamma h \geq 0\}$ is forward invariant under the slowest recession that the first term of eq. (123) permits, $\ddot{h} = -2\gamma\dot{h} - \gamma^2 h$, whose solutions $h(t) = (h_0 + \psi_0 t)\, e^{-\gamma t}$ with $\psi_0 = \dot{h}_0 + \gamma h_0$ are drawn from four admissible starts: each reaches the origin without crossing $h = 0$. A start that closes faster than γh_0 (dashed) is not protected and enters the sphere. This is why the governor maintains the invariant set and not the half-space $h \geq 0$ alone.

sphere of radius d about the user. (ii) Differentiating h twice along the governed motion, with $\dot{\boldsymbol{n}} = \boldsymbol{v}_t / \|\boldsymbol{r}_g - \boldsymbol{c}\|$ in the notation above,

$$\ddot{h} = \boldsymbol{n}^{\mathsf{T}}(\boldsymbol{a} - \ddot{\boldsymbol{c}}) + \frac{\|\boldsymbol{v}_t\|^2}{\|\boldsymbol{r}_g - \boldsymbol{c}\|},$$

so the constraint $\boldsymbol{n}^{\mathsf{T}}\boldsymbol{a} \geq \ell_1$ with ℓ_1 the first term of eq. (123) is exactly $\ddot{h} \geq -2\gamma\dot{h} - \gamma^2 h$. Put $\psi = \dot{h} + \gamma h$; then $\dot{\psi} = \ddot{h} + \gamma\dot{h} \geq -\gamma\psi$, so $\psi(t) \geq \psi(0)e^{-\gamma t} \geq 0$, and then $\dot{h} \geq -\gamma h$ gives $h(t) \geq h(0)e^{-\gamma t} \geq 0$. This is the second-order barrier construction of Ames et al. [1], with both poles at $-\gamma$ to match the filter's own $k_p = \gamma^2$, $k_d = 2\gamma$. (iii) Expand $\|\boldsymbol{r}_g + \boldsymbol{v}_g\,\mathrm{d}t + \frac{1}{2}\boldsymbol{a}\,\mathrm{d}t^2 - \boldsymbol{c} - \dot{\boldsymbol{c}}\,\mathrm{d}t - \frac{1}{2}\ddot{\boldsymbol{c}}\,\mathrm{d}t^2\|$ to first order in the tangent plane about $\boldsymbol{n}$. (iv) Triangle inequality: $\|\boldsymbol{r} - \boldsymbol{c}\| \geq \|\boldsymbol{r}_g - \boldsymbol{c}\| - \|\boldsymbol{r} - \boldsymbol{r}_g\| \geq \varrho - \varepsilon = d_{\text{safe}} + \mathcal{R}$, and the reach bound eq. (61). □

Figure 45 draws part (ii) in the phase plane of h.

Three things the proposition does not say. The invariance in (ii) is a statement about the continuous-time reference; the engine enforces (iii) at update instants, and between them the reference moves under a constant acceleration, so (ii) is the limit the discrete scheme approaches, not what it executes. The guarantee lapses whenever the step is infeasible, which is why that is reported as a violation rather than absorbed. And $\boldsymbol{c}$ is the true head position: an estimator with latency or dropouts moves the sphere away from the head, and no estimator is modelled (assumption 2.7).

Figure 46 shows the governor on the most demanding mission of section 17, the turnaround.

Remark 12.12 (The governor's bandwidth must be high). The gains k_p, k_d in eq. (122) must place the filter's natural frequency well above the frequency content of the mission; otherwise the governor adds a steady lag on every curved path, which is a different and worse behaviour than yielding during a turn. With $k_p = 16$, $k_d = 8$ the filter is critically damped at 4 rad/s, and with the feedforward $\boldsymbol{a}_{\text{raw}}$ in eq. (122) it is exactly transparent whenever no cap binds. Without the feedforward it lags every acceleration by $\boldsymbol{a}_{\text{raw}}/k_p$, about 5 cm at 0.8 m/s^2, which was enough to push the reference into the keep-out sphere a third of the time on a mission with no turn in it.

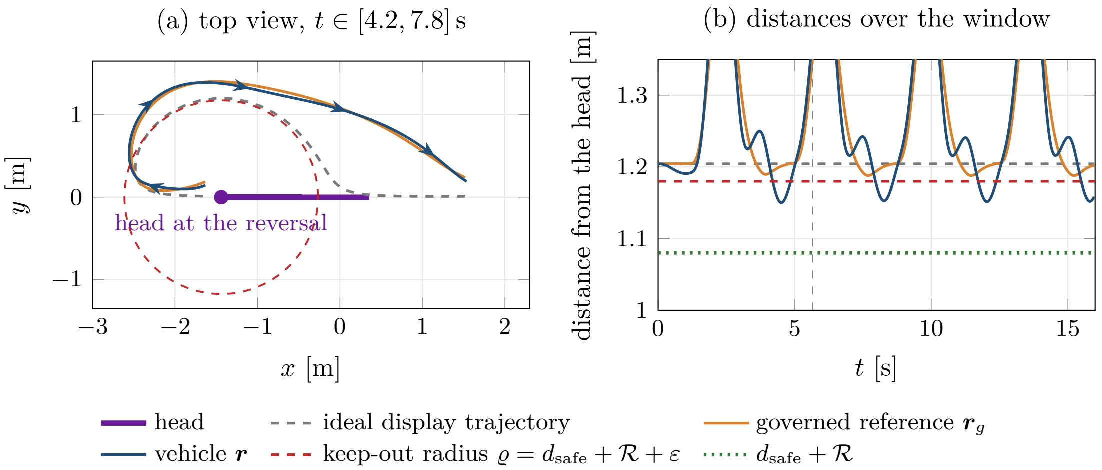


Figure 46: The turnaround flown by the engine with the governor of model 12.10 at $v_{\max} = 2.5\,\text{m/s}$, $a_{\max} = 2.5\,\text{m/s}^2$ and $\varepsilon = 0.10\,\text{m}$. (a) Top view around one reversal of the user. The ideal display trajectory is a semicircle of radius $d = 1.20\,\text{m}$ about the head, which proposition 12.9 shows is not flyable; the governed reference yields, swings wide and catches up after the turn, and the vehicle follows it, the arrows giving the direction of travel. The dashed circle is the keep-out sphere at the reversal, cut at the vehicle's altitude. (b) Distances from the head over the whole window, the vertical line marking the reversal shown in (a). The governed reference does not enter ϱ, as proposition 12.11(ii) and (iii) require; the vehicle falls below ϱ by at most its tracking error and stays above $d_{\text{safe}} + \mathcal{R}$, which is proposition 12.11(iv) with the tracking bound measured on this run. The peaks are the display lagging the user after each reversal.

Definition 12.13 (Two error measures)**.** With a governor in the loop, two distinct errors must be reported:

$$e_{\text{track}} = \|\boldsymbol{r}_g - \boldsymbol{r}\| \quad \text{(control performance)}, \qquad e_{\text{lag}} = \|\boldsymbol{r}_{\text{raw}} - \boldsymbol{r}\| \quad \text{(what the user sees)}. \tag{124}$$

Reporting only e_{track} would flatter the design, since the governor is free to make its own reference easy to follow. The peak of e_{track} over the whole window, start-up included, is the quantity that must stay below ε for proposition 12.11(iv).

12.7 Hover linearisation and controllability

Linearising eq. (104) about the hover trim by central differences gives $\dot{\delta x} = \boldsymbol{A}\,\delta x + \boldsymbol{B}\,\delta u$ on the $12 + N$ rigid-body and rotor states, after removing the quaternion scalar component which is not a free coordinate. The trim is computed, not assumed: with the display on a boom the centre of mass is off the rotor centroid and hover needs unequal rotor speeds. Controllability is assessed by the rank of the reachable subspace, computed by an orthogonal iteration $Q_{k+1} = \operatorname{range}[\boldsymbol{B} \;\; Q_k \;\; \boldsymbol{A}Q_k]$ on the matrices scaled by their 2-norms with a singular-value tolerance of 10^{-9}, which avoids forming the ill-conditioned powers of the Kalman matrix. Controllability requires the rank to equal the state dimension; for the design of section 17 it is 16 of 16. The eigenvalues cluster at the origin, reflecting the six open-loop integrators of eq. (106), which is precisely why the cascade rather than a single loop is required.

Part III
Coupling, Computation and Results

13 The coupled co-design problem

13.1 Statement

The algebraic layer of section 5 assumed the whole mission is spent hovering. The differential layer of section 11 assumed the mass was known. Neither is true independently. The honest statement couples them.

Problem 13.1 (Co-design)**.** Find a design vector p and a control law $u(\cdot)$ such that

$$\dot{x} = f(x, u, w; p), \qquad x(0) = x_0(p), \tag{125}$$

$$m = m_{\text{fix}} + m_{\text{str}}(p) + m_{\text{prop}}(p) + m_{\text{bat}}, \tag{126}$$

$$e_b(1 - \varsigma)\, m_{\text{bat}} \geq \int_0^{t_f} P\big(x(t), u(t); p\big)\, \mathrm{d}t, \qquad p_b\, m_{\text{bat}} \geq \max_{t \in [0, t_f]} P\big(x(t), u(t); p\big), \tag{127}$$

where in this section e_b denotes the usable specific energy before the landing reserve ς is removed and p_b the pack's specific power, subject to the path constraints

$$T(t) \leq \lambda m g, \quad \Omega_i(t) \leq \Omega_{\max}, \quad \ell(t) \geq d_{\text{safe}}, \quad \theta_{\text{read}}(t) \leq \theta_{\max}, \tag{128}$$

with $\ell(t)$ the distance from the user's eye to the nearest rotor disk, minimising a scalar objective

$$\mathcal{J}(p, u) = w_m m + w_P \bar{P} + w_N L_p + w_s\, \text{span} + w_e\, e_{\text{lag}}. \tag{129}$$

Equation (127) is the coupling. The right-hand side depends on the trajectory, which depends on the plant, which depends on m, which contains m_{bat}, which is the left-hand side. The first inequality keeps the reserve: starting from $E(0) = e_b m_{\text{bat}}$, the state of charge at t_f is at least ς. For a minimally sized continuous battery one of the two inequalities is an equality,

$$m_{\text{bat}} = \max\left\{ \frac{1}{e_b(1 - \varsigma)} \int_0^{t_f} P\ \mathrm{d}t,\ \frac{1}{p_b} \max_t P \right\}; \tag{130}$$

a discrete pack, built from whole cells, need not satisfy either with equality.

13.2 Reduction to a fixed point

Equations (126) and (127) can be reduced to a scalar fixed-point problem, which is what makes the system tractable.

Definition 13.2 (Inner and outer maps)**.** For a given battery mass μ, let

$$\mathcal{G}(\mu) \coloneqq \text{the solution } m \text{ of } m = m_{\text{fix}} + m_{\text{str}}(m) + m_{\text{prop}}(m) + \mu \tag{131}$$

be the gross mass consistent with that battery, and let

$$\mathcal{H}(\mu) \coloneqq \max\left\{ \frac{1}{e_b(1 - \varsigma)} \int_0^{t_f} P\big(x(t; \mathcal{G}(\mu), \mu), u(t)\big)\, \mathrm{d}t,\ \frac{1}{p_b} \max_t P\big(x(t; \mathcal{G}(\mu), \mu), u(t)\big) \right\} \tag{132}$$

be the battery mass that the resulting mission demands, by eq. (130).

Figure 47 shows how the engine evaluates the two maps.

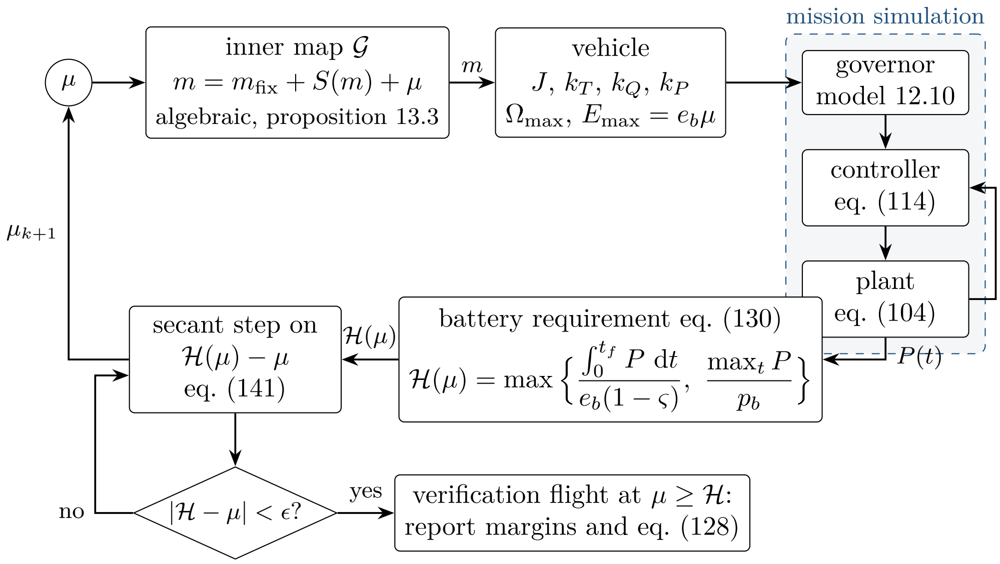


Figure 47: The co-design fixed point eq. (135) as the engine evaluates it. For a battery guess μ the inner map closes the mass algebraically; the vehicle is rebuilt from that mass (inertia tensor, rotor constants, pack energy); the mission is flown through the governor, the controller and the plant; the battery the flown mission needs is $\mathcal{H}(\mu)$, by eq. (130); and a guarded secant step on $\mathcal{H}(\mu) - \mu$ produces the next guess. Only the simulation is expensive, and it is run once per update. Convergence is followed by a verification flight at a battery no smaller than the last requirement, and only that flight's margins and path constraints are reported.

Proposition 13.3 (Well-posedness of the inner map)**.** *Let* $S(m) := m_{\text{str}}(m) + m_{\text{prop}}(m)$ *map* $[0, \infty)$ *into* $[0, \infty)$ *(masses are non-negative) and satisfy the Lipschitz bound*

$$|S(m_2) - S(m_1)| \leq L\,|m_2 - m_1| \quad \textit{for all } m_1, m_2 \geq 0, \qquad L < 1. \tag{133}$$

Then for every $\mu \geq 0$*, eq.* (131) *has exactly one solution* $\mathcal{G}(\mu)$*, the iteration* $m \leftarrow m_{\text{fix}} + S(m) + \mu$ *converges to it from any start at rate* L*, and* $\mathcal{G}$ *is Lipschitz with constant* $1/(1 - L)$*. If* S *is non-decreasing, so is* $\mathcal{G}$*. If* S *is differentiable,*

$$\mathcal{G}'(\mu) = \frac{1}{1 - S'(\mathcal{G}(\mu))}. \tag{134}$$

Proof. The map $\Phi(m) = m_{\text{fix}} + S(m) + \mu$ sends $[0, \infty)$ into itself because $S \geq 0$, and is a contraction with constant L by eq. (133), so the Banach fixed-point theorem [20, Ch. 5] gives existence, uniqueness and convergence of the iteration. For μ_1, μ_2 with solutions m_1, m_2, $m_2 - m_1 = S(m_2) - S(m_1) + \mu_2 - \mu_1$, so $|m_2 - m_1| \leq L|m_2 - m_1| + |\mu_2 - \mu_1|$, which is the Lipschitz bound. If S is non-decreasing and $\mu_2 > \mu_1$ but $m_2 < m_1$, then $m_2 - m_1 - (S(m_2) - S(m_1)) \leq -(1 - L)(m_1 - m_2) < 0$, contradicting $m_2 - m_1 - (S(m_2) - S(m_1)) = \mu_2 - \mu_1 > 0$. Finally $\Xi(m, \mu) = \Phi(m) - m$ has $\partial_m \Xi = S' - 1 \leq L - 1 < 0$, so the implicit function theorem applies and gives eq. (134). □

Remark 13.4 (Why a slope bound and not a growth bound)*.* Continuity, monotonicity and $\limsup S(m)/m < 1$ give existence but not uniqueness. The smooth, increasing, bounded $S(m) = 4/[1 + e^{-4(m-3)}]$ with $m_{\text{fix}} + \mu = 1$ makes $m = 1 + S(m)$ hold at $m \approx 1.0013$, 3 and 4.9987. The slope condition is what excludes this. For the lumped models $S' = f_s + \gamma_m$. For the component models of section 8, evaluated on the design of section 17, S' lies between 0.13 and 0.23 over 0.8 to 50 kg; the test suite asserts $0 \leq S' < 0.9$ on that interval. That is a check on one design over a bounded interval, not a proof for every parameter set. Figure 48 puts the counterexample beside the component models.

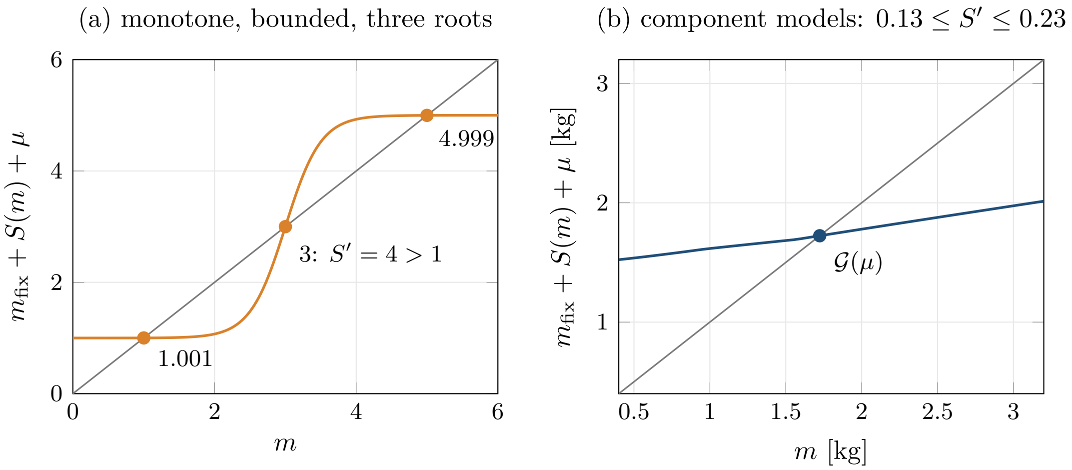


Figure 48: Uniqueness of the inner map eq. (131) needs a slope bound, not a growth bound. (a) The counterexample of the remark: S is smooth, increasing and bounded, yet $m = 1 + S(m)$ has three solutions, because $S'(3) = 4 > 1$ lets the curve cross the diagonal from below. (b) The component models of section 8 at the design battery of section 17: $m_{\text{fix}} + S(m) + \mu$ crosses the diagonal once, at the design mass, with slope between 0.13 and 0.20 on the plotted range, so proposition 13.3 applies and the inner iteration contracts at that rate.

Theorem 13.5 (Existence of a closed design)**.** *The system eqs.* (126) *and* (127) *is equivalent to the scalar fixed-point equation*

$$\mu = \mathcal{H}(\mu). \tag{135}$$

If $\mathcal{H}$ is continuous and maps a compact interval $[0, \mu_{\max}]$ into itself, a solution exists by Brouwer's theorem. If in addition $\mathcal{H}$ is a contraction, $\operatorname{Lip}(\mathcal{H}) = L < 1$*, the solution is unique and the iteration $\mu_{k+1} = \mathcal{H}(\mu_k)$ converges geometrically with rate L from any starting point.*

Proof. Equivalence is by construction. The existence and uniqueness statements are Brouwer's fixed-point theorem and the Banach fixed-point theorem respectively [20, Ch. 5]. □

Figure 49 shows $\mathcal{H}$ for the design, computed with the simulator: it is nearly affine, well inside the contraction regime, and the fixed point is found in two updates.

Proposition 13.6 (Contraction criterion)**.** *Suppose the mission power is well approximated by a mission-dependent multiple of the hover power, $\bar{P} = \varkappa P_{\text{hover}}(m)$ with $\varkappa$ independent of m, and that the energy term attains the maximum in eq.* (132) *in a neighbourhood of μ. Then*

$$\mathcal{H}'(\mu) = \frac{\varkappa t_f}{e_b(1-\varsigma)} \frac{\mathrm{d}P_{\text{hover}}}{\mathrm{d}m} \mathcal{G}'(\mu), \tag{136}$$

and in the fixed-rotor case with the lumped models, where $\mathcal{G}' = 1/(1 - f_s - \gamma_m) = 1/\alpha$,

$$\mathcal{H}'(\mu) = \frac{\varkappa}{\alpha} \cdot \frac{3}{2} \beta \sqrt{m}, \tag{137}$$

with β as in eq. (34)*, whose specific energy is net of the reserve, $e_b(1-\varsigma)$ in the notation of this section. The iteration therefore contracts precisely when $\frac{3}{2}\varkappa\beta\sqrt{m} < \alpha$, which at $\varkappa = 1$ is the condition $F'(m) > 0$ of corollary 6.9, i.e. $m < m^*$. On the power-limited branch $\mathcal{H}' = p_b^{-1}(\mathrm{d}P_{\text{peak}}/\,\mathrm{d}m)\mathcal{G}'$ instead, and where the two branches exchange the maximum $\mathcal{H}$ has a kink.*

Proof. Differentiate the energy term of eq. (132) by the chain rule. From eqs. (26) and (34), $P_{\text{hover}}(m) = \beta\, e_b(1-\varsigma)\, m^{3/2}/t_f + P_{\text{aux}}$, so $\mathrm{d}P_{\text{hover}}/\,\mathrm{d}m = \frac{3}{2}\beta\, e_b(1-\varsigma)\, m^{1/2}/t_f$. Substituting into eq. (136) with $\mathcal{G}' = 1/\alpha$, the factors t_f and $e_b(1-\varsigma)$ cancel and eq. (137) remains. □

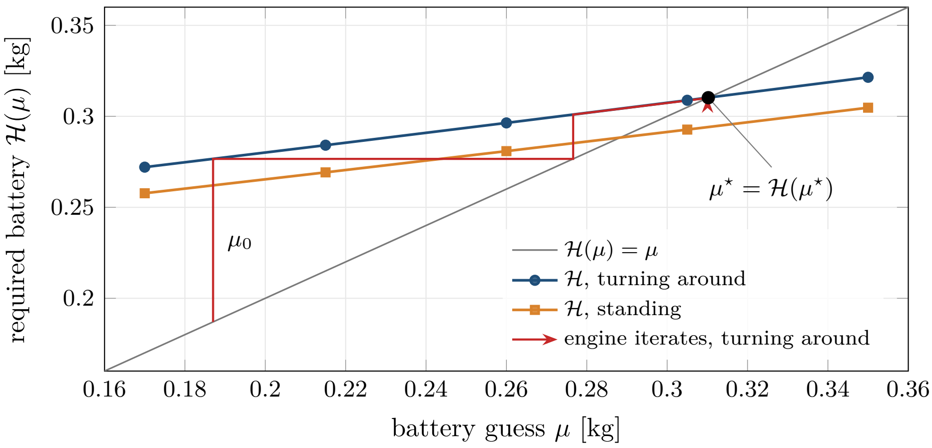


Figure 49: The outer map $\mathcal{H}$ for the design of section 17, sampled with the simulator at five battery masses for two missions, with the engine's iterates for the turnaround. $\mathcal{H}$ is nearly affine with slope 0.27 turning and 0.26 standing, the contraction factor of proposition 13.6; the fixed point $\mu^\star$ is where it crosses the diagonal. The first update is a plain fixed-point step, vertically to $\mathcal{H}$ and horizontally back to the diagonal; the second is a secant step that lands on $\mu^\star$. The two verification flights that follow are not drawn.

Remark 13.7 (What the overhead does, and does not, do). Proposition 13.6 connects the algebraic and the simulated pictures: the condition for the simulation-in-the-loop iteration to converge is the light-branch condition with β multiplied by $\varkappa$. It is tempting to say that overhead "degrades α". That is a reading of one inequality, not an identity. Multiplying the whole power by $\varkappa$ turns the closure into

$$\alpha m - \varkappa\beta\, m^{3/2} = m_{\text{fix}} + \varkappa\,\frac{P_{\text{aux}} t_f}{e_b}, \tag{138}$$

which scales the energy coupling and the auxiliary part of M_0 together, and by eq. (47) lowers the critical load by the factor $1/\varkappa^2$. A change of α would do neither. Figure 50 shows both effects on the design.

13.3 Residual formulation

Rather than iterate eq. (135) naively, the engine treats the pair $y = (m, m_{\text{bat}})$ and drives the residual

$$\boldsymbol{\mathcal{R}}(y) = \begin{bmatrix} m - \big(m_{\text{fix}} + m_{\text{str}}(m) + m_{\text{prop}}(m) + m_{\text{bat}}\big) \\ m_{\text{bat}} - \max\left\{\dfrac{E_{\text{mission}}(m, m_{\text{bat}})}{e_b(1-\varsigma)}, \dfrac{P_{\text{peak}}(m, m_{\text{bat}})}{p_b}\right\} \end{bmatrix} = \mathbf{0}. \tag{139}$$

The first component is algebraic and cheap; the second requires a trajectory simulation. The engine solves $\mathcal{R}_1 = 0$ at every outer step by the iteration of proposition 13.3, reducing the problem to the scalar $\mathcal{R}_2 = 0$ at one simulation per update. When $|\mathcal{R}_2|$ falls below tolerance the battery is rounded up to the last requirement and the design is flown again; the run is declared converged only when that verification flight requires no more battery than it carries. The energy margin $e_b(1-\varsigma)m_{\text{bat}} - E_{\text{mission}}$, the power margin $p_b m_{\text{bat}} - P_{\text{peak}}$ and the path constraints eq. (128) of the final flight are then reported separately. Closing the mass and energy balance and satisfying the path constraints are different outcomes: a converged battery loop can still be an unacceptable flight.

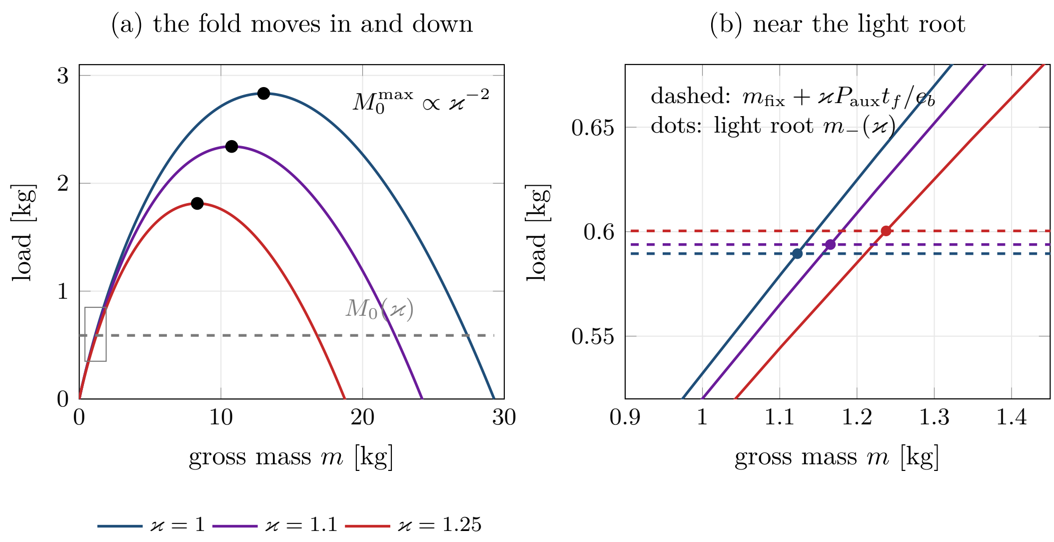


Figure 50: A mission overhead $\varkappa$ applied to the lumped closure of the design, eq. (138). (a) Multiplying all power by $\varkappa$ multiplies β by $\varkappa$: the fold moves to smaller mass and the critical load falls as $\varkappa^{-2}$, from 2.83 kg at $\varkappa = 1$ to 1.81 kg at $\varkappa = 1.25$; the load, dashed, rises only slightly because the auxiliary part is small. The rectangle marks panel (b). (b) Near the light root the design mass moves from 1.12 kg to 1.24 kg over the same range, about 10 %, while the margin to the fold shrinks by more than a third.

13.4 The outer optimisation

With the closure solved, eq. (129) becomes a function of the design vector alone, and problem 13.1 reduces to a constrained finite-dimensional program

$$\min_{p \in \mathcal{P}} \mathcal{J}(p) \quad \text{s.t.} \quad g_j(p) \leq 0, \; j = 1, \ldots, J, \tag{140}$$

with $\mathcal{P}$ a box and g_j the constraints of eq. (128) together with the modelling-validity constraints of assumptions 4.7 and 4.8 and eq. (23). The objective is non-convex, cheap to evaluate on the algebraic layer and expensive on the simulated one, and has large infeasible regions where the closure has no solution at all. This combination dictates the choice of a population-based method with exterior penalties, discussed in section 15.

Remark 13.8 (Why not adjoint optimal control). Problem 13.1 in full generality is a control co-design problem and could be attacked with adjoint-based methods over the trajectory. We do not, for two reasons. First, the objective is dominated by steady hover, for which the trajectory contributes a scalar $\varkappa$ near unity; the sensitivity of the design to the control law is small compared with its sensitivity to the rotor geometry. Second, and more importantly, the constraints that actually bind, acoustic and geometric, are algebraic functions of p and do not involve $u(\cdot)$ at all. Spending effort on trajectory optimisation would refine the part of the problem that is not deciding the answer.

14 The engine

14.1 Design principles

The implementation follows three rules, each of which was adopted because violating it caused a specific error during development.

Table 3: Modules and the sections of this paper they implement.

Module	Sections	Responsibility
`params.py`	2	parameter schema, derived quantities, named presets
`sizing.py`	3 to 7	momentum theory, closure, fold, sensitivities, fixed-point trace
`components.py`	4, 8, 10	rotor blade model, beam-sized arms, motor and propeller mass, inertia, near-field corrections
`acoustics.py`	9	source model, sound power, room field, A-weighting, blade passage
`electrical.py`	8	pack selection, K_V sizing, currents, C-rate checks
`dynamics.py`	11	state, forces, moments, allocation, hover linearisation
`control.py`	12	cascaded controller, authority, gain selection
`mission.py`	11, 12	human trajectories, wind, reference governor with keep-out
`simulate.py`	11	fixed-step integrator, telemetry, mission energy
`closure.py`	13	residual formulation, secant-accelerated battery loop, verification flight, margins
`numerics.py`	15	bounded mass-root search with explicit failure status, controllability rank
`optimize.py`	13	penalised objective, differential evolution, trade studies
`tether.py`	17	the closure with energy off-board
`sweep.py`	7	feasibility maps, exponent sweeps, endurance walls
`payloads.py`	17	display catalogue, readability, bill of materials

One source of truth for parameters. Every assumption is declared once, in a schema carrying its symbol, unit, admissible range, grouping and help text. Validation, the command line interface and the browser controls are all generated from that schema. Nothing about the model is written down twice. The alternative, a model that reads parameters and a user interface that mirrors them, guarantees eventual divergence between the two.

Analytic and numerical layers coexist. The closed forms of sections 6 and 7 are implemented alongside the numerical solvers that supersede them, and the test suite asserts agreement. The closed forms are not documentation; they are the oracle against which the code is verified. This is what caught the errors reported in section 16.

Assumptions are outputs wherever possible. The lumped constants f_s, S_m and FM remain as inputs to the algebraic layer, but the first-principles layer computes them and the engine reports both side by side. A model whose assumptions can be audited against its own detailed submodels is much harder to fool oneself with.

14.2 Module structure

Figure 51 shows how the modules depend on one another, and fig. 52 traces one evaluation of the coupled problem through them: the order of operations that the rest of this paper derives

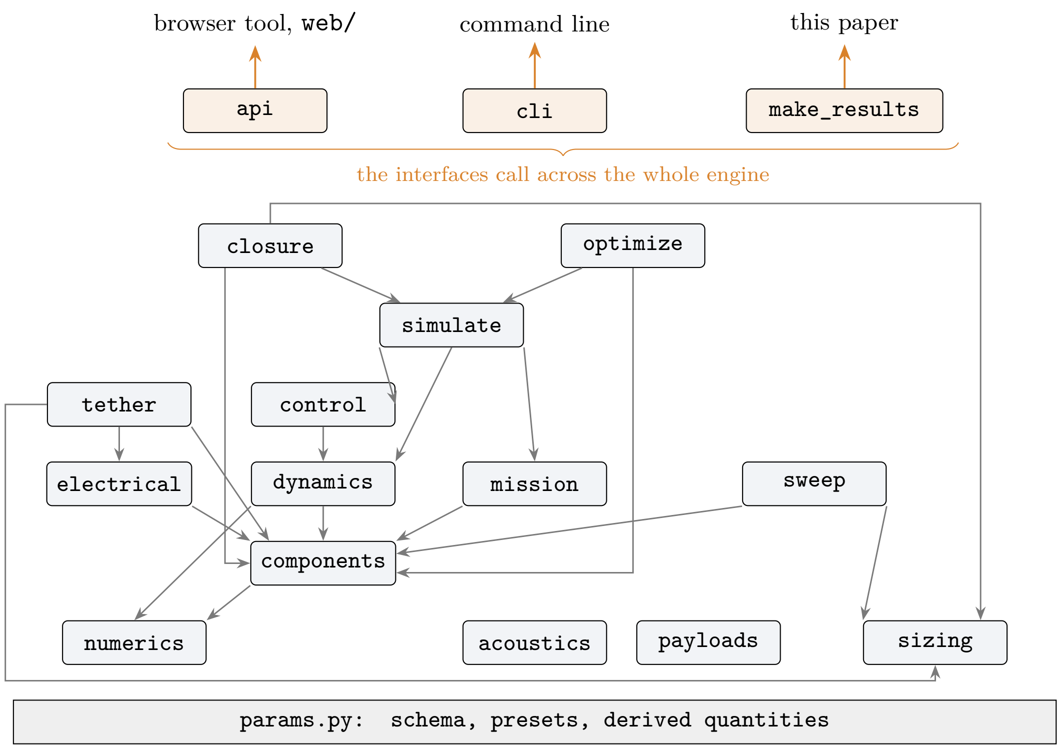


Figure 51: Import graph of the engine: an arrow from A to B means that module A imports B. Every module also imports `params.py`, drawn as the base. The layers read from the bottom up as the layers of the paper: the algebraic and component models, the dynamics and control, the simulation, and the two modules that close the coupled problem, `closure.py` and `optimize.py`. `acoustics.py` and `payloads.py` import nothing but the parameters. The three interfaces at the top call across the whole engine and their individual edges are omitted.

piece by piece.

14.3 Interfaces

Three front ends share the engine: a command line for scripted studies, an HTTP API, and a browser tool that generates its controls from the schema and recomputes on every parameter change. The tool exists because the questions that matter here, "what does this assumption cost", are answered by moving one number and watching six others, which is a different activity from running a batch job.

The browser tool renders the assembly in three dimensions from the same solved dimensions used by the closure: arm length and wall thickness from eqs. (60) and (66), rotor diameter and chord from section 4, motor body size from eq. (69), and the individual battery cells from the pack selection of section 8. The check that this is a drawing of the machine rather than a picture beside it is that the drawn parts sum to the closure's gross mass.

14.4 Reproducing the results

Every number in section 17 is produced by one script, `paper/make_results.py`, which writes the numbers together with the configuration that produced them (preset, payload, window, step, settling interval, seed, governor limits, solver tolerances and optimiser settings) to `paper/results/results.json`, and emits the table rows and macros that the LaTeX sources include. No figure in the design study is typed by hand. `paper/build.py` then compiles this document.

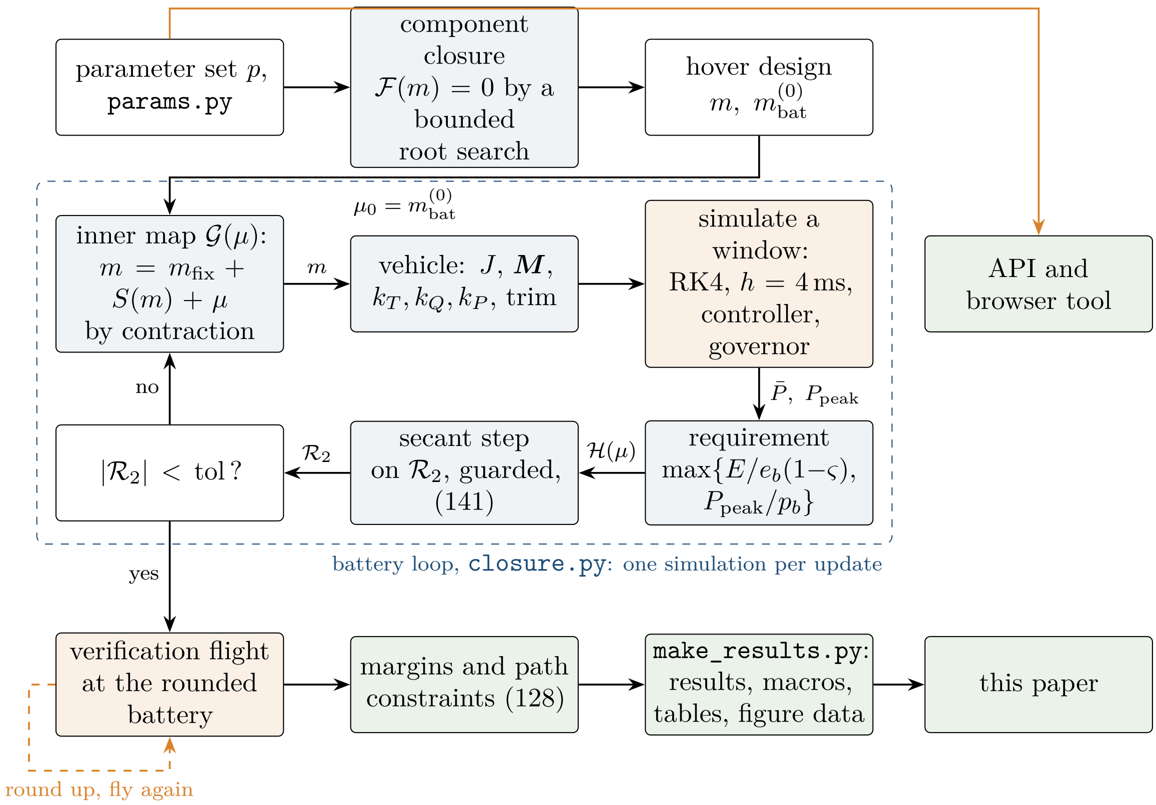


Figure 52: One evaluation of the coupled problem as the engine carries it out. A bounded root search on the component residual eq. (71) gives the hover design, which seeds the battery loop. Each update of the loop solves the inner mass map by contraction (proposition 13.3), rebuilds the vehicle, flies one simulation window and turns its mean and peak power into a battery requirement by eq. (130); a guarded secant step then moves the battery. When the residual is within tolerance the battery is rounded up and flown again, and the loop ends only on a verification flight that needs no more battery than it carries. The results pass through `make_results.py` into this paper, and the same engine serves the API and the browser tool.

15 Numerical methods

15.1 Solving the closure

Equation (39) could be solved by radicals at $q = 3/2$ or by Newton's method in general. Neither is used, for reasons that are structural rather than matters of taste.

Newton's method applied to $F(m) - M_0$ has a critical defect here: by lemma 6.1, $F'(m^*) = 0$, so the Newton step is unbounded near the fold, which is precisely the region of interest. Worse, from a starting point on the wrong side it converges to m_+, the unstable branch, without any indication that it has done so.

The engine instead exploits the shape established in lemma 6.1:

1. Compute m^* and $M_0^{\max}$ in closed form from eqs. (44) and (45). Feasibility is decided *analytically* by comparing M_0 with $M_0^{\max}$, never by whether an iteration converged.
2. If feasible, bracket m_- on $(0, m^*]$, where F is a strictly increasing bijection, and bisect. Bracket m_+ on $[m^*, \infty)$ by geometric expansion and bisect.

Bisection converges linearly with rate $1/2$ and requires $\lceil \log_2((b-a)/\epsilon) \rceil$ evaluations, about 40 for double precision. It is slower than Newton and it cannot fail. Given that a single evaluation of F costs microseconds, and given that the failure modes of Newton here are silent rather than loud, this is the correct trade.

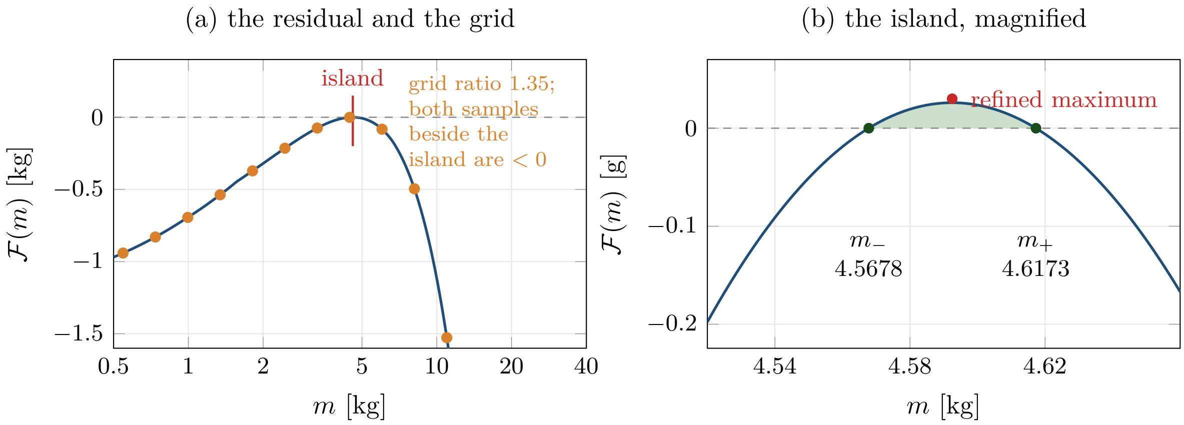


Figure 53: The component residual $\mathcal{F}$ of eq. (71) for the design of section 17 at $t_f = 4280.05\,\mathrm{s}$, just short of the fold. (a) On the scale of the whole search $\mathcal{F}$ appears merely to touch zero; the geometric grid of ratio 1.35 samples it at the dots, and the samples on either side of the positive stretch, at 4.460 kg and 6.021 kg, are both negative. (b) Magnified, the positive island is 1.1 % wide in mass and 0.03 g high, bounded by the roots $m_\pm$. The engine finds it by refining every sampled local maximum with a bounded maximisation in $\ln m$; the grid alone reported that no mass closes.

Remark 15.1 (Feasibility is analytic, not numerical). This is the single most important implementation decision in the sizing layer, and it follows from the critical slowing-down remark after theorem 6.8. A solver that reports non-convergence after K iterations cannot distinguish a near-feasible design from an infeasible one, because rate $= 1 - F'(m_-) \to 1$ as $M_0 \uparrow M_0^{\max}$. Deciding feasibility by $M_0 \lessgtr M_0^{\max}$ removes the ambiguity entirely.

For the component closure eq. (71) no closed form exists, and the shape of $\mathcal{F}$ is observed rather than proved (section 8), so the search cannot borrow the analytic feasibility test. The engine samples $\mathcal{F}$ on a geometric grid of ratio 1.35 from 10^{-3} kg upward, stopping at the first non-negative sample. Every sampled local maximum is then refined by bounded minimisation of $-\mathcal{F}$ in $\ln m$, because a positive stretch of $\mathcal{F}$ narrower than one grid interval can sit between two negative samples. The first bracket with a sign change, in increasing mass, is closed by Brent's method to a relative tolerance of 10^{-12}, which selects the light branch. The search reports one of four outcomes: a root, a near-tangent root resolved only to residual tolerance, a non-finite residual, or an exhausted search. The last is reported as a failure of a bounded search and is never described as infeasibility. The case that motivated the refinement is kept as a regression test: the design of section 17 at $t_f = 4280.0502\,\mathrm{s}$ has a positive island about 1 % wide in mass, with roots at 4.5678 kg and 4.6173 kg, against a grid spacing of 35 %. The grid alone straddled it and reported that no mass closes; fig. 53 shows the case.

15.2 The battery loop with the simulator inside it

Equation (135) is solved by iteration, but naive iteration converges only at the linear rate $\mathcal{H}'(\mu)$ of proposition 13.6, which near the fold approaches unity. Since each evaluation of $\mathcal{H}$ costs a full trajectory simulation, this is expensive.

The engine applies a secant step to the residual $\mathrm{gap}(\mu) = \mathcal{H}(\mu) - \mu$:

$$\mu_{k+1} = \mu_k - \mathrm{gap}(\mu_k)\,\frac{\mu_k - \mu_{k-1}}{\mathrm{gap}(\mu_k) - \mathrm{gap}(\mu_{k-1})}, \tag{141}$$

which uses only quantities already computed and therefore costs no additional simulations.

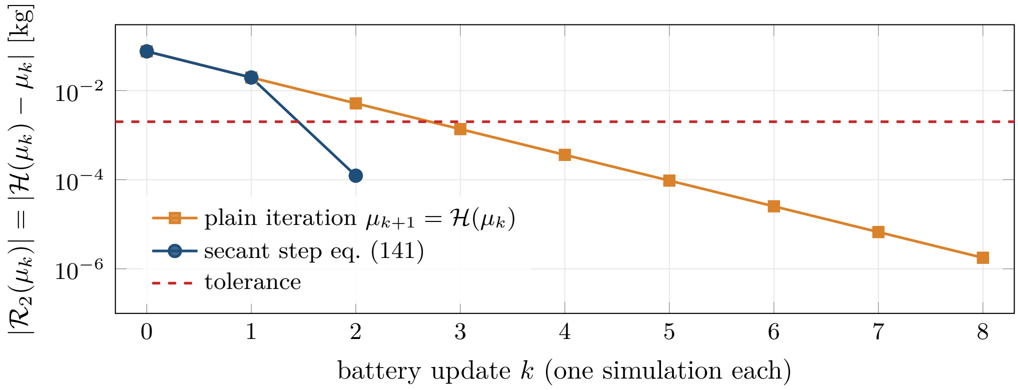


Figure 54: Convergence of the battery loop on the pacing mission, flown with an 8 s window so that plain iteration can be run to eight updates at reasonable cost. Plain iteration $\mu_{k+1} = \mathcal{H}(\mu_k)$ contracts linearly, the residual falling by a factor of about 0.263 per update, which is $\mathcal{H}'$ of proposition 13.6 at this design. The guarded secant step eq. (141) takes the same first step and is below the tolerance on the third update. Each point costs one full simulation.

Proposition 15.2 (Convergence order)**.** *If* gap $\in C^2$ *near a simple root* $\mu^\star$ *with* $\mathrm{gap}'(\mu^\star) \neq 0$*, the secant iteration eq.* (141) *converges locally with order* $\varphi = (1+\sqrt{5})/2 \approx 1.618$*, against order* 1 *for the fixed-point iteration.*

Proof. Classical; see Ortega and Rheinboldt [20, §8.2]. The error satisfies $|\epsilon_{k+1}| \approx C|\epsilon_k||\epsilon_{k-1}|$, whose characteristic exponent is the golden ratio. □

The step is guarded: it is accepted only when it moves in the same direction as the gap and by no more than six times its magnitude, otherwise the plain fixed-point step is taken. Because $\mathcal{H}$ takes the larger of an energy and a power requirement (eq. (132)) it has a kink where the two exchange, and proposition 15.2 does not apply across it; the guard is what keeps the iteration sane there. When the gap is within tolerance the battery is rounded up to the last requirement and the design is flown once more, and the loop reports convergence only if that flight requires no more battery than it carries. On every mission of section 17 this took three updates and two verification flights, five simulations in all, against eight or more for plain iteration. Figure 54 compares the two on one mission.

15.3 Trajectory integration

Equation (104) is integrated with classical fourth-order Runge-Kutta at fixed step. Fixed step rather than adaptive, because the cost must be predictable for an interactive tool and because the controller is evaluated once per step, making the scheme effectively a sampled-data system whose sample rate should not wander.

Local truncation error is $O(h^5)$ and global error $O(h^4)$. The step is chosen from the fastest closed-loop mode. With $\tau_m = 45\,\mathrm{ms}$, $K_R \approx 41/\mathrm{s}^2$ and hence $\omega_n = \sqrt{K_R} \approx 6.4\,\mathrm{rad/s}$, the fastest pole is the motor at $1/\tau_m \approx 22\,\mathrm{rad/s}$; at $h = 4\,\mathrm{ms}$ the product $h\lambda \approx 0.09$ sits deep inside the RK4 stability region, whose real-axis extent is $|h\lambda| \leq 2.79$ (fig. 55).

The quaternion is renormalised after each step, $q \leftarrow q/\|q\|$. This does not cost the order. The exact flow preserves $\|q\| = 1$, so the RK4 step returns $\tilde{q}$ with $\|\tilde{q}\| = 1 + O(h^5)$; the map $\tilde{q} \mapsto \tilde{q}/\|\tilde{q}\|$ is smooth near $\mathbb{S}^3$ with $\tilde{q}/\|\tilde{q}\| = \tilde{q} + O(\|\tilde{q}\| - 1)$, so the projection perturbs each step by $O(h^5)$, the same order as the local truncation error, and the global order $O(h^4)$ is retained.

Two different things are being integrated and the distinction matters. The controller is evaluated once per step and its output held, so RK4 integrates the plant under a piecewise-constant input at fourth order. The closed loop is therefore a sampled-data system at 250 Hz,

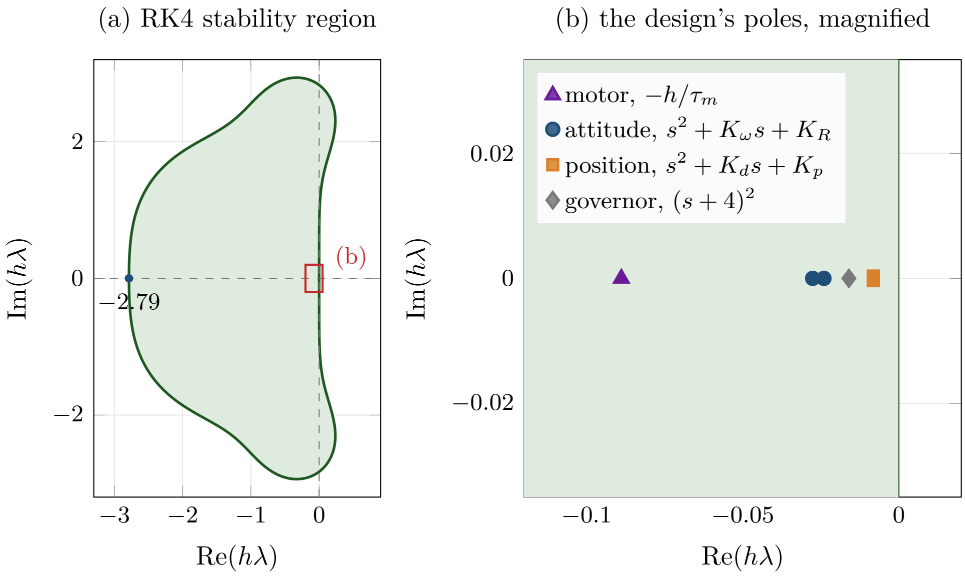


Figure 55: (a) The stability region of classical RK4, $\{z : |1 + z + z^2/2 + z^3/6 + z^4/24| \leq 1\}$, which reaches -2.79 on the negative real axis. (b) The products $h\lambda$ for the poles of the design's loops at $h = 4$ ms: the motor lag, the attitude loop $s^2 + K_\omega s + K_R$, the position loop $s^2 + K_d s + K_p$ and the governor's filter. The fastest, the motor, sits at $h\lambda = -0.0889$, about thirty times inside the boundary, so the step is set by accuracy and by the sampling of the controller, not by stability.

Table 4: Integration convergence on the turnaround at a fixed controller rate of 250 Hz: the plant is integrated with one to eight RK4 substeps per control period under the same held commands. Every reported quantity is unchanged to the digits shown.

Substeps	RK4 step	Mean power	Peak e_{track}	e_{lag} rms	Rotor clearance
1	4.00 ms	111.893 27 W	63.688 mm	1075.579 mm	612.939 mm
2	2.00 ms	111.893 27 W	63.688 mm	1075.579 mm	612.939 mm
4	1.00 ms	111.893 27 W	63.688 mm	1075.579 mm	612.939 mm
8	0.50 ms	111.893 27 W	63.688 mm	1075.579 mm	612.939 mm

which is what a digital autopilot is; the $O(h)$ difference between it and a continuous-feedback idealisation is a property of the system being simulated, not an integration error. Figure 56 shows the structure of one step.

Table 4 separates the two. On the turnaround, the hardest mission, the controller is held at 250 Hz while the plant is integrated with one to eight RK4 substeps per control period under the same held commands. The mean power changes by a relative 3.4×10^{-9}, and the peak tracking error and the minimum rotor clearance by less than 4.3×10^{-5} mm and 3.4×10^{-6} mm: at the design step the integration is converged, and the figures of section 17 are properties of the sampled-data loop, not of the integrator. Halving the controller period instead changes the loop being simulated, not the accuracy of simulating it. It moves the mean power by 0.08 % and the rotor clearance by 0.03 mm, but raises the peak tracking error from 64 mm to 85 mm, still inside the margin ε. The peak tracking error is therefore the one reported quantity that is sensitive to the controller rate, and its distance to ε is the number to recheck for an autopilot running at a different rate.

Remark 15.3 (Windowed mission energy). Integrating a full 1800 s mission at $h = 4$ ms would

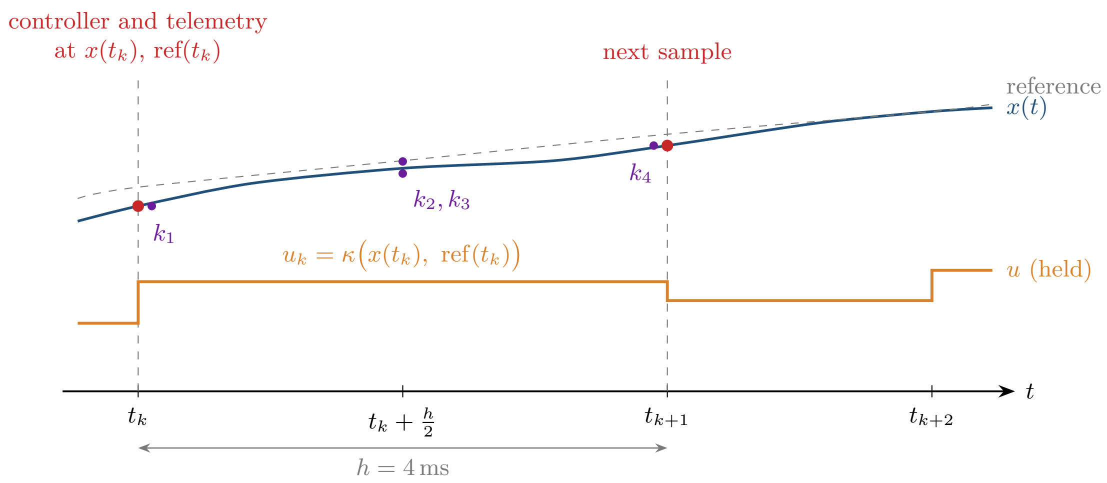


Figure 56: The sampled-data structure of every simulation. At t_k the controller reads the state and the references at t_k, and the telemetry records both at that instant; the controller's output u_k is then held over the step while RK4 evaluates the plant four times, at t_k, twice at $t_k + h/2$ and at t_{k+1}. The integration error is $O(h^4)$ for the plant under the held input. The difference between this sampled loop and a continuous-feedback idealisation belongs to the controller, which is sampled at 250 Hz as a digital autopilot would be.

require 4.5×10^5 steps per evaluation, which is incompatible with the outer loop. The engine instead integrates a representative window of 12 to 20 s, measures the mean electrical power after a settling interval, and extrapolates: $E_{\text{mission}} = \bar{P}\, t_f$. This is exact for a periodic mission whose window spans a whole number of periods and is otherwise an approximation whose error is the difference between the window mean and the mission mean. It is recorded as a limitation.

Remark 15.4 (Initial conditions). The initial state is taken to be the reference position *and the reference velocity*. Starting at rest beside a person already walking is not a gentler initial condition but a worse one: on a curved path the display sits in the person's path, so they close on it while it accelerates. This artefact contaminated every mission statistic until it was found; see section 16.

15.4 Linearisation

$\boldsymbol{A}$ and $\boldsymbol{B}$ are formed by central differences,

$$\boldsymbol{A}_{:,j} \approx \frac{f(x_0 + h\boldsymbol{e}_j) - f(x_0 - h\boldsymbol{e}_j)}{2h}, \qquad h = \varepsilon \max\{1, |x_{0,j}|\}, \tag{142}$$

with $\varepsilon = 10^{-6}$. Central differences have truncation error $O(h^2)$ against $O(h)$ for one-sided, and the total error is minimised near $h \sim \epsilon_{\text{mach}}^{1/3} \approx 6 \times 10^{-6}$, so the choice is near-optimal. Controllability is assessed by the reachable-subspace iteration of section 12 with a singular-value tolerance of 10^{-9} on the norm-scaled matrices, and requires full rank.

15.5 The outer optimisation

Equation (140) is solved by differential evolution [22], a population-based stochastic method, with constraints handled by an exterior quadratic penalty

$$\tilde{\mathcal{J}}(p) = \mathcal{J}(p) + \varpi \sum_j \left(\frac{\max\{0, g_j(p)\}}{s_j} \right)^2, \tag{143}$$

with per-constraint scales s_j chosen so that a violation of engineering significance contributes order unity.

Three properties of the problem dictate this choice. The objective is non-convex and its landscape contains large regions where the closure has no solution at all, so gradient methods have nowhere to start. Several design variables are integer (rotor count, blade count), so smoothness fails. And an evaluation of the algebraic objective costs milliseconds, so a population of a few hundred is affordable.

The penalty is exterior rather than a barrier precisely because the infeasible region must remain traversable: a barrier method cannot cross from one feasible island to another through infeasible territory, and here the islands are separated by the fold.

15.6 Numerical hygiene

Three details are worth recording because each caused a visible defect.

Non-finite values in transport. `NaN` and $\pm\infty$ are not representable in JSON. Infeasible cells in a sweep, and quantities such as $M_0^{\max}$ when no fold exists, are genuinely undefined rather than zero. The API maps every non-finite float to `null` and the front end renders it as a dash, so an undefined quantity is never silently displayed as 0.

Solidity and Reynolds guards. Without assumption 4.7 and eq. (24) the optimiser drives solidity to 0.55 and tip speed toward zero, exploiting the fact that proposition 4.3 has no notion of blade interference and proposition 4.5 none of Reynolds number. Both constraints exist to keep the optimiser inside the domain where the model is a model of something.

Boundary comparisons. Constraints active exactly at their bound were being reported as violations because of floating-point comparison. All constraint checks carry a relative tolerance of 2×10^{-3}, chosen so that a design sitting on a bound reads as satisfying it.

16 Verification, calibration and errors found

A model of this kind is only as good as the evidence that it computes what it claims. This section records the verification strategy, the calibration against measured hardware, and, because they are instructive, the errors that the strategy caught.

16.1 Verification against the closed forms

The closed forms of sections 6 and 7 exist in the implementation alongside the numerical solvers and serve as an oracle. The test suite comprises 66 assertions; those bearing on the theory are:

- *Momentum theory.* $T = 2\rho A v_i^2$ holds identically to machine precision; with $C_{d_0} \to 0$, $\kappa \to 1$ and the near-field corrections disabled, the computed FM $\to 1$ to within 10^{-5}, confirming theorem 3.2 and eq. (20).
- *Coefficients.* α and β from definition 5.6 agree with hand computation to four figures.
- *Fold.* m^* and $M_0^{\max}$ agree with eq. (47) to 10^{-12} relative; both roots satisfy eq. (36) to 10^{-8}; $m_- < m^* < m_+$ as theorem 6.2 requires; at $M_0 = M_0^{\max} - 10^{-9}$ the two roots agree to 10^{-3} relative, exhibiting the square-root merging of eq. (50); and for $M_0 > M_0^{\max}$ no solution is returned.
- *Iteration.* The naive iteration converges to the analytic m_- to 10^{-4} relative from a distant start, and is classified as diverging when $M_0 > M_0^{\max}$, confirming theorem 6.8.
- *Scaling.* $M_0^{\max}(2t_f) = M_0^{\max}(t_f)/4$ to 10^{-9}, and the numerically computed logarithmic sensitivities reproduce theorem 7.3 to three decimal places, including the non-obvious $S_g = -3.770$ which depends on γ_m/α.
- *Exponent.* The endurance wall is finite for $p = 0$ and absent for $p = 1.4$, confirming theorem 6.11.

Table 5: Calibration of the acoustic anchor. Hover rotor speeds are manufacturer figures; tip speed and thrust per rotor follow from geometry and mass. The four aircraft span a factor of 3.6 in mass and agree on C to 1.5 dB.

Aircraft	m	T/rotor	V_{tip}	L_{WA}	$L_p(1\,\text{m})$	C
DJI Mini 3	0.249 kg	0.61 N	71.6 m/s	79.0	71.0	82.8
DJI Air 2S	0.595 kg	1.46 N	67.1 m/s	82.0	74.0	83.8
DJI Mavic 3	0.895 kg	2.19 N	68.8 m/s	83.0	75.0	82.3
DJI Mavic 2 Pro	0.907 kg	2.22 N	67.9 m/s	83.0	75.0	82.6
					mean	**82.9**

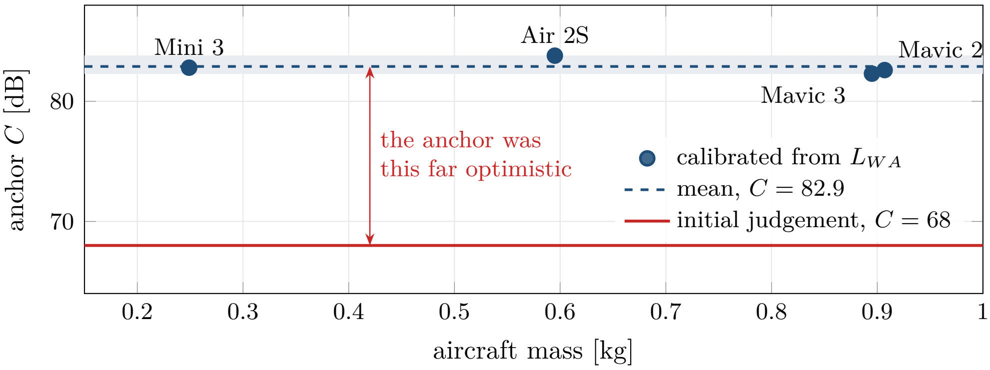


Figure 57: The acoustic anchor C recovered from the declared sound power of four aircraft through eqs. (73) and (75), against aircraft mass. The four values lie in the narrow band shaded across a factor of 3.6 in mass, about the mean $C = 82.9$. The value first assigned by judgement, $C = 68$, lies below every one of them by the length of the arrow, an error larger than the effect of any design choice in section 17.

- *Dynamics.* With zero thrust the vertical acceleration is exactly $-g$; at the hover trim the allocation returns $T = mg$ and $\boldsymbol{\tau} = \mathbf{0}$ to 10^{-9}; $\dot{E}$ equals the negative of the computed electrical power identically; rotations satisfy $RR^{\mathsf{T}} = I$ and $\det R = 1$ to 10^{-10} through the quaternion round trip; and the hover linearisation is controllable at full rank.
- *Review regressions.* The near-fold root pair of section 15 is found; the quadrotor authority equals the balanced differential eq. (117) on every axis; the slab inertia is assigned to the panel axes; a forward boom produces a product of inertia and a trimmed hover with unequal rotor speeds; telemetry at t compares the state at t with the references at t; the governor keeps a reference out of a head that walks toward it; the battery loop's reported battery survives its verification flight; and the inner-map slope of proposition 13.3 lies in $[0, 0.9)$ over 0.8 to 50 kg for the shipped component models.

16.2 Calibration of the acoustic anchor, and a fourteen-decibel error

Model 9.2 contains an anchor C which was initially assigned by judgement, at $C = 68$. Calibration against published declarations exposed this as badly wrong.

Manufacturers declare a sound *power* level L_{WA} under EU regulation 2019/945. Converting to pressure at 1 m by eq. (75) and inverting eq. (73) gives C for each aircraft.

Table 5 gives $C = 82.9$ with a spread of 1.5 dB. The consistency across a 3.6× mass range is strong evidence that the functional form eq. (73) is right, and that the original anchor was 14 dB optimistic. Figure 57 shows the four values against the one first assigned.

Table 6: Motor mass model against catalogue peak torque. The model is conservative at the large end, which is the safe direction for a design study; the designs here use motors of 50 to 60 g where the error is small.

Motor	Actual mass	Peak torque	Model mass	Error
T-Motor MN2806	66 g	0.30 N m	63 g	−5%
T-Motor MN3110	96 g	0.63 N m	111 g	+16%
T-Motor MN505-S	190 g	2.03 N m	273 g	+44%

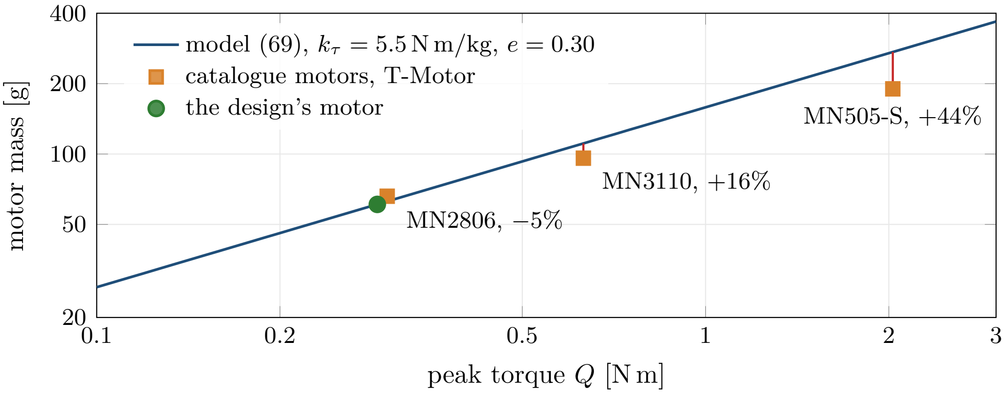


Figure 58: The calibrated motor-mass model eq. (69) against catalogue motors, on logarithmic axes on which the model is a straight line of slope $1/(1+e) = 0.77$. The red connectors are the model errors of table 6: small at the size of motor this design uses (the green point, the design's 61 g motor at its torque for maximum thrust) and conservative at the large end.

Fourteen decibels is not a refinement. It is the difference between a machine one can think beside and one that must leave the room, and it exceeds the entire range spanned by every design decision available in the study. The error had two components: an uncalibrated anchor, and the eight-decibel confusion of proposition 9.5 between a declared sound power and a pressure at one metre. It was found only because the model was made to reproduce measured hardware.

The test suite now asserts that eq. (73) reproduces the declared levels of three of these aircraft to within 1.5 dB.

16.3 Calibration of the motor model

k_τ in eq. (69) was likewise assigned by judgement at 3.0 N m/kg and found to be pessimistic by roughly a factor of two.

With $k_\tau = 5.5$ N m/kg at $m_{\mathrm{ref}} = 0.1$ kg and $e = 0.30$, table 6 results. The recalibration mattered structurally, not just numerically: with the pessimistic constant the motor mass penalty overwhelmed the induced-power benefit of a larger rotor, and the optimum diameter sat on the lower bound of the search box. After recalibration a genuine interior optimum appears, which is the physically expected behaviour, and the test suite now asserts its existence. Figure 58 shows the calibrated law against the catalogue.

16.4 Errors found by the model itself

Three defects were found not by inspection but because a computed quantity was inconsistent with a hand calculation.

The tip-speed design rule. Holding tip speed fixed while growing the rotor made power *increase* with diameter, contradicting observation 3.6. The cause was that at fixed tip speed the profile power eq. (19) scales with disk area while the induced power falls only as $A^{-1/2}$. The resolution is corollary 4.6: a rotor is designed at constant blade loading, not constant tip speed, and with V_{tip} derived from eq. (21) the expected monotonicity is recovered.

Safety measured to the wrong point. The simulator's minimum-distance check measured from the vehicle centre of mass to the user. By eq. (61) the blades extend $\mathcal{R}$ from that point, and for these designs $\mathcal{R} = 0.63\,\mathrm{m}$, so the check was optimistic by the entire reach of the machine. Recomputing to the nearest blade tip showed the nominal configuration violating its own safety distance by a factor of three. This is discussed in section 17.

A startup transient masquerading as a failure. The simulation began with the vehicle at rest at the reference position while the human was already at full walking speed. On a curved path the display sits directly in the person's path, so the separation collapsed from 1.100 m to 0.603 m within 0.62 s before recovering. Initialising the vehicle and the governor with the reference velocity, as noted in section 15, removed it; the minimum envelope gap then agrees with the static tip gap of eq. (144) to within a centimetre, the difference being the conservative sphere against the actual blade tips. The same mistake reappeared one level up: the vehicle was initialised level while the reference on a curved path already needed a tilt and a body rate, and the attitude loop spent the first second catching up. That transient set the whole-window peak tracking error, which is the quantity the keep-out margin is checked against. The initial attitude, body rate and thrust are now the trim for the reference at $t = 0$.

Remark 16.1. The common feature of all of these is that they were caught by comparison against an independently computed quantity, not by reading the code. This is the argument for keeping the analytic layer alive in the implementation even after it has been superseded.

16.5 Errors found by independent review

An independent review of the engine and of an earlier draft of this paper found five defects that the tests above had not, each now covered by a regression test.

1. The component closure bracketed the first sign change of $\mathcal{F}$ on a geometric grid of ratio 1.35 and called a grid with no positive sample infeasible. Near the fold the positive stretch is narrower than the grid, and a design with two roots was reported as having none. The paper had also described that failure as infeasibility. Section 15 gives the replacement.
2. The torque authority divided the balanced differential by N, a factor of four on every axis, which propagated into the gain rule and into every layout comparison. Equation (116) is now a linear program and appendix B the closed form it reduces to.
3. Telemetry compared the state advanced to $t + \mathrm{d}t$ with references evaluated at t, labelled t.
4. The display slab's inertia was assigned to the wrong body axes, and the offsets were not moved to the centre of mass, so a boom produced no product of inertia. Section 8 now builds one tensor from one geometry.
5. After the review, and found by the tracking statistics it demanded: the attitude loop used a zero desired angular rate, so on a curved path it lagged the rotating thrust direction by a standing error. The rate feedforward of eq. (115) removed a 0.27 m tracking excursion and a rotor pass at 0.34 m from the eye.

17 Design study

Every number in this section is produced by `paper/make_results.py` from the engine, together with the configuration that produced it (section 14). Unless stated otherwise the flights use a window of 16 s at a step of 4 ms, statistics taken after a settling interval of 3 s, gust seed 12345, and the governor of model 12.10 at $v_{\max} = 2.5\,\mathrm{m/s}$, $a_{\max} = 2.5\,\mathrm{m/s^2}$, $\varepsilon = 100\,\mathrm{mm}$, heading rate cap 1.5 rad/s.

17.1 The requirement, worked backwards

Four human-facing requirements bound the design.

(R1) *Readability.* Two quantities, and they pull in opposite directions. Pixel density in pixels per degree, $1920/(2\arctan(w/2d))$ for a panel of width w at distance d, measures sharpness and rises with distance. Text size is an angular height: ANSI/HFES 100 [9] asks for a character height of at least $16'$, with 20 to $22'$ preferred, and it falls with distance. A desktop meets the second by scaling its interface, which divides the usable workspace by the scale factor.

(R2) *Noise.* A quiet office is 45 dBA, ordinary speech 60 dBA. Working beside a source for half an hour requires something in between.

(R3) *Downwash.* Loose paper begins to move at roughly 5 m/s. By eq. (15) the wake velocity is $2v_i = 2\sqrt{\mathrm{DL}/2\rho}$, so this bounds disk loading.

(R4) *Size and safety.* The machine must fit beside a desk, and by eq. (61) its blades extend $\mathcal{R}$ from its centre.

All four pull the same way, toward large slowly turning lightly loaded rotors, and (R2) to (R4) conflict with the span limit through $\mathcal{R} = L + R$.

17.2 The binding constraint is acoustic

Applying eq. (53) to the fixed load of the design below (546 g of panel, electronics and gimbal), the energetic wall is comfortable at short sessions: at 20 min the critical effective load is 6.37 kg against a requirement of 0.57 kg, a utilisation of 9 %. Half an hour of flight costs about a quarter of a kilogram of cells. Energy is not what stops this product.

Noise is. Observation 9.3 says the only lever with authority is tip speed, and corollary 4.6 says solidity is how one buys it. Optimising under (R1) to (R4) drives the design to four rotors of 492 mm diameter with wide, low-aspect-ratio blades turning at 1036 rpm, a tip speed of 26.7 m/s against roughly 68 m/s for a commercial camera drone.

17.3 Which display

A resolution is not a payload. Table 7 evaluates the candidates at the standoff the safety analysis below will impose, 1.20 m, each with the aircraft re-optimised around it by differential evolution over rotor diameter, blade loading, aspect ratio and blade count under a span limit of 1.45 m, a free-field level of 58 dBA at 1 m, a wake of 7.5 m/s and a gross mass of 6.0 kg. Every 1920-wide panel is sharp at that distance; none delivers a full desktop there, for the reason in (R1). Areal density is what discriminates between the aircraft: a desktop monitor panel is more than twice as heavy per unit area as a laptop panel, because it carries thicker glass, a heavier backlight and a metal chassis.

Figure 59 shows the same comparison graphically.

Table 7: Display options at 1.20 m, each with the aircraft re-optimised under identical limits. "Logical" is the desktop width in logical pixels once the interface is scaled to a 16′ character height. Areal density in kg/m^2, n/a for a whole device. Levels in dBA, free field at 1 m and at the user's ear in a $5 \times 4 \times 2.4$ m room with $\bar{\alpha} = 0.20$ by proposition 9.8. "Over limits" means the closure exists but the best design found violates at least one of the four limits.

Display	px/deg	Logical	Areal	Gross	Power	L_p(1 m)	At ear	Verdict
13.3 in laptop panel	137	590	4.1	1.502 kg	85.8 W	50.8	53.6	closes
15.6 in laptop panel	117	692	4.2	1.723 kg	109.4 W	52.8	55.6	closes
iPad Pro 13 in, whole	206	565	9.6	2.184 kg	134.1 W	56.2	59.0	closes
iPad Pro 13 in display only	206	565	5.0	1.640 kg	98.5 W	52.9	55.6	closes
21.5 in desktop panel	86	955	9.8	4.409 kg	327.1 W	62.4	65.2	over limits
24 in desktop panel	77	1065	12.3	6.229 kg	517.1 W	66.9	69.7	over limits
15.6 in laptop lid assembly	117	692	7.0	2.074 kg	137.9 W	55.7	58.5	closes
13.3 in ultrabook, whole	188	574	n/a	4.142 kg	280.7 W	61.7	64.4	over limits
15.6 in laptop, whole	117	692	n/a	5.478 kg	408.4 W	65.0	67.8	over limits

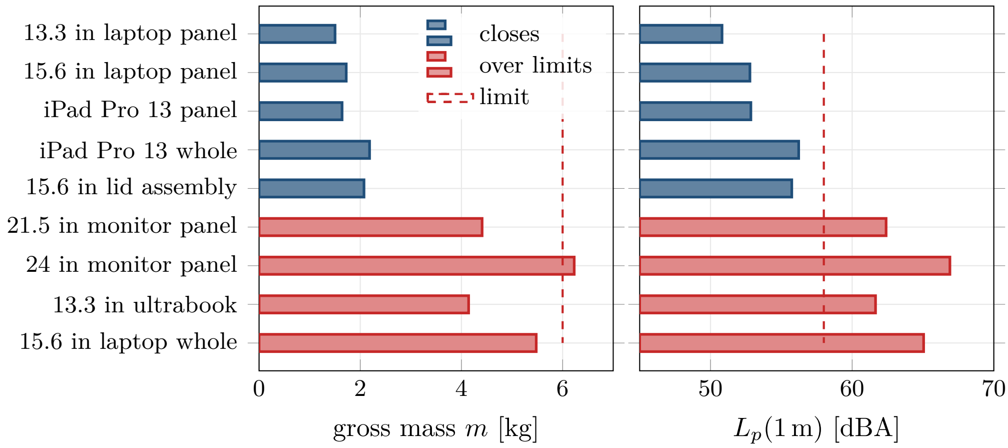


Figure 59: The display candidates of table 7, each with its aircraft re-optimised. Left: gross mass, against the 6.0 kg limit. Right: free-field level at 1 m, against the 58 dBA limit. Every candidate that fails exceeds the noise limit, and only one of them exceeds the mass limit: a heavier display needs a more heavily loaded rotor, and by observation 9.3 that is paid for in decibels long before it is paid for in kilograms.

The full 1920×1080 *panel is available, provided it is a laptop panel and not a monitor panel. Flying a whole laptop is not: the panel is a sixth of a 15.6-inch machine's mass and the rest is keyboard, battery, cooling and chassis, none of which is visible while hovering. A tablet is different in kind because a tablet is almost entirely screen, which is why the iPad flown whole still closes while the laptop flown whole does not. What no panel of this size delivers at* 1.20 m *is a full desktop's worth of text: see section 17.5.*

17.4 Geometry relative to the user

The standoff a user specifies is measured to the *display*. The rotors extend $\mathcal{R} = L + R$ back toward them from the machine's centre.

Proposition 17.1 (Blade clearance). *Let the display be at horizontal standoff d from the user's eye, carried a distance $b \geq 0$ ahead of the rotor centroid on a boom, with the rotor plane*

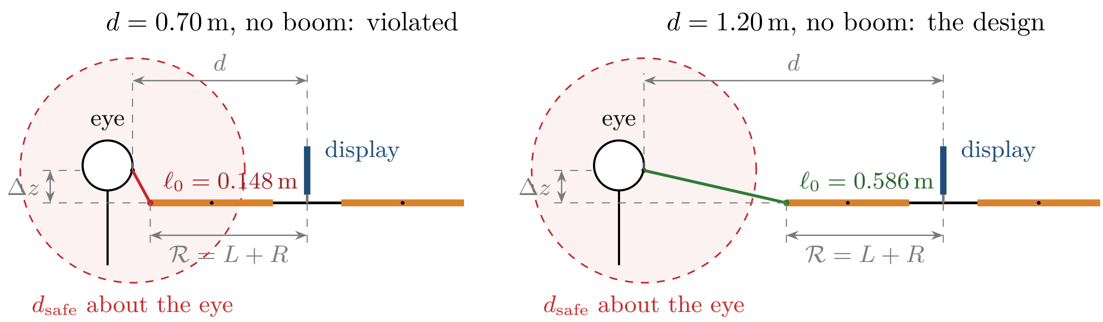


Figure 60: Proposition 17.1 in side view, to scale. The display is held at eye height at standoff d; with the screen above the rotors the rotor plane lies $|\Delta z| = 0.130\,\mathrm{m}$ below the eye; the nearest blade tip is the reach $\mathcal{R} = L + R$ back toward the user from the rotor centroid. At $d = 0.70\,\mathrm{m}$ the tip is inside the sphere of radius d_{safe} about the eye, although the display is well outside it; a check made at the vehicle centre would pass this design. At $d = 1.20\,\mathrm{m}$ the tip is outside.

Table 8: Two ways to recover blade clearance. The boom keeps the display close and pays in pitch inertia and authority; standing back is free on every axis and quieter, and pays in text size. "Turn" is the minimum distance from the eye to the nearest rotor disk during the turnaround mission, against the 0.45 m required. J_{yy} in $\mathrm{kg\,m^2}$, $\alpha_{\mathrm{pitch}}^{\max}$ in $\mathrm{rad/s^2}$, K_R in $1/\mathrm{s}^2$, screen angle is the panel's apparent width, logical width as in table 7.

Option	Tip gap	Gross	Power	J_{yy}	$\alpha_{\mathrm{pitch}}^{\max}$	K_R	Screen	Logical	Turn
$d = 0.70\,\mathrm{m}$, no boom	0.148 m	1.724 kg	106.0 W	0.0612	60	41	27.7°	1186	0.56 m
$d = 0.95\,\mathrm{m}$	0.347 m	1.724 kg	106.0 W	0.0612	60	41	20.6°	874	0.55 m
$d = 1.20\,\mathrm{m}$	0.586 m	1.724 kg	106.0 W	0.0612	60	41	16.4°	692	0.61 m
boom 0.20 m	0.301 m	1.742 kg	107.5 W	0.0731	41	41	27.7°	1186	0.58 m
boom 0.38 m	0.470 m	1.812 kg	113.0 W	0.1053	23	26	27.7°	1186	0.55 m
boom 0.50 m	0.586 m	1.927 kg	122.2 W	0.1414	14	16	27.7°	1186	0.45 m

a height Δz above the eye. Then the nearest blade tip lies at

$$\ell_0 = \sqrt{\big(\max\{d + b - \mathcal{R}, 0\}\big)^2 + \Delta z^2}, \tag{144}$$

and the requirement $\ell_0 \geq d_{\mathrm{safe}}$ is met either by

$$d \geq \mathcal{R} + \sqrt{d_{\mathrm{safe}}^2 - \Delta z^2} - b \qquad \text{or} \qquad b \geq \mathcal{R} + \sqrt{d_{\mathrm{safe}}^2 - \Delta z^2} - d. \tag{145}$$

At a nominal $d = 0.70\,\mathrm{m}$ with $\mathcal{R} = 0.63\,\mathrm{m}$ and $|\Delta z| = 0.13\,\mathrm{m}$, eq. (144) gives $\ell_0 = 0.148\,\mathrm{m}$ against a safe distance of 0.45 m: a violation by a factor of three, invisible to any check evaluated at the vehicle centre.

Figure 60 draws the two standoffs to scale.

Table 8 compares the two remedies, each option closed, its gains capped by model 12.7 for its own authority (attitude and position gains together, since the cascade needs its time-scale separation), and flown through the turnaround mission with the governor active. For every option but the last standoff the ideal display position lies inside the keep-out radius $\varrho = d_{\mathrm{safe}} + \mathcal{R} + \varepsilon$, by up to 47 cm; those missions start on the sphere and the governor holds the reference there, so their "turn" clearance measures the keep-out sphere and not the option. The sphere is conservative for a boom, whose reach is set by the rear rotors while the front rotors are what approach the user.

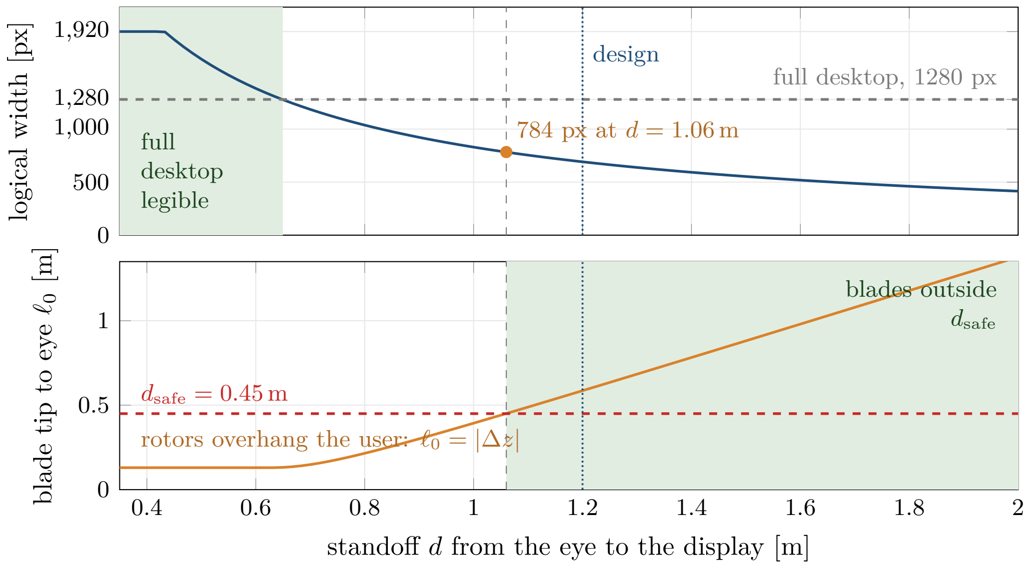


Figure 61: Readability against blade clearance, over the standoff d from the eye to the display. Top: the logical desktop width once the interface is scaled to a $16'$ character height; it is the full 1920 pixels only within 0.43 m, where unscaled text already meets the minimum, and it falls as $1/d$ beyond. A 1280-pixel desktop needs $d \leq 0.65$ m (green). Bottom: the nearest blade tip ℓ_0 of eq. (144); it stays at $|\Delta z|$ while the rotors overhang the user and clears d_{safe} only for $d \geq 1.06$ m (green). The two green regions do not overlap, and at the first safe standoff the desktop is already down to 784 pixels.

A boom of 0.38 m *multiplies the pitch inertia by 1.72, reduces the pitch authority to 0.38 of the no-boom value and the admissible attitude gain to* $K_R = 26.0$*, and costs* 89 g *of gross mass and* 59.8 dBA *at the ear against* 56.6 dBA*. Standing back to* 1.20 m *achieves the same static clearance at no cost in mass, power or authority, is the only option whose ideal position is outside the keep-out sphere, and keeps its rotors* 0.61 m *from the eye through the turnaround. The price is the one (R1) names, and it is paid below.*

17.5 What the user can read

At 1.20 m the 15.6-inch panel subtends 16.4° at 117 px/°: sharper than any desktop monitor. But a 16 px body-text character, cap height 11 px, subtends only $5.8'$ at that distance, against the $16'$ minimum of [9]. Meeting the minimum takes an interface scale of 278 % (347 % for the preferred $20'$), which leaves a logical desktop of 692×389 pixels. That is a tablet's workspace on a laptop's panel. At 0.70 m the same panel gives $9.9'$ and a logical width of 1186, a laptop's workspace, and its blades 0.15 m from the eye.

The two safe ways to fly a panel this size, standing back or a boom, and the requirement that the user read a full desktop on it, are not simultaneously satisfiable at 1920×1080 *in* 345 mm*. A safe flying screen at reading distance either carries a physically larger panel, which table 7 shows the acoustics will not allow, or presents a smaller workspace than the resolution suggests. The resolution is not the constraint; the angular size of a character is.*

Figure 61 shows the whole trade over standoff.

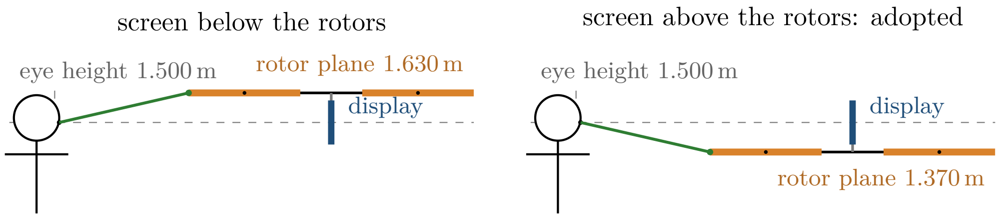


Figure 62: The two placements of the display at the design standoff, to scale, with the display held at eye height in both. The nearest-tip clearance (green) is the same either way, because eq. (144) depends on Δz only through Δz^2. What differs is where the rotor plane sits relative to the head: with the screen below it is at the top of the head, with the screen above it drops below the chin, and the panel is then the object between the blades and the face. The heights marked are rotor plane heights above the floor.

17.6 Screen above or below the rotors

Because the display is vertical and the rotor flow is vertical (section 11), placing the panel above or below the rotor plane is nearly free aerodynamically: 1.720 kg against 1.724 kg, with the same noise. What changes is geometry. With the screen below, the rotor plane sits at 1.63 m, level with the user's head; with it above, the rotors drop to 1.37 m and the panel itself lies between the blades and the user's face. The latter is adopted. Tracking is marginally better in still air, because by the remark following eq. (100) the drag moment becomes weathercock-stable, and marginally worse in a steady breeze for the same reason.

Figure 62 compares the two at the design standoff.

17.7 The design

Table 9 gives the resulting machine with a buildable payload: a bare laptop panel on a matched controller board, a Raspberry Pi running a remote desktop client, a separate flight controller, and an on-sensor inference camera. Figure 63 draws it to scale.

17.8 Flight performance

Table 10 reports the design of table 9, with its hover-sized battery, flown through each mission with the governor of model 12.10 active. Two clearances are reported. The envelope gap is the distance from the eye to a sphere of radius $\mathcal{R}$ about the centre of mass, the conservative quantity that proposition 12.11(iv) bounds. The rotor clearance is the distance from the eye to the nearest oriented rotor disk, which is what a blade could actually touch.

The six motions are not defined elsewhere, so they are stated here. Each is a smooth trajectory of the person's position and heading over the 16 s window, and the ideal vehicle position follows from it by holding the display at the standoff in front of the eyes. Standing is postural sway of a few centimetres; pacing is walking back and forth along one line while facing the display; sit to stand lowers the eyes by 0.45 m; walking a loop and jogging follow a circle of radius 1.44 m, jogging adding a vertical bob at the stride frequency; and turning around reverses the heading at each end of a walk. Figure 64 draws them, together with the acceleration each one asks of the display.

The overhead $\varkappa$ ranges from 1.000 standing to 1.055 on the turnaround. The peak tracking error stays below ε on every mission (worst 64 mm), so by proposition 12.11(iv) the envelope guarantee holds on these runs; the nearest any rotor disk comes to the eye is 0.61 m, on the turnaround. Disturbance rejection is not what limits the machine indoors: a 3 m/s breeze with 1.5 m/s gusts, far stronger than any room, produces 198 mm of lag and 0.65° of screen tilt

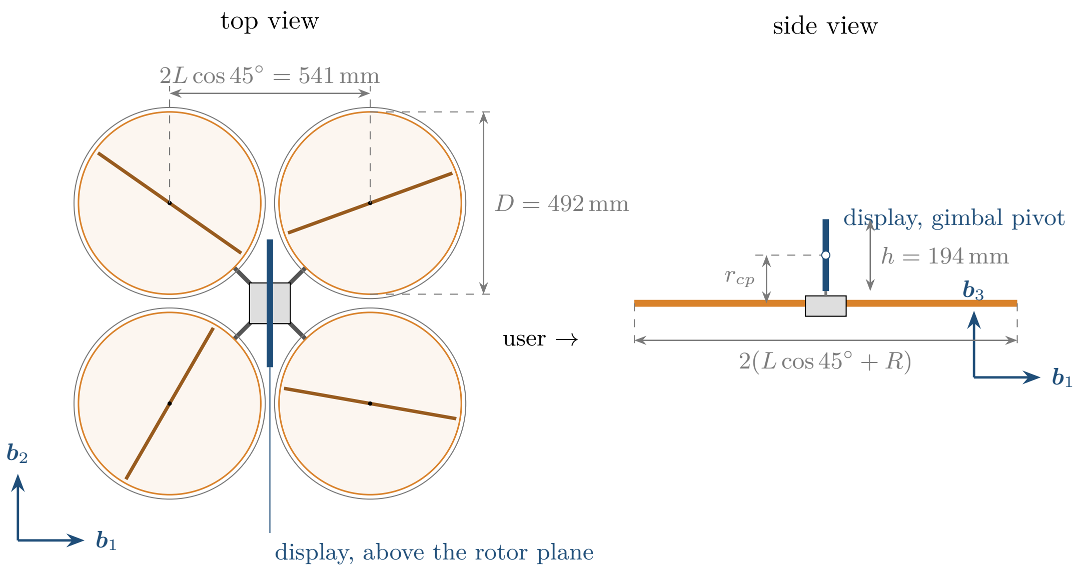


Figure 63: DESK-1/W to scale. Top view: four rotors of diameter D in the × layout, the arm length L set by the tip gap rule of section 8 so that adjacent guards (outer circles) nearly touch, the display edge on above the hub, and $\boldsymbol{b}_1$ pointing at the user. Side view: the display on its gimbal a height r_{cp} above the rotor plane, which is what puts the panel, and not the blades, between the rotors and the user's face. The largest dimension, tip to tip across the diagonal, is the span $2(L + R) = 1.26$ m.

while the airframe leans 14°, which is the gimbal doing what it exists for. Figure 65 shows the hardest mission, the turnaround, as time histories.

What the table also shows is that a battery sized for hover is not sized for the mission: on every moving mission the pack lands below its reserve, at 7 % on the turnaround. That is the reason the coupled loop exists.

Table 9: DESK-1/W with the thin-client payload, screen above the rotors, standoff 1.20 m. Sized for hover by eq. (71).

	Quantity	Value	Note
Geometry	rotors	4×492 mm	2 blades, 82 mm chord, AR 3.0
	solidity	0.212	below the 0.25 cascade limit
	span	1.26 m	reach 0.63 m
	display	345×194 mm	15.6 in, 117 px/°, 692×389 logical
Mass	fixed payload	546 g	panel 280 g, electronics 181 g, gimbal 85 g
	frame	400 g	arms 81 g, guards 155 g
	propulsion	489 g	motors 244 g, props 221 g
	battery	288 g	5S1P 18650, 63.0 W h
	gross	1.724 kg	FM = 0.725 implied
Rotor	speed	1036 rpm	1390 rpm at $T_{\max}$
	tip speed	26.7 m/s	35.8 m/s at $T_{\max}$
	blade loading	0.120	stall at 0.14
	Reynolds	102 000	$C_{d_0} = 0.023$ corrected
Power	hover	106.0 W	of which 16.0 W auxiliary
	session, hovering	30.0 min	to the 12 % reserve; 34.1 min to empty
	session, worst mission	28.4 min	turnaround, to the reserve
Human factors	L_p free field 1 m	53.8 dBA	
	L_p at the ear, in room	56.6 dBA	proposition 9.8, $r_c = 0.91$ m
	L_p elsewhere in room	54.8 dBA	does not fall with distance
	wake velocity	6.0 m/s	Beaufort 3
Control	J_{xx}, J_{yy}, J_{zz}	0.0649, 0.0612, 0.1141	kg m^2, about the centre of mass
	authority roll/pitch/yaw	56/60/4.5	rad/s^2, eq. (116)
	gains	$K_R = 41.4$, $K_\omega = 12.9$	below the bound 44.4 of model 12.7
	blade tip to face, static	0.586 m	against 0.45 m required

Table 10: Missions flown on the hover-sized battery. $\varkappa$ is the mission overhead of proposition 13.6; "session" is how long the 288 g pack lasts at that power to the 12 % reserve. "Track" is the peak tracking error over the whole window, start-up included, to be compared with $\varepsilon = 100$ mm; lag is the r.m.s. distance from the ideal display position, eq. (124).

Mission	Power	$\varkappa$	Session	Track	Lag rms	Screen tilt	Sat.	Envelope	Rotor
Standing	106.0 W	1.000	30.0 min	0 mm	0 mm	0.00°	0.0 %	0.57 m	0.73 m
Pacing	106.4 W	1.003	29.9 min	16 mm	10 mm	0.03°	0.0 %	0.56 m	0.72 m
Walking a loop	107.2 W	1.011	29.7 min	18 mm	16 mm	0.52°	0.0 %	0.56 m	0.73 m
Jogging	110.6 W	1.043	28.8 min	46 mm	66 mm	1.89°	0.4 %	0.57 m	0.73 m
Turning around	111.9 W	1.055	28.4 min	64 mm	1076 mm	2.68°	0.2 %	0.52 m	0.61 m
Sit to stand	106.0 W	1.000	30.0 min	2 mm	0 mm	0.00°	0.0 %	0.57 m	0.73 m

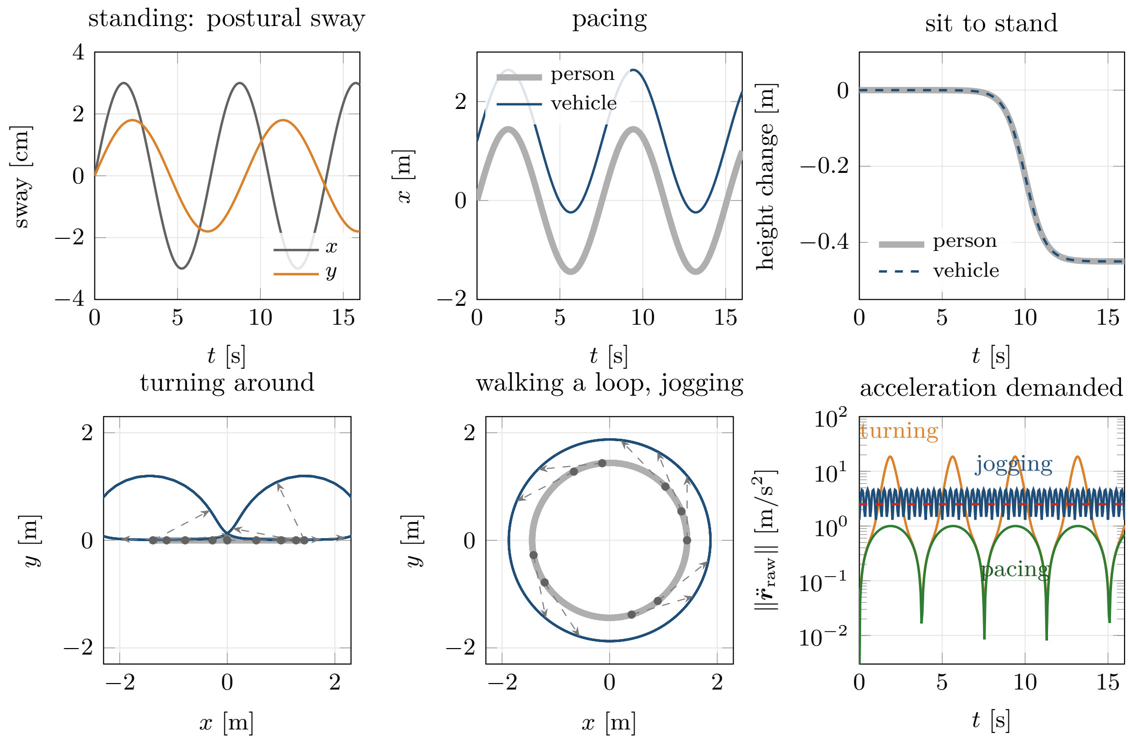


Figure 64: The six motions flown in table 10, over one window. Grey is the person, blue the ideal vehicle position $\boldsymbol{r}_{\mathrm{raw}}$ that holds the display at the standoff in front of their eyes, and dashed arrows join the two at intervals. Pacing is mirrored by the vehicle at a fixed offset; sit to stand is followed in height; on the loop the vehicle runs on an outer circle. Turning around is different in kind: at each reversal the display must swing through a half circle of radius d about the person (proposition 12.9). The last panel shows the price. The raw reference asks for an acceleration an order of magnitude above the governor's cap $a_{\max}$ (dashed) at every reversal, jogging's vertical bob alone exceeds it at the stride frequency, and ordinary walking stays below it.

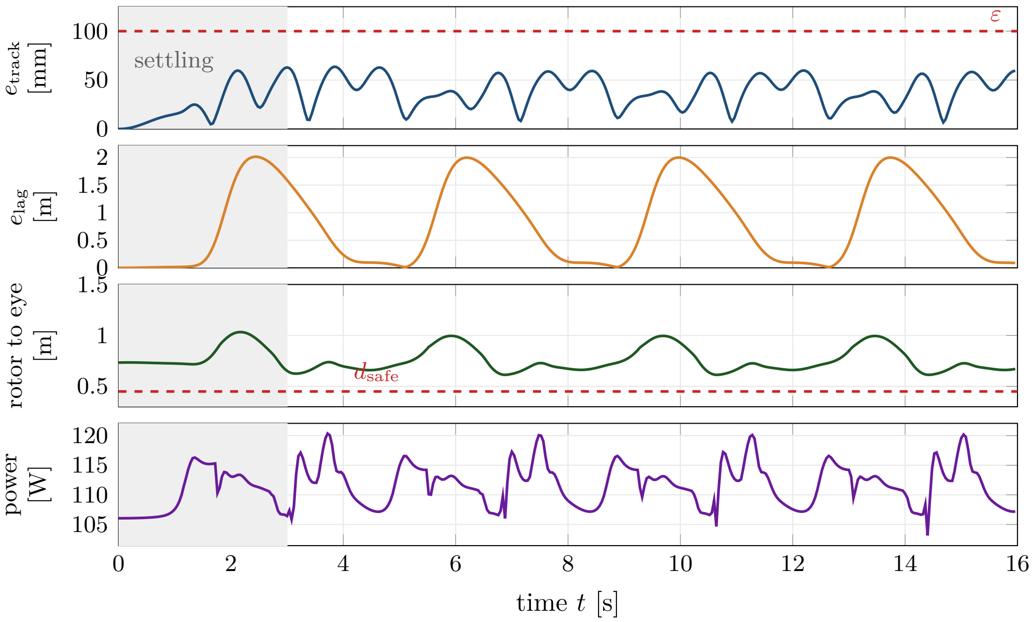


Figure 65: The turnaround flown with the governor. (a) The tracking error against the governed reference stays below the margin ε, which is the hypothesis of proposition 12.11(iv) checked on this run. (b) The lag behind the ideal display position grows to about 2 m at each reversal: the governor lets the display fall behind rather than swing it round the head at the acceleration of fig. 64. (c) The distance from the eye to the nearest rotor disk never approaches d_{safe}. (d) The electrical power, whose window mean after the settling interval (grey) is the $\bar{P}$ of the energy integral; the brief spikes are the rotor speeds responding to the jerk of the reference at the start and end of each swing.

17.9 The fully coupled loop, following the user

The design of table 9 was sized for hover and then flown. That is not problem 13.1. The coupled problem sizes the battery around the motion actually performed: guess m_{bat}, close frame and propulsion around it by proposition 13.3, rebuild the vehicle including its inertia tensor and rotor constants, fly the mission with the governor and keep-out, integrate the electrical power, and ask what battery that mission required, by eq. (130). Table 11 gives the result for each motion. Every row converged in 3 battery updates by the secant iteration eq. (141) followed by two verification flights, 5 full 6-DOF simulations in all, and every row's final flight met the reserve, the peak-power requirement, the rotor clearance and the readability limit.

Figure 66 shows the table's battery column against what the hover sizing would have done. Three observations follow.

The standing row reproduces the hover sizing to within 0.1 g of battery. This is a consistency check, not a validation: the simulation and the closure share the rotor model of section 4, and agreement between them says that the substitution of model 5.5 at $\varkappa = 1$ is implemented correctly, nothing more.

Following a person is cheap; turning with them is not. Ordinary motion costs between 1 g and 17 g of battery. Repeated turnarounds cost 22 g of battery and 28 g of gross mass, almost all of it the mission overhead $\varkappa = 1.055$ propagated through the fixed point: by eq. (138) the overhead multiplies β, and here it moves the design 1.6 % further up the closure curve. Sizing the battery for the most demanding motion is a worst-case operating policy, and it is the one adopted here. For a defined session with known phases the energy is the time-weighted sum over the phases, with the peak-power requirement enforced separately by eq. (130); a session of occasional turns needs less than table 11's last row, and one of continuous turning needs exactly it.

The inertia coupling is weak for this layout. The cycle eq. (2) runs through J, but the battery sits near the centre of mass, so J_{yy} moves only from 0.0612 to 0.0616 kg m^2 between the lightest and heaviest converged designs. The coupling that matters is the energy one; the dynamic one is present and small.

Dynamic clearance. The static clearance of table 9, 0.586 m, is not the clearance flown. During the turnaround the governed reference is held outside $\varrho = d_{\mathrm{safe}} + \mathcal{R} + \varepsilon$ of the head by

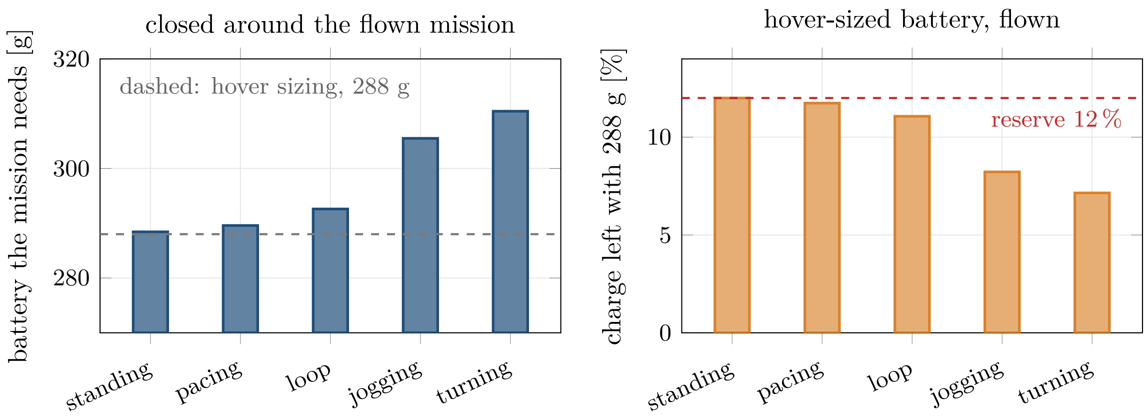


Figure 66: The coupled closure against the hover sizing. Left: the battery each motion needs when eq. (139) is solved around the flown mission; the dashed line is the 288 g of the hover sizing. Right: the charge left at t_f when that hover-sized pack is flown through each motion instead. Every moving mission lands below the 12 % reserve. The two panels are the same fact from either side: the right is the violation, the left is its cure.

Table 11: The fully coupled closure eq. (139) on the final design, one row per motion. "Hover sizing" is the first-principles closure of eq. (71) with the mission assumed to be hover, i.e. the design of table 9. "Updates/sims" counts battery updates and total simulations including the verification flights; "reserve" is the state of charge left at t_f on the final flight; "rotor" the minimum eye-to-disk distance on it; "feasible" whether that flight met every path constraint.

Motion	Gross	Battery	vs. hover	Mean power	$\varkappa$	Upd./sims	Reserve	Rotor	Feasible
hover sizing	1.724 kg	288 g	n/a	106.0 W	n/a	n/a	12.0 %	n/a	n/a
standing	1.724 kg	288 g	+0 g	106.1 W	1.000	3/5	12.03 %	0.73 m	yes
pacing	1.725 kg	290 g	+1 g	106.5 W	1.003	3/5	12.03 %	0.72 m	yes
walking a loop	1.729 kg	293 g	+4 g	107.6 W	1.011	3/5	12.03 %	0.73 m	yes
jogging	1.745 kg	305 g	+17 g	112.3 W	1.043	3/5	12.03 %	0.73 m	yes
turning around	1.751 kg	310 g	+22 g	114.2 W	1.055	3/5	12.03 %	0.61 m	yes

eq. (123), the vehicle tracks it to within 64 mm, and the nearest rotor disk passes the eye at 0.61 m (envelope gap 0.52 m). An earlier version of the governor, with a keep-out on position only and a velocity test that ignored the head's own motion, let the vehicle inside the safe distance by about 4 cm on this mission; the margin ε and the moving-boundary barrier are what closed that gap, and proposition 12.11(iv) states exactly the hypothesis under which they do.

17.10 The tether

Taking the battery off the aircraft removes the mission time from the closure. What replaces it is a conductor whose mass follows from Ohm's law: at bus voltage V delivering power P over length L with fractional drop δ,

$$I = \frac{P}{V}, \quad R_{\text{cond}} = \frac{2\rho_e L}{A_c}, \quad IR_{\text{cond}} \le \delta V \implies A_c \ge \frac{2\rho_e LP}{\delta V^2}, \quad m_c = 2\rho_m L A_c, \tag{146}$$

with A_c taken as the largest of the resistive requirement, the continuous current density limit $I/J_{\max}$ with $J_{\max} = 10\,\text{A/mm}^2$, and a handling floor of $0.13\,\text{mm}^2$. Figure 67 shows how the three requirements trade over power.

Observation 17.2. Equation (146) contains no mission time, so endurance ceases to be a design variable and becomes an operating choice. It does not delete the fold. A carried conductor sized by resistance has mass proportional to $P \propto m^{3/2}$, which is the same term as the battery's with a coefficient smaller by orders of magnitude at these lengths and voltages: the fold moves far away, it does not vanish. The $m^{3/2}$ term is absent only when the carried cable mass stops depending on power, because the tether is supported from above or because a gauge floor binds. For the case below the lumped closure is gauge-limited (yes), which is why its β_{tether} is zero.

Redesigning the aircraft for a 5 m lead at 48 V, with the same component models as the battery closure, gives 1.617 kg at 97.9 W and 55.5 dBA at the ear. This is not the battery design with the battery removed: propulsion and structure are resized around the lighter aircraft, and it still carries a 184 g landing reserve, sized by the peak power it must deliver (power-limited) rather than by energy. The current is 2.28 A; the conductor is set by ampacity at $0.23\,\text{mm}^2$ per leg and 37 g for the lead. 48 V remains below the 60 V DC safety-extra-low-voltage threshold, and the design meets the convergence and current-density checks the engine applies.

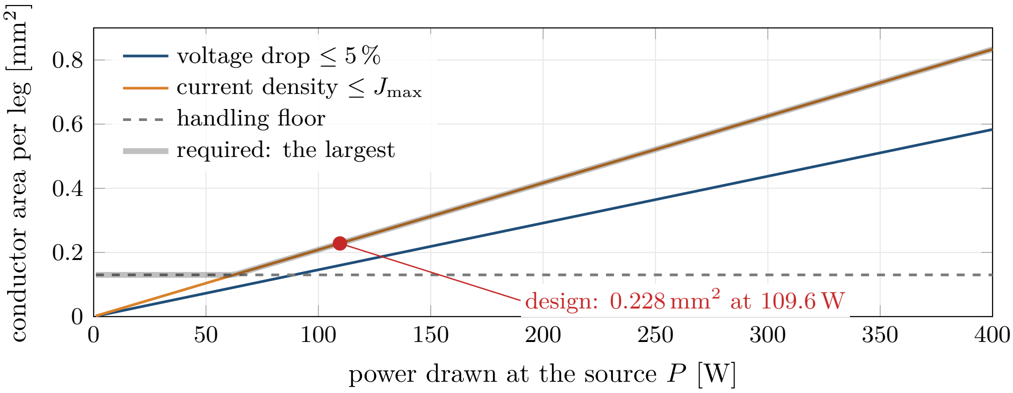


Figure 67: Conductor area per leg for a 5 m lead at 48 V, against the power drawn at the source. Each requirement of eq. (146) is linear in P or constant, and the conductor is sized by their maximum (grey band): the handling floor at low power, the current density limit above about 60 W. At this length and voltage the voltage-drop requirement, the only one that grows with L^2, never binds. The design point lies on the ampacity line, so the carried conductor mass is proportional to power, which is the $m^{3/2}$ term that keeps the fold.

18 Limitations

Every result in section 17 is the output of a model. This section states where the model is known to be inadequate, ordered by how much it would change the conclusions.

18.1 Recirculation

Model 3.1 assumes quiescent inflow. By the exchange-time calculation of eq. (86) (fig. 34) the machine passes a room's volume of air through its disks every 21 s, so within a minute of a thirty-minute session the assumption is false. The correction is not a coefficient; it requires solving the room flow with a moving boundary. Two consequences are unquantified: the degradation of rotor performance in its own recirculated wake, and the persistent room-scale draught of order 0.1 m/s. This is the largest open item.

18.2 Rotor aerodynamics and the flight power model

The rotor model is momentum theory plus a single profile term plus three near-field corrections. There is no blade element momentum theory, so the radial distribution of inflow and loading is not resolved and the effects of twist and taper enter only through lumped constants. There is no wake model, no blade-vortex interaction, and no unsteady inflow: by assumption 2.6 the inflow responds instantaneously, whereas its true time constant is of order $R/v_i \approx 0.1$ s, comparable with the attitude loop. The descent regime is excluded entirely, as noted after eq. (16).

The flight simulation compounds this. Its power law eq. (98) is calibrated at hover and has no inflow term, so it returns the same power for a rotor at a given speed whether the vehicle is hovering or translating. The axial correction derived in section 3 is therefore not in the mission energies of tables 10 and 11. Those tables close the energy balance around a near-hover approximation of the power; that is a computational coupling, and it is exercised honestly, but it is not a validation of manoeuvring energy. The figures are appropriate for indoor following at walking speed and vertical rates of a few tenths of a metre per second, and are not to be read as predictions for take-off, descent or aggressive flight. Coupling an axial and oblique inflow model consistently to both thrust and power is the next step, and it must be done to both at once: a climb term in the power alone, against a hover-calibrated thrust, would be inconsistent.

18.3 Acoustics

Three separate weaknesses. The source term eq. (73) is a broadband scaling law fitted to four aircraft of one manufacturer; the exponents are theoretically motivated but the anchor is empirical and the aircraft are all conventional two-blade designs, so extrapolation to $\sigma = 0.212$ paddle blades is an extrapolation. The room model eq. (77) uses a single frequency-averaged absorption coefficient and assumes a diffuse field, which fails below the Schroeder frequency eq. (81), i.e. around 200 Hz, which is exactly where the first blade harmonics lie. And tonal content is computed but not summed into the reported level, so a design with a strong 35 Hz fundamental and audible beating between rotors could be unacceptable at a level the model calls fine. A measured prototype is the only way to settle this.

18.4 Uncertainty

Every result is a single point estimate. The quantities that carry the most uncertainty are the figure of merit of a low-Reynolds paddle rotor, the motor mass constant, the screen drag coefficients, the electrical efficiency, the cell specific energy at the discharge rate used, and above all the acoustic anchor, whose calibration spread alone is 1.5 dB. No propagation of these uncertainties through the closure was performed, so the paper does not show that the design ranking of tables 7 and 8 is robust to them, nor that the candidate satisfies its constraints across their plausible ranges. The margins reported are margins of a nominal model. A proper treatment samples the parameter ranges, re-closes the design at each sample, and reports the fraction of samples for which each constraint holds.

18.5 Structures

Model 8.1 covers bending stress, tip deflection and the first bending mode of an idealised cantilever. It does not cover buckling, joints, bonded interfaces, fatigue, or crash loads. For a machine intended to operate beside a person, crash and containment analysis is not optional in practice, and none is performed here. Guard rings are modelled only as a mass; their ability to actually arrest a blade is asserted, not computed.

18.6 Failure modes

The controller of section 12 has no failure mode: the allocation eq. (119) assumes every rotor is available, and after the loss of one the quadrotor described here falls. That is a statement about this controller and not about quadrotors. Mueller and D'Andrea [19] show that a quadrotor can be controlled after the loss of one, two or even three propellers in a relaxed-hover mode in which the vehicle spins continuously about a tilted axis. That capability is not implemented here, and a continuously spinning 1.7 kg machine at arm's length from a person is not thereby shown to be acceptable.

A hexacopter is not the answer either, or not simply. With the engine's alternating-spin regular layout, removing one rotor and its opposite leaves an allocation matrix of rank 3 (section 16): the four remaining rotors cannot produce thrust, roll, pitch and yaw independently, so the standard allocation loses a wrench direction, and lift arithmetic alone would call for $\lambda \geq 1.5$ to hover on four rotors of six, not the $\lambda \geq 2$ an earlier draft stated. Re-optimised under the same limits as the quadrotor, the hexacopter (six rotors of 384 mm) comes out at 1.935 kg, 128.7 W and 57.6 dBA at the ear against 1.747 kg, 106.6 W and 56.5 dBA for the quadrotor (table 12), and its authority is lower on every axis: 43/52/3.3 against 56/59/4.4 rad/s^2. Its

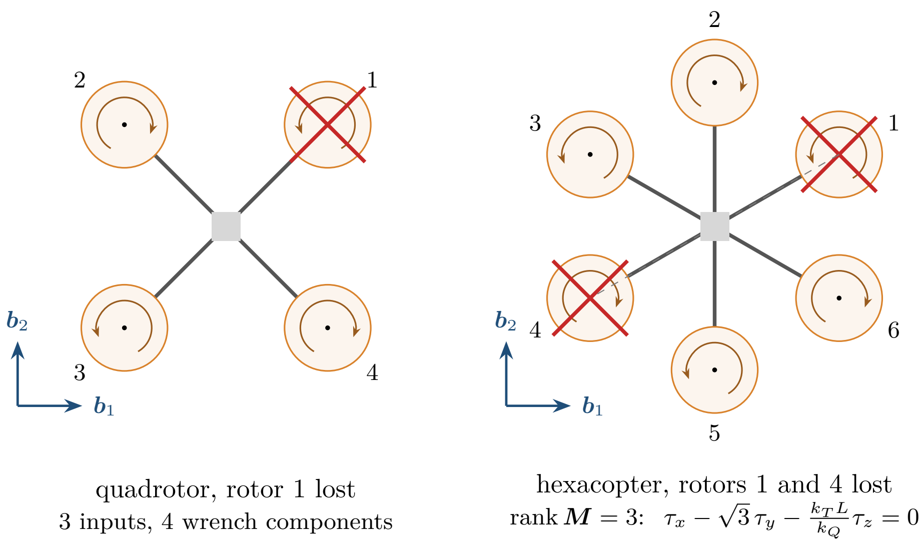


Figure 68: Rotor failure in the engine's layouts, top view, arcs showing each rotor's spin. Left: a quadrotor that loses a rotor has three inputs for four wrench components; with the remaining spins unbalanced it cannot hover at zero yaw rate, which is why the relaxed-hover controllers of [19] let it spin. Right: a hexacopter that loses the opposite pair 1 and 4 keeps four rotors, but their allocation columns satisfy $\boldsymbol{M}_{:,2} + \boldsymbol{M}_{:,5} = \boldsymbol{M}_{:,3} + \boldsymbol{M}_{:,6}$, so rank $\boldsymbol{M} = 3$ and every moment the four can produce satisfies the relation printed below it: roll, pitch and yaw can no longer be commanded independently.

Table 12: Quadrotor and hexacopter re-optimised under the same limits as table 7. Authorities by eq. (116); the last column is the ratio of roll to pitch torque authority.

Layout	D	Gross	Power	At ear	Span	α_{roll}	α_{pitch}	α_{yaw}	$\tau_{\text{roll}}/\tau_{\text{pitch}}$
quadrotor	498 mm	1.747 kg	106.6 W	56.5	1.27 m	56	59	4.4	1.00
hexacopter	384 mm	1.935 kg	128.7 W	57.6	1.23 m	43	52	3.3	0.87

weakest axis in angular acceleration is roll, not pitch: the roll torque authority is the smaller of the two (table 12, last column), because with two rotors on the pitch axis the yaw constraint couples into the roll differential, and the display slab makes J_{xx} exceed J_{yy}. The row sums $\sum_i |\boldsymbol{M}_{j+1,i}|$, which an earlier draft used and which predict pitch as the weak axis, mislead here (appendix B). The failure case therefore needs a different answer than more rotors, and none is offered here. Figure 68 shows both failures.

18.7 Dynamic clearance

Proposition 12.11 is a conditional guarantee, and each of its conditions is a limitation. It is enforced at update instants on the governed reference, not in continuous time and not on the vehicle. It transfers to the vehicle only through the hypothesis $\sup_t \|\boldsymbol{r} - \boldsymbol{r}_g\| \leq \varepsilon$, which table 10 checks for the missions flown and which nothing proves for missions not flown. It lapses whenever the caps and the keep-out cannot be satisfied together, which the engine reports as a violation rather than hides; on the reported runs this did not occur, but only because a mission whose ideal position lies inside the sphere is started on the sphere rather than inside it, a choice that removes the conflict from the start-up and not from the product. And it protects a sphere of radius $\mathcal{R}$ about the centre of mass; the rotor clearance reported alongside it is the physical quantity, and the difference between the two is the conservatism of

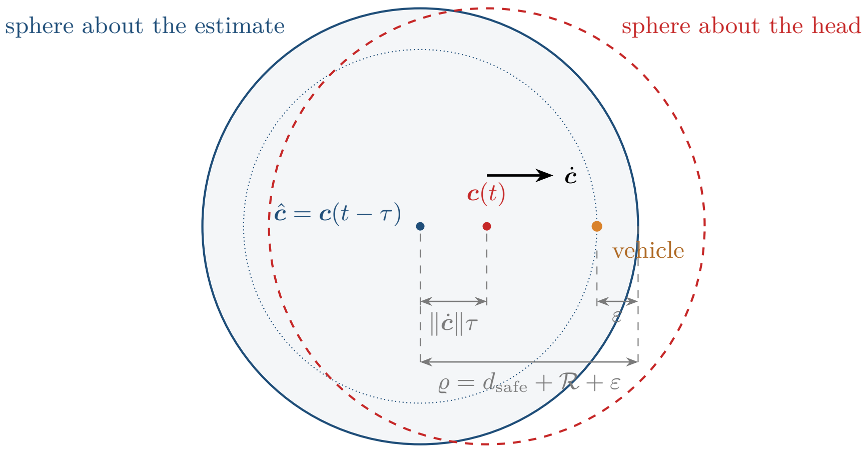


Figure 69: Why estimator latency erodes the keep-out guarantee, not to scale. The governor holds the reference outside the sphere of radius ϱ about the estimated head $\hat{\boldsymbol{c}} = \boldsymbol{c}(t-\tau)$, and the vehicle stays within ε of the reference. If the head has moved $\|\dot{\boldsymbol{c}}\|\tau$ since it was seen, the sphere is displaced by that much, and in the direction of motion the vehicle can be closer to the true head than proposition 12.11(iv) allows. At 1.4 m/s and 100 ms the displacement is 0.14 m, larger than the whole margin $\varepsilon = 0.10$ m.

the envelope, not a margin the design has earned.

18.8 Estimation and autonomy

Assumption 2.7 gives the controller exact knowledge of the state and of the user's position. Real vision-based human tracking has latency, dropouts, and occlusion, and the reference governor of model 12.10 is precisely where those errors would surface: a keep-out sphere centred on an estimated head position is only as safe as the estimate, and a latency of 100 ms at a walking speed of 1.4 m/s is 14 cm of sphere in the wrong place, more than the whole tracking margin ε. No estimator is modelled and no sensor noise is injected.

Figure 69 draws the effect.

18.9 Mission energy and the operating policy

The windowed extrapolation $E_{\text{mission}} = \bar{P}t_f$ of section 15 is exact only for a mission periodic within the window. For a session with distinct phases it is an approximation of unquantified error, and the battery of table 11 is sized for the worst single phase repeated for the whole session, which is a policy rather than a law. Reported endurance figures also assume constant e_b; real cells lose capacity at high discharge and low temperature, and the internal-resistance voltage sag that would reduce available thrust at low state of charge is not modelled.

18.10 Numerics

The integration is converged at the design step (table 4). The controller rate is a different matter: it defines the sampled-data loop, and doubling it moves the peak tracking error by 21.1 mm, so the rate simulated must be the rate the real autopilot runs at. The differential evolution of eq. (140) is stochastic and offers no optimality certificate: the reported designs are the best found, not the best that exists, and the exterior penalty of eq. (143) means a returned point may violate a constraint slightly, which table 7 reports as such. Runs are seeded for reproducibility, which makes results repeatable without making them global.

18.11 Scope

Nothing was built. The paper computes what a set of models implies, and the models were calibrated against published data for aircraft that are not this aircraft. The most valuable next step is not a better model but a measured rotor: a single 492 mm two-blade paddle propeller on a thrust stand in an anechoic space would settle table 5, proposition 4.5 and eq. (24) simultaneously, and those three carry most of the uncertainty in the result.

A Nomenclature

Dimensionless quantities carry the unit 1.

Table 13: Symbols.

Symbol	Unit	Meaning
Geometry and mass		
m	kg	gross mass
m_{fix}	kg	fixed payload and avionics
m_{str}	kg	structural mass
m_{prop}	kg	motors, speed controllers, propellers
m_{bat}	kg	battery pack
$S(m)$	kg	$m_{\text{str}}(m) + m_{\text{prop}}(m)$, the mass that scales with m
N	1	number of rotors
N_b	1	blades per rotor
D, R	m	rotor diameter, radius
A	m^2	total actuator disk area, $N\pi D^2/4$
L	m	arm length, hub to rotor centre
$\mathcal{R}$	m	reach, $L + R$, centre to nearest blade tip
ℓ	m	distance from the user's eye to the nearest rotor disk
c	m	blade chord
σ	1	solidity, $N_b c/(\pi R)$
AR	1	blade aspect ratio, R/c
J	$\text{kg}\,\text{m}^2$	inertia tensor about the centre of mass
I	m^4	second moment of area
Aerodynamics		
T	N	total thrust
v_i	m/s	induced velocity at the disk
w	m/s	far-wake velocity, $2v_i$
DL	Pa	disk loading, T/A
V_{tip}	m/s	tip speed, ΩR
C_T	1	thrust coefficient
C_T/σ	1	blade loading coefficient
C_{d_0}	1	blade profile drag coefficient
κ	1	induced power factor
κ_{int}	1	rotor interference factor
FM	1	figure of merit
ρ	kg/m^3	air density

Symbol	Unit	Meaning
$C_{D,n}, C_{D,t}$	1	screen drag coefficients, normal and tangential
Power and energy		
P_{ideal}	W	ideal induced power, Tv_i
$P_{\mathrm{ind}}, P_{\mathrm{prof}}$	W	induced and profile power
$P_{\mathrm{sh}}, P_{\mathrm{el}}$	W	shaft and electrical power
P_{aux}	W	non-propulsive electrical load
η	1	motor times speed-controller efficiency
e_b	J/kg	battery specific energy available to the mission (model 5.5)
p_b	W/kg	usable battery specific power
E	J	remaining battery energy
t_f	s	mission duration
ς	1	state-of-charge reserve at landing
$\Theta_p(m)$	s	longest mission a mass m closes, eq. (57)
$\hat{m}$	kg	mass at which Θ_p is largest
ϑ	1	propulsive share of total power at $\hat{m}$, eq. (59)
Closure		
α	1	thrust coupling, $1 - f_s - \gamma_m$
β	$\mathrm{kg}^{-1/2}$	energy coupling
M_0	kg	effective fixed load
$M_0^{\max}$	kg	critical fixed load at the fold
m^*	kg	fold mass
f_s	1	structural mass fraction
γ_m	1	propulsion mass fraction, $\lambda g/S_m$
γ_b	1	battery mass fraction
λ	1	thrust margin, $T_{\max}/mg$
S_m	N/kg	propulsion specific thrust
p	1	disk-area scaling exponent, $A \sim m^p$
q	1	power exponent, $(3-p)/2$
$\mathcal{F}$	kg	component closure residual, eq. (71)
$\mathcal{G}, \mathcal{H}$	kg	inner and outer maps of the co-design fixed point
Dynamics and control		
$\boldsymbol{r}, \boldsymbol{v}$	m,m/s	position, velocity
R, q	1	attitude as rotation matrix, quaternion
$\boldsymbol{\omega}$	rad/s	body angular velocity
$\boldsymbol{\omega}_d, \dot{\boldsymbol{\omega}}_d$	rad/s,rad/s^2	desired body rate and its derivative
$\boldsymbol{j}_d$	m/s^3	reference jerk
$\boldsymbol{h}_\omega$	rad/s	rate of the desired thrust direction, eq. (115)
Ω_i	rad/s	speed of rotor i
k_T, k_Q, k_P	1	rotor thrust, torque, power coefficients
$\boldsymbol{M}$	1	allocation matrix
Δ	rad^2/s^2	balanced squared-speed differential, eq. (116)
τ_m	s	rotor speed time constant
$\boldsymbol{e}_R, \boldsymbol{e}_\omega$	1	attitude and rate errors
K_R, K_ω	1/s^2,1/s	attitude gains
K_p, K_d, K_i	1	position gains

Symbol	Unit	Meaning
$\tau_j^{\max}, \alpha_j^{\max}$	$\mathrm{N\,m}$,$\mathrm{rad/s^2}$	torque and angular acceleration authority
$\boldsymbol{r}_g, \boldsymbol{v}_g$	m,m/s	governed reference
$\boldsymbol{c}$	m	user's head position
ϱ	m	keep-out radius, $d_{\text{safe}} + \mathcal{R} + \varepsilon$
ε	m	tracking margin inside the keep-out radius
$e_{\text{track}}, e_{\text{lag}}$	m	error against the governed and the raw reference
$\varkappa$	1	mission power overhead, $\bar{P}/P_{\text{hover}}$
Acoustics		
L_p, L_W	dB	sound pressure, sound power level
C	dB	acoustic anchor
Q	1	directivity factor
$\mathcal{R}_{\text{room}}$	$\mathrm{m^2}$	room constant
$\bar{\alpha}$	1	average absorption coefficient
r_c	m	critical distance
f_{BPF}	Hz	blade passage frequency

B Supplementary derivations

B.1 The cubic form of the closure

At $q = 3/2$, substituting $u = \sqrt{m} > 0$ into eq. (36) gives

$$\beta u^3 - \alpha u^2 + M_0 = 0. \tag{147}$$

Proposition B.1. *Equation* (147) *has exactly one negative real root for all* $\alpha, \beta, M_0 > 0$. *Its two remaining roots are real and positive if and only if* $M_0 \leq M_0^{\max}$*, in agreement with theorem 6.2.*

Proof. Write $\Pi(u) = \beta u^3 - \alpha u^2 + M_0$. By Vieta, the product of the three roots is $-M_0/\beta < 0$, so either one or three roots are negative. Since $\Pi(0) = M_0 > 0$ and $\Pi(u) \to +\infty$ as $u \to +\infty$, and $\Pi'(u) = 3\beta u^2 - 2\alpha u = u(3\beta u - 2\alpha)$ has roots $u = 0$ and $u = 2\alpha/(3\beta)$, the function decreases on $(0, 2\alpha/(3\beta))$ and increases thereafter, so it has at most two positive roots. Hence exactly one root is negative. The local minimum value is

$$\Pi\left(\frac{2\alpha}{3\beta}\right) = \beta\frac{8\alpha^3}{27\beta^3} - \alpha\frac{4\alpha^2}{9\beta^2} + M_0 = \frac{8\alpha^3 - 12\alpha^3}{27\beta^2} + M_0 = M_0 - \frac{4\alpha^3}{27\beta^2} = M_0 - M_0^{\max},$$

using eq. (47). Two positive real roots exist iff this is non-positive, i.e. iff $M_0 \leq M_0^{\max}$. Note also that $u^* = 2\alpha/(3\beta)$ satisfies $(u^*)^2 = 4\alpha^2/(9\beta^2) = m^*$, as required for consistency. ☐

Figure 70 draws Π in the three cases.

B.2 Derivative of the closure solution with respect to endurance

Differentiating $F(m(t_f)) = M_0(t_f)$ implicitly, with $\beta = \beta_0 t_f$ and $M_0 = m_{\text{fix}} + P_{\text{aux}} t_f / e_b$,

$$F'(m)\,\frac{\mathrm{d}m}{\mathrm{d}t_f} = \frac{\partial M_0}{\partial t_f} + m^{3/2}\frac{\partial \beta}{\partial t_f} = \frac{P_{\text{aux}}}{e_b} + \beta_0 m^{3/2}, \tag{148}$$

so

$$\frac{\mathrm{d}m}{\mathrm{d}t_f} = \frac{P_{\text{aux}}/e_b + \beta_0 m^{3/2}}{\alpha - \frac{3}{2}\beta\sqrt{m}}. \tag{149}$$

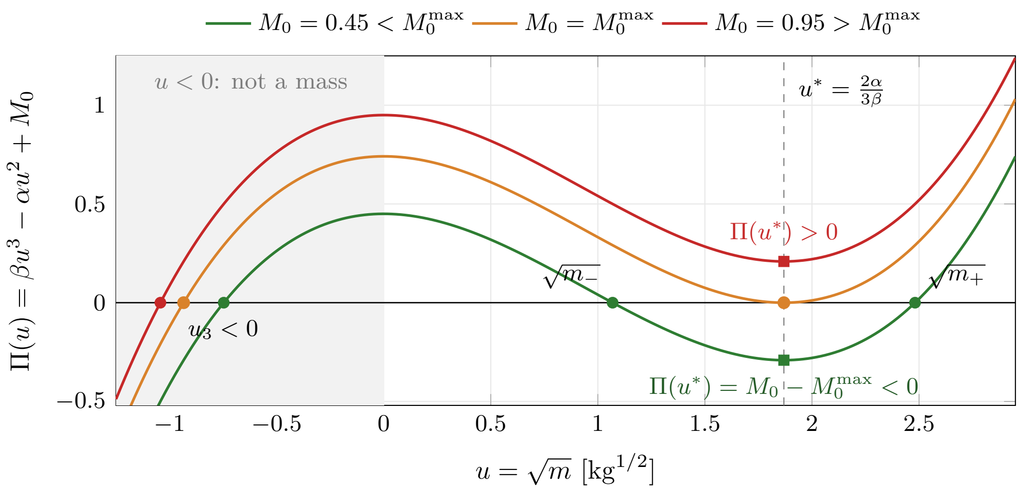


Figure 70: The cubic eq. (147) in $u = \sqrt{m}$ for the worked example of section 5. $\Pi(0) = M_0 > 0$ is a local maximum and Π has its local minimum at $u^* = 2\alpha/(3\beta)$, where $(u^*)^2 = m^*$ and $\Pi(u^*) = M_0 - M_0^{\max}$. One root is always negative and is not a mass. The two positive roots $\sqrt{m_\pm}$ exist exactly when the minimum is not above the axis, and they merge at u^* when $M_0 = M_0^{\max}$.

The denominator is $F'(m)$, which vanishes at the fold; eq. (149) therefore diverges there, quantifying the remark following theorem 6.6. Both terms in the numerator are positive. The denominator is positive on the light branch and negative on the heavy branch (lemma 6.1), so the light root increases with endurance and the heavy root decreases: as t_f grows the two branches approach each other and merge at the fold.

B.3 Rotor speed at maximum thrust

For a fixed-pitch rotor at fixed collective, $T = k_T\Omega^2$, so the speed required for the sizing thrust $T_{\max} = \lambda mg/N$ relative to hover $T_h = mg/N$ is

$$\frac{\Omega_{\max}}{\Omega_h} = \sqrt{\frac{T_{\max}}{T_h}} = \sqrt{\lambda}. \tag{150}$$

Hence hover occurs at $1/\sqrt{\lambda}$ of full rotor speed: at $\lambda = 1.8$ this is 74.5%, which is the figure quoted in section 17 and which also fixes the voltage headroom the pack must provide.

B.4 Torque authority of the symmetric quadrotor

Holding total thrust at mg and the other two moments at zero, the largest moment about body axis j is the value of the linear program

$$\max\ \tau_j \quad \text{s.t.} \quad \boldsymbol{M}\boldsymbol{u} = (mg,\ \tau_j\boldsymbol{e}_j), \qquad 0 \le u_i \le \Omega_{\max}^2, \tag{151}$$

and its counterpart for $-\tau_j$; the authority is the smaller of the two. The engine solves eq. (151) directly for every layout. For the $\times$ quadrotor with alternating spins it has a closed form.

Let $a_i = \boldsymbol{M}_{j+1,i}$. For each of the three moment rows $|a_i| = a$ is the same for every rotor ($k_T L/\sqrt{2}$ for roll and pitch, k_Q for yaw), and the sign vector $s_i = \operatorname{sign} a_i$ is orthogonal to the thrust row and to the other two moment rows. Write $\boldsymbol{u} = \Omega_h^2\mathbf{1} + \boldsymbol{d}$. The thrust constraint is $\sum_i d_i = 0$, and $\tau_j = a\sum_i s_i d_i = 2aD$ with $D = \sum_{s_i=+1} d_i = -\sum_{s_i=-1} d_i$. Each of the two increments is at most $\Omega_{\max}^2 - \Omega_h^2$ and each of the two decrements at most Ω_h^2, so $D \le 2\Delta$ with

$$\Delta = \min\{\Omega_h^2,\ \Omega_{\max}^2 - \Omega_h^2\}. \tag{152}$$

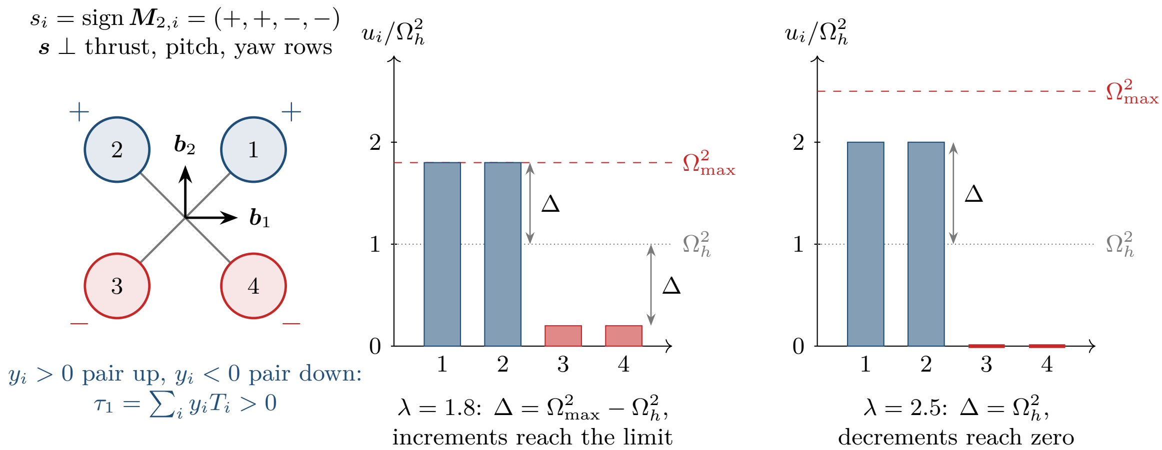


Figure 71: The balanced differential that attains the torque authority of the $\times$ quadrotor, drawn for roll. The sign vector $\boldsymbol{s}$ of the roll row is orthogonal to the thrust, pitch and yaw rows, so $\boldsymbol{u} = \Omega_h^2 \mathbf{1} + \Delta \boldsymbol{s}$ changes the roll moment and nothing else. Δ is the smaller of the room to speed rotors up and the room to slow them down: at $\lambda = 1.8$ the increments reach $\Omega_{\max}^2$ first, at $\lambda = 2.5$ the decrements reach zero first. Either way each of the four rotors contributes Δ, which is why eq. (153) carries no factor $1/N$.

The bound is attained by $\boldsymbol{d} = \Delta \boldsymbol{s}$, which satisfies every constraint by the orthogonality, hence

$$\tau_j^{\max} = 4a\Delta = \Delta \sum_{i=1}^{4} |\boldsymbol{M}_{j+1,i}|, \tag{153}$$

which is eq. (116). There is no factor $1/N$: the thrust sum is held by pairing increments with decrements, not by dividing the moment. For $\lambda \leq 2$, $\Delta = (\lambda - 1)\Omega_h^2$. Figure 71 shows the two cases of the minimum.

The closed form uses the equal-magnitude and orthogonality properties of this layout and should not be extended blindly. For the hexacopter with $\psi_0 = \pi/6$ the roll row has $\sum_i |\sin \psi_i| = 4$ and the pitch row $\sum_i |\cos \psi_i| = 2\sqrt{3}$, because two rotors sit on the pitch axis and contribute nothing to pitch; the magnitudes are unequal and the LP must be solved. The same holds for any layout with a failed rotor.

B.5 Sound power to pressure with the reference constants

The reference intensity implied by $p_0 = 20\,\mu\text{Pa}$ is $I_0 = p_0^2/(\rho_0 c_0)$. At $\rho_0 = 1.225\,\text{kg/m}^3$ and $c_0 = 343\,\text{m/s}$, $\rho_0 c_0 = 420\,\text{Pa s/m}$ and $I_0 = (2 \times 10^{-5})^2/420 = 9.52 \times 10^{-13}\,\text{W/m}^2$, against the nominal $W_0/1\,\text{m}^2 = 1 \times 10^{-12}\,\text{W/m}^2$. The discrepancy is $10 \log_{10}(1/0.952) = 0.21\,\text{dB}$, which is why eq. (74) is stated as an equality: the reference constants are chosen to make it one to within a fifth of a decibel at standard conditions.

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